

The Space Economy at a Glance 2026



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Foreword

Space activities and applications are increasingly relevant to the economic and policy priorities globally. They support critical infrastructures, enable new sources of data and services, and contribute to innovation, resilience and international co-operation. As the space economy evolves, governments need robust evidence to understand changing capabilities, market dynamics and policy challenges, and to ensure that space activities deliver broad-based benefits while remaining safe, secure and sustainable.

This publication contributes to the OECD's programme of work on science, technology and innovation, carried out under the Committee for Scientific and Technological Policy (CSTP). It draws on the longstanding activities of the OECD Space Forum, which brings together governments and space agencies to improve the international evidence base on the space economy and to support policy dialogue on emerging issues. Building on earlier OECD work on the measurement, impacts and governance of space activities, *The Space Economy at a Glance* provides a concise and comparative overview of recent developments across countries and value chains.

The purpose of this report is to help policy makers assess the factors shaping competitiveness in the space economy, including investment, industrial capabilities, research and innovation partnerships, entrepreneurship, supply chains and access to finance. It also points to areas where new capabilities may emerge, and to the policy questions raised by the expanding scale and diversity of space activities, including orbital congestion, space traffic management, debris and environmental externalities.

By bringing together indicators, analysis and policy insights, this report aims to support stakeholders in the space sector, and in particular governments, in designing policies and actions that strengthen innovative, resilient and responsible space ecosystems. It also contributes to the OECD's broader objective of supporting countries in harnessing science, technology and innovation for economic growth and prosperity for all.

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Editorial

Space has entered a new phase of economic and strategic importance. Long associated with scientific ambition, national prestige and security, it now underpins an expanding range of services, infrastructures and capabilities that are increasingly embedded in the functioning of modern economies and societies. As public and private investments grow, and as more countries seek to build or strengthen their presence in the sector, space is becoming more firmly established as both an area of economic opportunity and a domain of public policy relevance.

This evolution brings with it a new set of policy considerations. Reliable access to orbit, the development and operation of strategic capabilities, and the ability to maintain secure, resilient and sustainable space-based systems are no longer concerns confined to specialised communities. They are becoming more closely connected to broader questions of industrial competitiveness, technological capacity, infrastructure resilience and long-term economic performance. At the same time, the growing scale and diversity of space activities raise important challenges, including governance related to orbital environments and the need for effective international co-operation.

These developments are closely relevant to OECD's wider work on science, technology and innovation. Across its analysis of productivity, industrial transformation, critical technologies and resilience, the OECD has consistently highlighted the importance of understanding how technological domains reshape economic structures and policy priorities. In the case of space, this requires stronger evidence, improved measurement and a clearer understanding of how space-related investments, capabilities and applications diffuse across economies and societies.

The Space Economy at a Glance contributes to this effort. Drawing on the OECD's long-standing analytical and statistical work in the field, it provides a concise overview of key trends, actors and policy issues shaping the space economy today. In doing so, it aims to support governments and stakeholders in assessing the role of space in innovation, growth, resilience and international co-operation, and in situating the space economy within wider economic policy objectives. The report draws on the longstanding activities of the OECD Space Forum, which brings together governments and space agencies to improve the international evidence base on the space economy, as well as the collective work on science, technology and innovation policy carried out under the Committee for Scientific and Technological Policy (CSTP).

Jerry Sheehan

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Reader's guide

Rationale and use of *The Space Economy at a Glance*

The space economy is evolving rapidly, with implications that extend beyond the space sector itself. Governments are seeking to understand how public investment, technological capabilities, commercial activity and growing reliance on space-based infrastructure shape economic performance, strategic autonomy, public-service delivery and resilience. This report responds to that need by providing a concise, comparative, and evidence-based overview of recent developments in the space economy. Its coverage is selective rather than exhaustive, focusing on areas where new indicators and policy-relevant evidence are available, while pointing to areas where further measurement work will be needed.

The publication is designed primarily for policy makers, space agencies, statistical offices, regulators and public institutions involved in science, technology, innovation, industrial, digital, environmental and resilience policies. It may also be useful to stakeholders in industry, research organisations and international bodies seeking a clearer view of how space activities are changing and how their benefits and risks are distributed across economies and value chains.

The report brings together new OECD indicators and analysis across several dimensions of the space economy, including public space budgets, civil space R&D, policy instruments, private finance, scientific publications, patents, orbital activity, satellite-enabled services, market concentration and supply-chain dependencies. It also draws on emerging measurement approaches, including thematic accounts and indicators of satellite-service uptake, to improve understanding of how space activities contribute to national economies and public value.

The report's contribution is to connect several dimensions that are often analysed separately: public investment and market-shaping policies; scientific and technological capabilities; changing market structures and commercial dynamics; and the risks linked to growing dependence on space infrastructure. By doing so, it aims to help governments assess not only how to support a more innovative and competitive space economy, but also how to preserve open access to essential data and signals, safeguard orbital resources, manage strategic dependencies and embed resilience across the wider economy.

Structure and contents of the report

The Space Economy at a Glance is organised around four thematic chapters that together examine the strategic foundations of the space economy, the evolution of innovation capabilities, changing market structures and the vulnerabilities that accompany growing dependence on space systems. The report is complemented by a series of country profiles showcasing selected OECD Space Forum members and illustrating how national space economies, policy priorities and institutional arrangements differ across countries.

Statistical coverage

The indicators presented in *The Space Economy at a Glance* cover different aspects of the space economy, which the *OECD Handbook on Measuring the Space Economy* defines in the following way:

The space economy is the full range of activities and the use of resources that create and provide value and benefits to human beings in the course of exploring, understanding, managing and utilising space. Hence, it includes all public and private actors involved in developing, providing and using space-related products and services, ranging from research and development, the manufacture and use of space infrastructure (ground stations, launch vehicles and satellites) to space-enabled applications (navigation equipment, satellite phones, meteorological services, etc.) and the scientific knowledge generated by such activities. It follows that the space economy goes well beyond the space sector itself, since it also comprises the increasingly pervasive and continually changing impacts (both quantitative and qualitative) of space-derived products, services and knowledge on economy and society.

This provides the starting point for adapted definitions in the fields mentioned below, which may be further refined depending on data availability. Capturing and delimiting statistics associated with the exploitation of space data and signals remains a recurring challenge.

Government space budgets

Government space budgets cover activities whose primary objective is civilian or dual-use (i.e. not exclusively military). For example, the US Global Positioning System is considered dual use because it is available both for defence and civilian uses. Military space programmes play an important role in the space economy and are mentioned several times in this report, but because they are only rarely identifiable in government budgets, they have been excluded from international indicators to facilitate international comparability.

The number of activities funded by government space budgets varies greatly across countries and can include all or some of the following:

- the manufacture and operation of space infrastructure
- the use of satellite data, signals and related equipment (navigation equipment, satellite phones, communication services, geospatial intelligence, etc.)
- associated research and development, both basic and applied
- exploitation of scientific knowledge generated by the above activities
- contributions to international space-related programmes.

The indicator uses revised budget estimates for 2025 and budget expenditure for previous years. It only includes resources met from national/federal government sources, excluding local/regional government budgets and the private sector. In cases where fiscal years do not correspond to calendar years, fiscal years starting in the period 1 January-30 June are converted to the first calendar year in the budget period, e.g. 2025 for fiscal year 2025/26, while those starting 1 July-31 December are attributed to the second year, e.g. 2026 for fiscal year 2025/26.

Space-related patents

The space-related patent analysis is based on updated data from previous patent studies jointly developed and conducted by the European Patent Office, the European Space Agency and the European Space Policy Institute. The following technology categories are included, identified thanks to a combination of International Patent Classification codes and keywords: propulsion, spacecraft electrical power, space system control, life support, space debris and quantum in space. The studies and queries can be found on the European Patent Office's space innovation platform.

Space-related scientific publications

The bibliometric analysis of space-related scientific publications draws on 254 journals and conference proceedings indexed in Elsevier's Scopus database. The dataset covers 491 496 fractional counts of scientific papers published between 2009 and 2024. The journal selection is based on a combination of All Science Journal Classification codes and journal names (primarily including astrophysics, space science, aerospace engineering and remote sensing). Only journals with a SCImago Journal Rank indicator (a measure of journal prestige) are included in the analysis.

Space-related official development assistance

Space-related official development assistance indicators rely on projects in the Creditor Reporting System database of the OECD Development Co-Ordination Directorate. Projects were identified thanks to keywords and were tagged as "space-related" if they 1) use or support space science, space infrastructure and/or satellite data and signals; or 2) provide space-related administrative support or training. The analysis is based on 2 712 projects between 2010 and 2023 and their associated commitments.

Comparability over time

The indicators in *The Space Economy at a Glance* are the result of a continuous process of methodological improvement aimed at improving the robustness and international comparability of the indicators. As a result, data across different editions of *The Space Economy in Figures* and *The Space Economy at a Glance* may not be comparable. To analyse time trends, it is preferable to use the data for different time periods from the latest edition of *The Space Economy at a Glance*. All comparisons over time presented in this report are based on annual revisions of historical data and the methodological improvements which have been implemented in this edition.

Country coverage

This report features data on the space economy from OECD countries and selected non-OECD economies (e.g. Brazil, India, Thailand) with space programmes and transparent reporting of budgets and activities. The following OECD countries provided budget data (Australia, Canada, France, Germany, Italy, Korea, Luxembourg, the Netherlands, Norway, Switzerland and United Kingdom), in addition to Thailand. Data for the remaining countries have been collected from both national and international budget sources, including financial data from the European Space Agency, Eumetsat and the European Union, which for many countries represent the bulk of government civilian space budgets.

Acronyms and abbreviations

ASI	Italian Space Agency (Italy) <i>Agenzia Spaziale Italiana</i>
ASJC	All Science Journal Classification
BRIICS	Brazil, Russian Federation, India, Indonesia, China, South Africa
CAD	Canadian dollar (currency)
CNES	National Centre for Space Studies (France) <i>Centre National d'Études Spatiales</i>
CEOS	Committee on Earth Observation Satellites
CHF	Swiss franc (currency)
COTS	Commercial Orbital Transportation Services
CSA	Canadian Space Agency
DLR	German Aerospace Center <i>Deutsches Zentrum für Luft- und Raumfahrt</i>
EGNOS	European Geostationary Navigation Overlay Service
EO	Earth observation
ESA	European Space Agency
EUMETSAT	European Organisation for the Exploitation of Meteorological Satellites
EUR	Euro (currency)
GBARD	Government budget allocations for research and development
GBP	British pound sterling (currency)
GDP	Gross domestic product
GEO	Geostationary orbit
GNSS	Global Navigation Satellite system
GPS	Global Positioning System
ICT	Information and communication technologies
IP	Intellectual property
ITU	International Telecommunication Union
JAXA	Japan Aerospace Exploration Agency
JPY	Japanese yen (currency)
KASA	Korea AeroSpace Administration

KRW	Korean won (currency)
LEO	Low-earth orbit
NASA	National Aeronautics and Space Administration
NEA	Nuclear Energy Agency
NICFI	Norway's International Climate and Forest Initiative
NOK	Norwegian krone (currency)
ODA	Official development assistance
OECD	Organisation for Economic Co-operation and Development
PCT	Patent Cooperation Treaty
PNT	Positioning, navigation and timing
PPP	Purchasing power parity
R&D	Research and development
RHU	Radioisotope heater unit
RTG	Radioisotope thermoelectric generator
STI	Science, technology and innovation
USD	US dollar
USSR	Union of Soviet Socialist Republics

Executive summary

Space activities' economic footprint is becoming larger and more visible

Once considered a niche domain, the space economy is now a growing industrial force supporting innovation, new services and public policy objectives. Global space-related revenues are estimated at USD 550-600 billion in 2025, depending on scope, approaching the size of the global semiconductor industry (value at about USD 620 billion in 2024).

Space is also a measurable component of national economies. Harmonised accounts show that in the United States the sector generated USD 142.5 billion in GDP in 2023, or 0.5% of the total, as well as USD 240.9 billion in gross output and 373 000 private-sector jobs. In Italy, the second country to develop such accounts, space activities generated 0.14% of GDP, EUR 8 billion in output and more than 23 000 jobs. The OECD support countries' measurement strategies and developing comparable space accounts.

The largest economic value lies downstream, where satellite data and services are embedded in everyday economic activity. Space-based systems support more than half of the most critical infrastructures and services across OECD countries, including transport, energy, communications, and food supply. This dependence is economically significant and in some advanced economies, close to one fifth of GDP is supported by satellite-enabled services (i.e. estimated to support around 18% of UK GDP). Satellites also provide more than 90% of the observations assimilated into weather forecasting systems.

This is amplified by greater data availability and expanding connectivity. Around 70% of data from active earth observation missions is now openly available, enabling businesses of all sizes to develop applications in agriculture, transport, finance, infrastructure management and emergency response. Satellite broadband represented only 0.7% of OECD fixed broadband subscriptions in 2023 but low-earth orbit constellations are accelerating connectivity growth by improving latency, speed and coverage.

The space economy is scaling through infrastructure expansion, wider international participation and increasing commercial activity

This role is underpinned by rapid expansion of space infrastructure. More than 14 000 operational satellites were in orbit by end-2025, rising to nearly 15 000 by mid-2026, following a doubling between 2020 and 2022; International Telecommunication Union filings and constellation projects point to a much larger pipeline of more than 2 million possible future satellites, though only a fraction is likely to be deployed. The Global Exploration Roadmap to 2050, developed with input from 27 space agencies, sets out the Moon as the near-term focus with dozens of planned missions, and Mars as a longer-term objective.

Space has become a truly global activity, but launch access remains concentrated. 109 countries had launched at least one satellite by 2025, as rideshare launches, CubeSats and commercial procurement reduced barriers to entry. Orbital launch activity in 2025 was still conducted by a few actors: 181 US launch attempts, 92 from China and 8 from Europe, even as around 220 small- and micro-launcher initiatives are tracked globally.

The space economy is increasingly commercially operated, but remains centred on a few key actors. Private operators accounted for 88% of satellites launched in 2025, up from 23% in 2010, and telecommunications for 82%; one operator alone accounted for around 60% of all active satellites. Growth does not yet translate evenly into profitability: some emerging manufacturers still invest more than 60% of revenues in R&D and capital expenditure.

Public demand, defence priorities and private capital are shaping growth

Governments remain central market shapers, combining rising budgets with a growing toolkit for commercialisation. OECD civilian space budgets rose 15%, from USD 40.5 billion in 2022 to USD 46.4 billion in 2025, while the OECD STIP Compass Space Economy Policies portal identified 212 space-related policy initiatives and 289 policy instruments in 2026 across 45 countries, the European Space Agency and the European Union. The number of countries with space regulations in place has increased by 50% over the past decade. Public support is also more geographically diverse, with new agencies and commercialisation support emerging across Asia-Pacific, Africa, Latin-America and the Middle East.

Public demand is increasingly shaped by defence and security requirements, although comparisons remain difficult as many programmes are classified or dual-use. In 2025, military programmes accounted for 46.3% of government space spending in the United States, 25% in France and 14.8% in Japan; Japan's defence-space spending rose almost sevenfold between 2022 and 2025, and Germany announced plans to invest EUR 35 billion in space-related defence capabilities by 2030.

Private financing is broadening the capital base. Investment, including venture capital, is estimated to have reached USD 11-13 billion in 2025, its highest level since 2021. An OECD sample of 40 recently listed space companies raised more than USD 100 billion over the past decade through new share and bond issuance.

Space innovation is accelerating, while the rapid development of the space economy is creating new risks and governance challenges

OECD space innovation remains robust, even as space science becomes more multipolar. Space-related scientific output has more than doubled since 2009, with participation in global research networks expanding from 80 countries in 2000 to 119 in 2024, with collaboration links nearly tripling. Technological development remains concentrated in leading OECD economies. The United States accounted for 34.2% of space-related patent applications over 2020-23, followed by the EU27, Japan and China. Space innovation is increasingly focused on AI-enabled analytics, quantum applications, debris mitigation, in-orbit systems and advanced power technologies. Spacecraft electrical power alone represented 46% of space-related patent applications in 2023.

Expansion is also creating resilience and sustainability risks, from orbital debris and traffic management to cyber threats, jamming and supply-chain vulnerabilities. Around 45 860 debris objects are regularly tracked in orbit, while more than 140 million smaller objects are too small to track but still potentially damaging. These risks are compounded by vulnerabilities in space-enabled services, such as GNSS interferences affecting European air transport rising more than 22-fold between 2021 and 2025, and by concentrated supply chains, as nearly half of 29 crucial space-related inputs depend on a single country for at least two-thirds of global production (titanium, gallium, germanium, and rare earth elements).

The policy challenge is evolving from supporting growth to governing scale and complexity. As space systems become embedded in critical infrastructure, governments will need to strengthen statistical measurement, promote competitive and resilient markets, improve traffic co-ordination and debris mitigation, address supply-chain dependencies, and keep space-enabled services secure and accessible.

1

Strategic capabilities and competitiveness in the space economy

This chapter examines why space has become a strategic economic and policy domain, and how governments shape national competitiveness in the space economy. It highlights the continued importance of public investment, R&D support, procurement, regulation and institutional frameworks, while also considering the growing role of private finance, firm-level R&D and entrepreneurship-oriented policy instruments. Together, these factors determine how countries build, retain and deploy the capabilities needed to benefit from space systems and services.

The case for space: strategic capabilities, economic benefits and the need for strong policy engagement

Space is no longer a specialised domain at the margins of the economy. Space systems underpin essential functions in modern societies, including communications, navigation, weather forecasting, disaster management, financial transactions, transport, agriculture and defence. This makes space both an economic activity and a strategic capability: countries need reliable access to space-based infrastructure not only to support growth and innovation, but also to safeguard resilience, autonomy and security.

The global space economy generates around half a trillion US dollars in annual revenues, across a wide range of activities and market segments and is growing. Despite differences in scope, coverage and methodology, available estimates converge on the same broad picture. Published estimates vary from USD 544.3 billion in commercial space revenues, up 13% (Space Foundation, 2026^[1]), to EUR 564 billion, up 7% (European Space Agency, 2026^[2]), reflecting strong growth dynamics despite methodological differences across measurement approaches. The space economy is approaching the scale of the global semiconductor industry, valued at approximately at about USD 620 billion in 2024.

Governments remain central to the development of the space economy. Many space R&D activities are characterised by high upfront investment, long development cycles and technological uncertainty, associated with small and regulated markets. Furthermore, space activities support public services and often generate broad public benefits, such as scientific discoveries, environmental monitoring, national security, and disaster alerts. Finally, despite successive waves of liberalisation since the 1980s, several space capabilities and products are subject to heavy government regulation and export controls. Markets alone are therefore unlikely to provide the volume, continuity or strategic direction of investment required. Public procurement, mission funding and R&D support help sustain critical capabilities, reduce technological and market uncertainty, create early demand and crowd in private investment.

Public support also has an industrial and competitiveness rationale. Space programmes contribute to high-technology manufacturing, advanced engineering, digital services, R&D investment and skilled employment. Space applications generate linkages across key sectors, including transport, energy, agriculture, insurance, environmental services and defence. The policy challenge is therefore not simply to “grow the space sector”, but to ensure that national economies are able to develop, retain and deploy the capabilities needed to benefit from space-enabled infrastructure, services and innovation.

New statistical thematic accounting approaches are making the economic footprint of space more visible in national economies. This matters because space activities are distributed across aerospace, electronics, telecommunications, software, R&D, public administration, trade and services, and can therefore only be partially isolated from more aggregated categories in official industry statistics. OECD work, including the *Handbook on Measuring the Space Economy*, has helped provide a common conceptual basis for this measurement agenda. Over the years, countries have used this handbook as a baseline for measurement, with some developing dedicated industry surveys that generate detailed evidence on the economic scale and impact of space activities. More recently, thematic accounting initiatives based on higher-level national statistics have added a complementary perspective, marking an important step forward in space economy measurement.

- The latest US Bureau of Economic Analysis (BEA) statistics provide an important benchmark: they show that the US space economy is deeply embedded across several strategic parts of the economy. In 2023, it generated USD 142.5 billion in GDP, equivalent to 0.5% of US GDP, USD 240.9 billion in gross output and 373 000 private-sector jobs. This makes it broadly comparable in economic scale to specialised components of the US economy, such as the air transportation industry or the motion picture and sound recording industry. Its policy importance therefore lies less in its aggregate GDP share than in its strategic role, high-technology content, links to public missions and embeddedness across manufacturing, R&D, telecommunications,

government and digital services. The US space economy is also probably larger than current official estimates indicate. While commercial satellite TV, long the dominant segment of the sector, has declined in recent years, increased activity in space manufacturing may not be fully captured, as satellites and launchers produced by companies for their own use may be understated in national accounts, as these activities do not necessarily involve observable market transactions (Highfill and Rao, 2025^[3]).

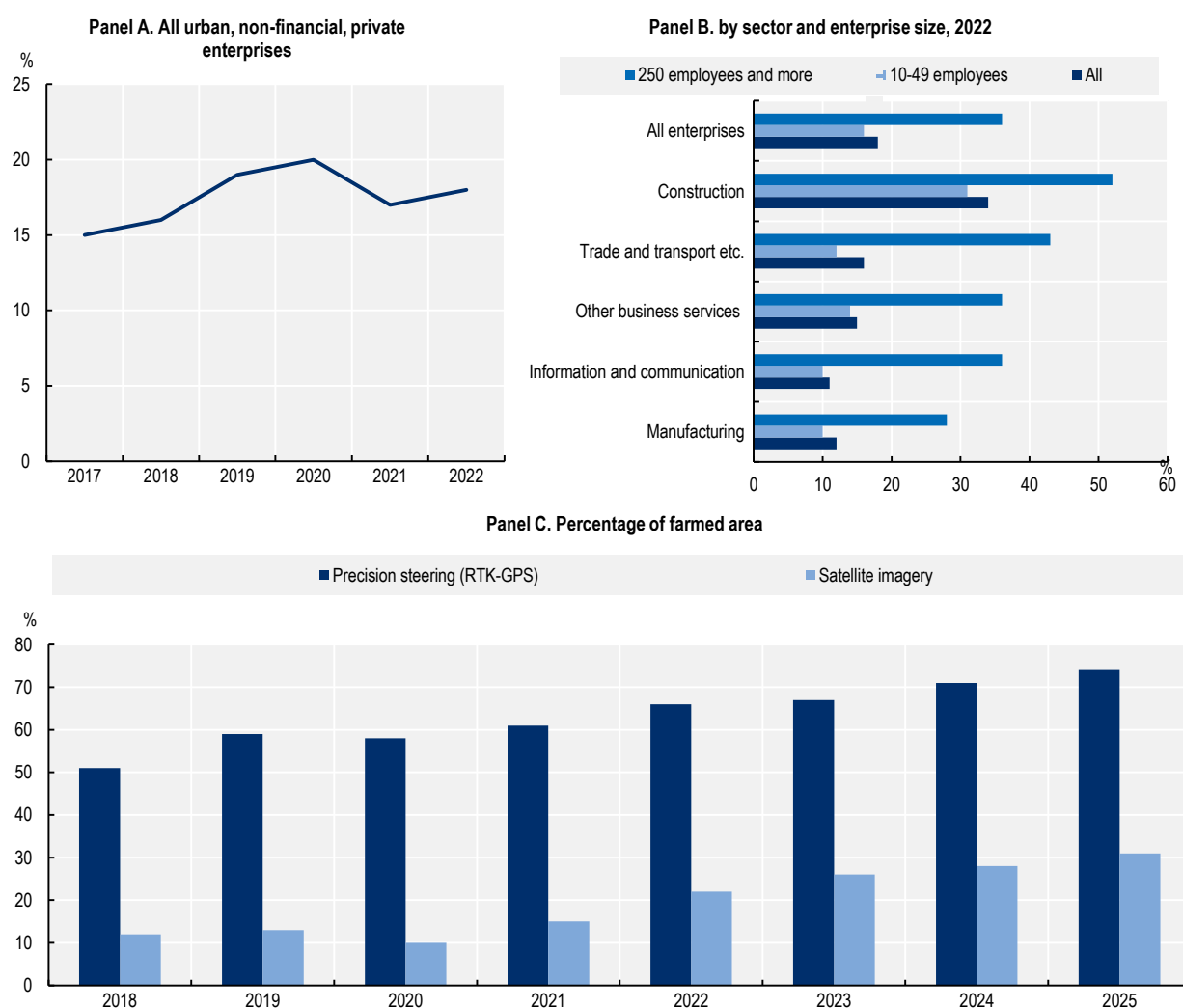
- European evidence points in the same direction. Italy's first thematic assessment estimates the space economy at around EUR 8 billion in output, EUR 2 billion in value added, 23 300 workers and about 0.1% of GDP in 2021.¹ Italy's space economy is a strategic high-value industrial and digital ecosystem, comparable to specialised niches in aerospace, electronics, telecommunications, software and advanced engineering. Upstream activities, mainly producing goods and services used in space, accounted for 67.3% of space value added, manufacturing for 50.9%, large firms for 78.2%, and multinational enterprises for 90.4%, underlining the sector's industrial concentration and international integration.

The OECD is supporting further work to make space economy statistics more consistent across countries and more relevant for policy making, first at national level and eventually globally. Eurostat, the Joint Research Centre and the European Space Agency are building the basis for a European Space Economy Thematic Account. Spain is developing its Space Economy Thematic Account, with publication planned for the first half of 2027, France is developing baseline evidence with INSEE and CNES, while Canada, Korea and Germany are exploring as well new approaches based on their long-standing space industry surveys.

Beyond these measures of the space economy, emerging evidence shows how satellite-enabled services are diffusing across the broader economy. These indicators capture a different dimension of value: not the direct size of the space sector, but the extent to which firms and consumers rely on satellite data and signals as inputs into everyday economic activity. UK Space Agency industry survey estimates suggest that satellite services play a significant and growing enabling role in the UK economy, underpinning around 18% of UK non-financial GDP in 2022, compared with 16% in 2021 (London Economics, 2025^[4]). The largest estimated contribution came from industries relying on positioning, navigation and timing services provided by navigation satellites, at 14% of GDP, followed by industries supported by meteorological satellite services at 11%, and by satellite communications and earth observation services, each supporting around 7% of GDP.

Official Danish statistics provide one of the few granular examples available to date, showing growing use of satellite-based services across enterprises and agriculture, with 18% of urban, non-financial private enterprises using satellite-based services in 2022, with uptake higher among larger firms and 74% of farmed area used RTK-GPS precision steering in 2025, while satellite imagery was used on 31% of farmed area, up from 12% in 2018 (Figure 1.1).

Figure 1.1. Growing use of satellite-based services in the Danish economy



Sources: Statistics Denmark (2025^[5]), "Enterprises use of satellites and Internet of Things (10 employees)", www.statbank.dk and Statistics Denmark (2025^[6]), "Precision agriculture 2025" and editions from previous years, <https://www.dst.dk/da/Statistik/udgivelser/NytHtml?cid=52564>.

These efforts, complementing more traditional space industry surveys and market reports, show why governments need both strong policy engagement and strong statistical foundations. Space is strategically important, but it can be challenging to capture its economic contribution to the economy using standard official statistics. Better measurement helps governments assess the returns from public budgets and R&D support, identify industrial strengths and dependencies, compare national capabilities, and design evidence-based policies that support competitiveness. Building on national initiatives, the OECD is working closely with countries on thematic-account approaches, with the longer-term objective of developing internationally comparable evidence and, eventually, global estimates of the space economy.

The rest of this chapter examines key policy levers through which governments shape space competitiveness. It first considers the scale and orientation of public space budgets, before looking at the role of R&D support, public procurement and mission funding in developing capabilities and creating demand. It then turns to the evolving contribution of private finance, and to the regulatory and institutional frameworks that influence market entry, innovation, investment and international positioning. Together,

these instruments determine how countries translate strategic ambition and public investment into durable industrial, technological and economic advantages.

Public space investment has expanded, but funding intensity varies widely across countries

Governments invest in space to advance strategic, scientific and economic objectives. Public funding underpins national capabilities in space science, security, defence and civilian applications, and remains a major source of demand for space systems and services. Over the period 2010-25, government demand has evolved significantly, with public programmes supporting space transportation, satellite manufacturing, earth observation, communications, navigation, exploration and other applications.

The broader role of space in government strategic objectives appears to be evolving, as reflected in the EC-OECD STIP Compass policy database. Space science and exploration programmes have traditionally served as instruments of international co-operation and diplomacy, and these remain important priorities for many countries. Since the 2010s, however, commercialisation objectives have become more explicit, for example in the 2010 space strategies of Germany and the United States, and have since become mainstream. Over the past decade, defence, security and resilience considerations have also gained prominence. Since 2019, Australia, France, Italy, Portugal and the United Kingdom have published defence space strategies, while similar themes are addressed in several civilian space plans, including those of Denmark, Norway and Poland, as well as in multiple US space policy directives.

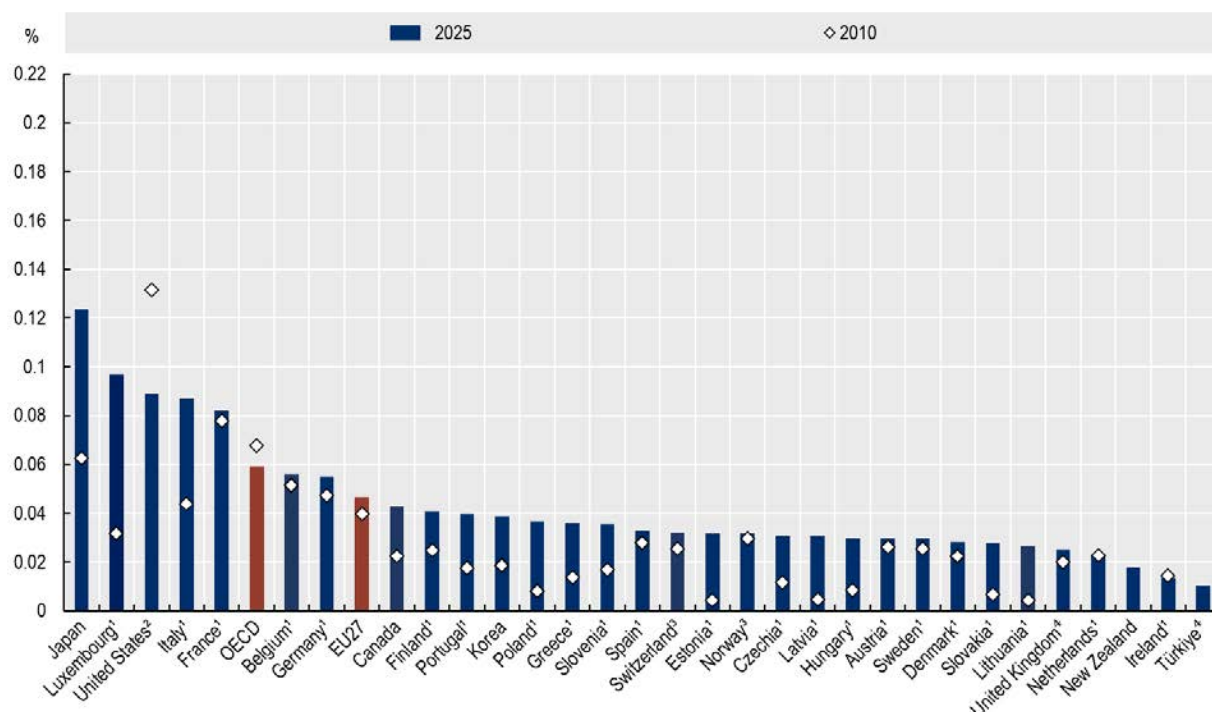
Public investment remains a major growth driver, with OECD civilian space budgets rising by nearly 15%, from USD 40.5 billion in 2022 to USD 46.4 billion in 2025, using an updated methodology (OECD, 2023^[7]) (Figure 1.2).²

Public support for commercial space activity has expanded, reflecting the growing role of private firms in space transportation, satellite manufacturing, earth observation, communications and other applications. This has contributed to shifts in policy instruments, including wider use of public procurement and service-based contracts, particularly in the United States and Europe; the transfer of technological know-how from public agencies to industry, as observed more recently in countries such as Korea and India; and improved access to public funding instruments, including in Europe and Japan.

Several major space-faring economies have increased or reoriented public investment over the past decade. In North America, the United States has continued to shape institutional demand through major civil and defence initiatives, including the Artemis lunar exploration programme and the consolidation of military space activities under the US Space Force. These developments have reinforced demand for commercial solutions in advanced launch, exploration, satellite communications, space situational awareness and national security capabilities. Canada's space activities are gaining renewed momentum through a series of public investments aimed at strengthening national capabilities. Significant investments in next-generation earth observation systems (RADARSAT), European Space Agency programmes, satellite systems, lunar technologies, and emerging launch capability are helping retain and advance niche Canadian expertise in areas such as synthetic aperture radar and space robotics. Canada's exploration profile was further strengthened in April 2026, when a Canadian astronaut became the first non-American to take part in a lunar flyby during the Artemis II mission.

Figure 1.2. Selected OECD civilian space budgets, 2010 and 2025

As a percentage of gross domestic product



Note: 1. Includes contributions to Eumetsat, the European Space Agency and the European Union Space Programme (estimation); 2. Includes the dual-use US Global Positioning System; 3. Includes contribution to Eumetsat and selected European Union programmes (e.g. Copernicus and/or Galileo/EGNOS); 4. Includes contributions to Eumetsat.

Sources: OECD analysis based on government budget documents.

In Europe, space activity has grown markedly. ESA's budget has risen substantially since 2010, supported by both member-state contributions and European Union-funded programmes. Several countries have reinforced national programmes and industrial capabilities, including in earth observation, launchers, secure communications and small satellites. At European Union level, large flagship programmes have become central to Europe's space infrastructure: Copernicus for earth observation, Galileo and EGNOS for civilian positioning, navigation and timing, and GOVSATCOM and IRIS² for governmental and dual-use secure connectivity.

Box 1.1. Science and space exploration as sources of institutional demand

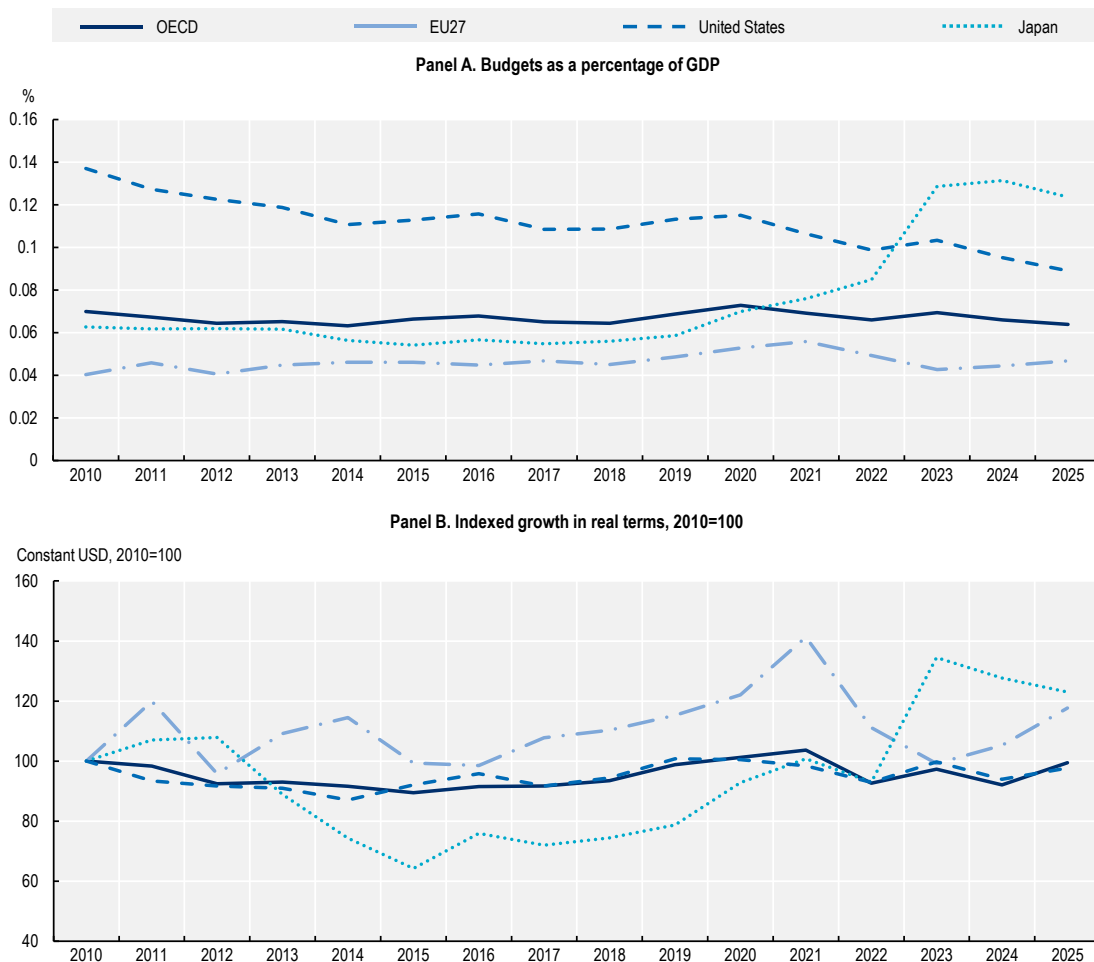
Science, exploration and human spaceflight remain major institutional markets in the space economy. As core civilian objectives of government space programmes, they generate demand for launch, spacecraft manufacturing, robotics, mission operations, life-support systems, communications, navigation and scientific instrumentation. Although their benefits are often difficult to quantify, these missions support scientific discovery, technological learning, specialised industrial capabilities, and public outreach and engagement.

Exploration programmes are also changing how public demand is organised. Governments continue to define objectives, fund missions and manage strategic risks, but increasingly procure selected services from private firms, such as crew and cargo transport, lunar delivery services, human landing systems,

private astronaut missions and planned commercial space stations. Institutional missions can therefore shape markets by creating early demand, validating technologies and supporting new capabilities, while preserving the public-good rationale for sustained public investment.

In Asia, public investment in space has also risen significantly. Japan strengthened institutional support through the Space Strategy Fund, established at JAXA in 2024 to provide multi-year support for private companies, universities and research institutes in satellites, exploration and space transportation. Korea expanded its ambitions through the creation of KASA in 2024, rising budgets, continued development of the Nuri launcher and plans for lunar exploration, building on the Danuri lunar orbiter. The People's Republic of China [hereafter 'China'] has continued to scale up space-based infrastructure in earth observation, navigation and communications, while intensifying activities in space science, lunar and planetary exploration, and human spaceflight. Space plays a growing role also in smaller Asian economies. For example, Thailand launched its THEOS-2 earth observation satellite in 2023, representing a historical investment of USD 237 million (2025 value). Thailand's Geo-Information and Space Technology Development Agency, established in 2000, received a 2026 budget allocation of USD 36 million.

Figure 1.3. Selected trends in civilian space budgets, 2010-2025



Note: EU27 budgets include national civil and dual-use programmes, contributions to Eumetsat, the European Space Agency and the European Space Programme.

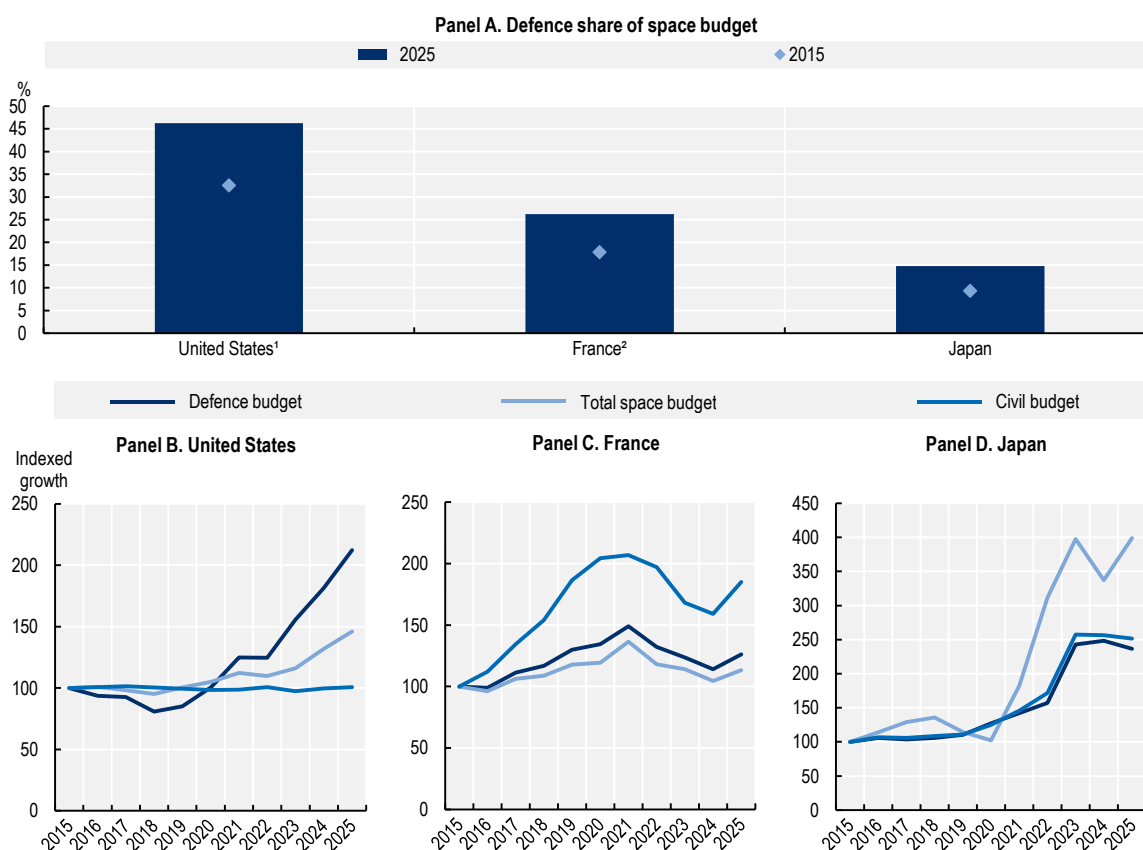
Source: OECD calculations based on government budget sources.

At the same time, national space budget data reveal large variations in public space funding intensity. Between 2010 and 2025, some countries invested heavily, including Japan, Korea and Luxembourg, while others increased spending as part of COVID-19 recovery programmes. Several European countries also increased their contributions to optional ESA programmes. At the OECD level, civil public space budgets have remained broadly stable as a share of GDP between 2010 and 2025, while keeping pace with above-average inflation rates (Figure 1.3). This aggregate trend masks significant variation across countries and regions.

Beyond OECD countries, a widening group of countries is investing in space capabilities. Activity has expanded in Africa, supported by the inauguration of the African Space Agency in 2025 as a continental co-ordination mechanism, as well as in the Middle East and Asia-Pacific. These investments typically remain below the levels seen in the largest OECD space economies and show significant year-on-year variation, ranging from tens to hundreds of millions of dollars. This volatility reflects the fact that public space budgets are often shaped by large, one-off programmes, including commercial satellite procurement, launch activities and infrastructure development.

Figure 1.4. Size and growth of selected OECD military space budgets, 2015-2025

Measured as share of the total budget (panel A) and indexed growth in real terms (Panels B-D, 2015=100)



1. This does not include funding for the GPS programme, considered dual-use and therefore included in the civilian budget; 2. Data for France are based on individual counts of programmes until and including 2019.

Note: Analysis only based on publicly available data, not classified programmes.

Source: OECD calculations based on government budget sources.

A full assessment of public space investment needs to account for defence-related expenditure, but military space budgets remain challenging to compare internationally. Many programmes are classified, embedded in broader defence budget lines or reported as dual-use activities. Even with these limitations, available evidence points to a clear rise in defence space spending. In 2025, military programmes accounted for 14.8% of total government space budgets in Japan, 25% in France and 46.3% in the United States, where the US Space Force budget now exceeds NASA's budget (Figure 1.4).

This shift is visible across a growing number of OECD countries. The United Kingdom has committed GBP 1.4 billion over ten years under its *Defence Space Strategy*, alongside other large programmes (e.g. Skynet military satellite communications upgrade). Germany has announced plans to invest EUR 35 billion in space-related defence capabilities by 2030, one of the largest recent European commitments. France's 2024-30 military programming law provides EUR 6.4 billion for military space, with a later announced additional EUR 4.2 billion effort to 2030. Canada's new Defense Industrial Strategy identifies key sovereign capabilities in the space domain, including space-based intelligence; surveillance and reconnaissance; space domain awareness; satellite communications; and space launch. Strategic investments in these areas have already begun, primarily in the launch sector, where the government announced a 10-year CAD 200 million agreement to lease a dedicated space launch pad as well as a CAD 105 million contest to launch Canadian payloads from Canadian soil. Japan's defence-space spending has accelerated sharply, rising almost sevenfold since 2022 to reach JPY 540 billion in 2025.

Defence demand is also reshaping the types of capabilities being procured. Several European countries, including Portugal, Poland, the Netherlands, Finland, Sweden and Germany, have announced plans to acquire commercial synthetic-aperture radar (SAR) satellite constellations for sovereign intelligence, surveillance and reconnaissance needs, while Greece is pursuing a related sovereign SAR capability through its *National Satellite Space Project*. These developments point to a broader move toward resilient communications, space situational awareness, sovereign intelligence capabilities, rapid access to launch and greater use of commercial satellite services for defence missions.

China's public space spending remains particularly difficult to estimate, as civilian, commercial and military activities are closely integrated under the country's military-civil framework, and several major infrastructures are linked to the People's Liberation Army. Rough estimates suggest that China is likely the world's second-largest government space spender, but these figures remain indicative. Operational indicators, including launch cadence, mass-to-orbit, the deployment of more than 1 000 satellites over the past decade and sustained support for commercial space ventures, point to strong growth in public and strategic space investment.

Overall, defence-related space spending is becoming an increasingly important, but still only partially observable, component of public space investment. This calls for caution in international comparisons, while also underlining a clear strategic trend: Space, which in recent decades has often been treated primarily as a civil, scientific and commercial domain by many economies, is once again becoming central to defence, resilience and strategic autonomy.

Civil space R&D remains significant, but growth has slowed

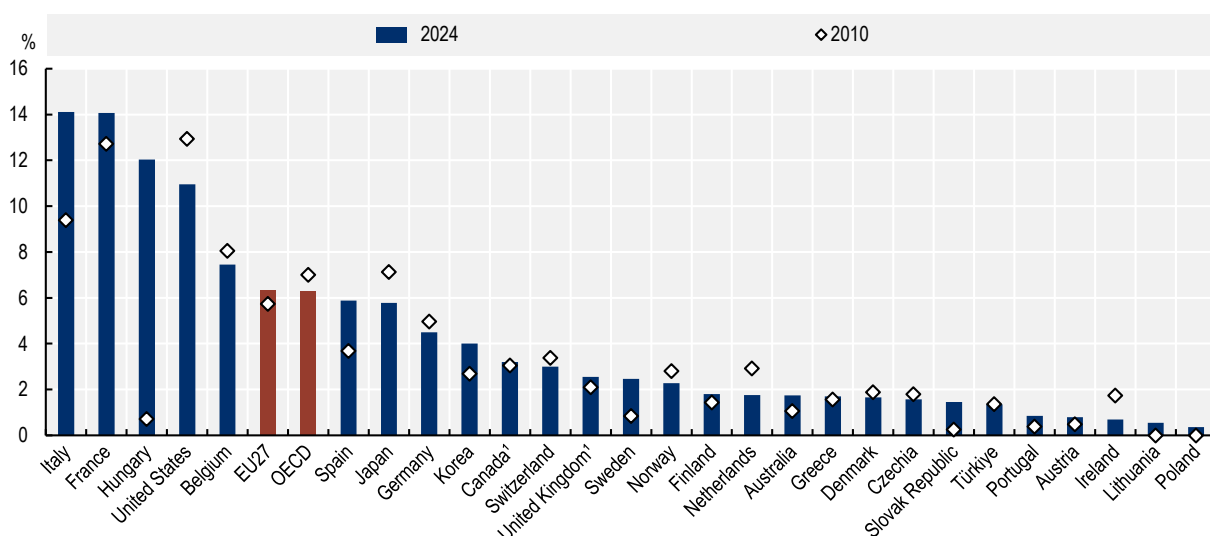
Research and development are central drivers of competitiveness in the space economy. Space activities require specialised capabilities in areas such as propulsion, robotics, materials, communications, earth observation, data processing, systems engineering and mission design. Public R&D funding helps sustain long-term technology development, supports scientific discovery and de-risks capabilities that may not yet have a clear commercial market.

Since the 1980s, OECD has collected data on government R&D allocations by socio-economic objective (GBARD), including civil space R&D. This differs from general space budgets because it covers only

government funding specifically allocated to civil space R&D, whereas general space budgets may also include defence space programmes, satellite operations, procurement, infrastructure, launch services, regulatory activities and other non-R&D spending.

In 2024, civil space accounted for 6.3% of total civil GBARD across the OECD area. France and Italy recorded the highest shares, at 14.1%, followed by Hungary (12%) and the United States (11%) (Figure 1.5). At the OECD level, this represents a slight decline from 2010, largely reflecting changes in the United States. In Europe, by contrast, civil space has increased as a share of civil GBARD in several countries. This growth has been driven in part by new memberships in the European Space Agency (ESA) and by post-COVID-19 recovery programmes, particularly in France, Italy, Spain and Portugal. From a measurement perspective, space R&D is also increasingly difficult to isolate as space-based applications become more widely embedded across sectors, including defence.

Figure 1.5. Civil space as a percentage of total civil Government Budget Allocations for R&D, 2010 and 2024



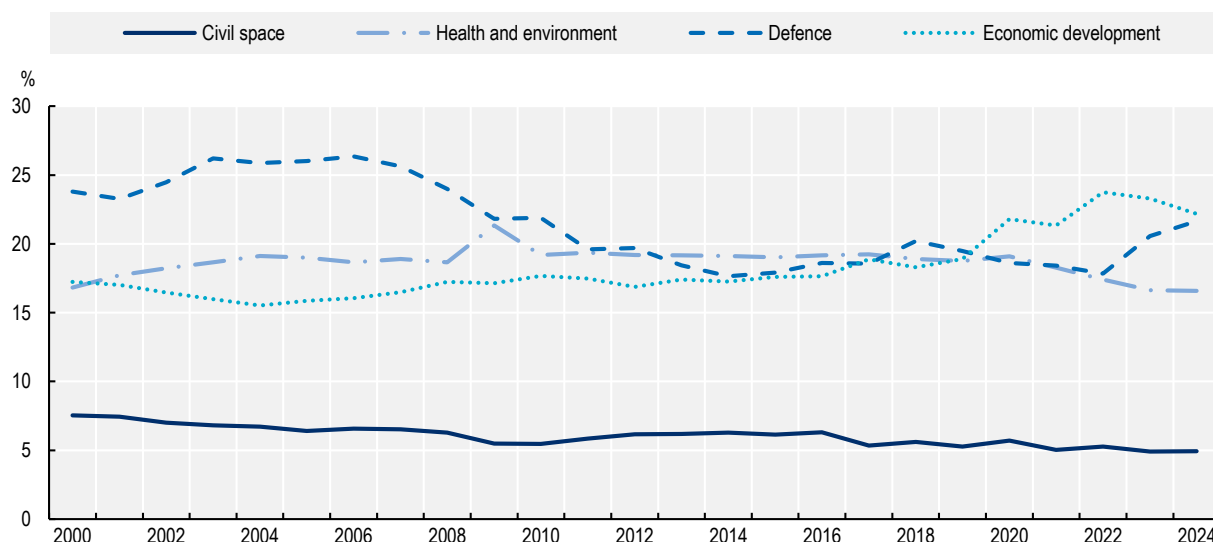
Note: 1. Data for Canada are from 2023.

Source: OECD (2026^[8]), [OECD Main Science and Technology Indicators \(MSTI\) database](#), database last updated 31 March 2026.

Civil space R&D remains significant, but its growth has slowed relative to other areas of public research spending. Based on OECD GBARD statistics, civil space R&D has declined as a share of total civil government R&D budget allocations since the early 2000s (Figure 1.6). Growth trends also point to a widening gap between civil space GBARD and broader civil and total GBARD, the latter including defence R&D (Figure 1.7). This suggests that, while public support for space R&D remains important, it has grown more slowly than other areas of government-funded research, particularly defence-related R&D which is picking up.

Figure 1.6. Civilian space's share of total government R&D budget allocations is decreasing, 2000-2024

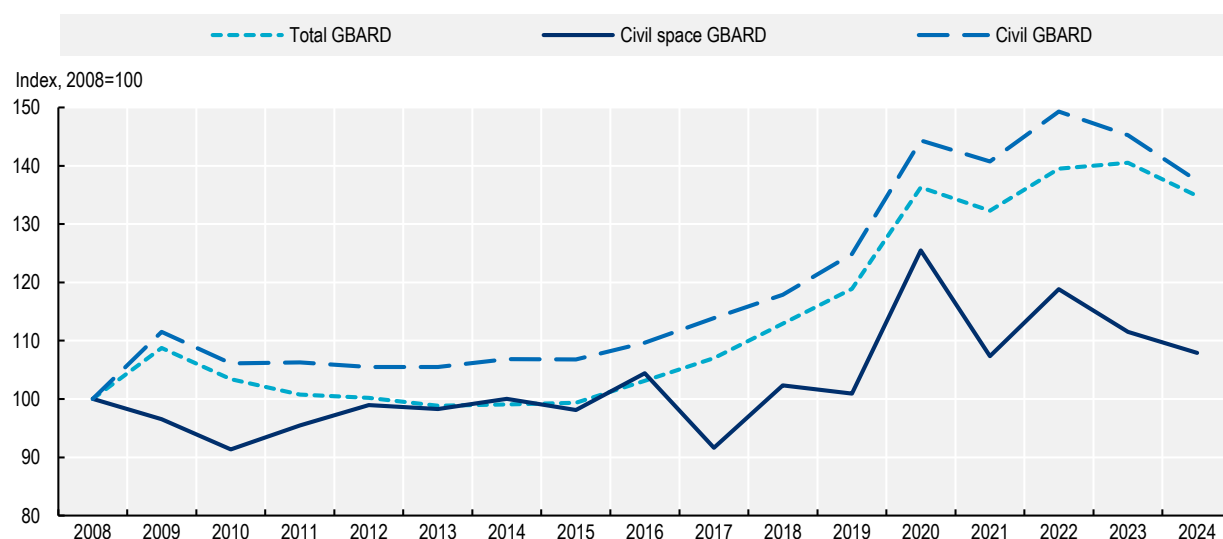
Selected NABS 2007 socio-economic objectives and their share in total GBARD



Source: OECD (2026^[8]), [OECD Main Science and Technology Indicators \(MSTI\) database](#), database last updated 31 March 2026.

Figure 1.7. Growth in civilian space R&D has not kept up with growth in other elements of GBARD

Indexed growth of civil space GBARD, total civil GBARD and total GBARD (including defence), OECD area



Source: Calculations based on data from [OECD Main Science and Technology Indicators \(MSTI\) database](#). OECD civil GBARD and civil space GBARD in constant USD PPP are estimations calculated from MSTI data.

The outlook for civilian space R&D is uncertain in many countries. OECD governments face fiscal constraints, while R&D support is increasingly delivered through tax incentives rather than direct funding. At the same time, defence-related priorities are growing, and governments face competing demands to invest in health, energy and environmental research. Measurement challenges have also intensified as space technologies

and applications become more deeply integrated across sectors, including defence activities. It would nevertheless be misleading to infer from public R&D budget data alone that the support for space innovation lacks momentum. Several developments point to a more diversified innovation landscape: greater private-sector engagement in financing space activities, increased self-funded R&D by some firms, the growing use of public procurement as a tool for innovation and market creation, stronger entrepreneurship-oriented policy instruments, and efforts by many countries to adapt legal and regulatory frameworks to the evolving space economy.

Civil space R&D should therefore be understood as one component of a broader innovation system. Public R&D remains essential for long-term missions, basic science, and frontier technologies. However, competitiveness increasingly depends on how public R&D connects with industrial ecosystems, procurement practices, private investment, start-ups, downstream users and regulatory frameworks.

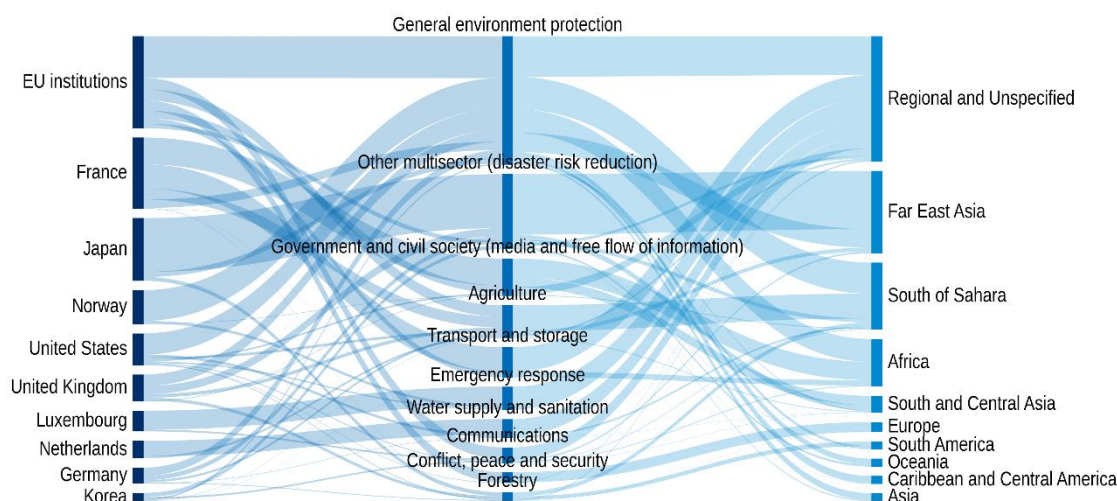
Space-related official development assistance spreads the benefits of government investment more widely

In the 2010s, several OECD countries launched initiatives to widen the benefits of domestic space capabilities and knowhow to lower-income countries, such as the International Partnership Programme in the United Kingdom; and the Geodata for Agriculture and Water project in the Netherlands. The European Space Agency and the European Commission are also working with international partners to improve the uptake of earth observation data in development projects. Furthermore, multiple efforts are improving access to relevant data and training sets, such as the Africa Regional Data Cube and the global Open Data Cube (OECD, 2020^[9]). In 2020, Norway launched the Satellite Data Programme as part of its International Climate and Forest Initiative, purchasing commercial high-resolution satellite imagery of tropical forest regions for universal access and use.

These efforts can be traced in official development assistance (ODA) statistics. Identified space-related ODA covers projects using or supporting space science, space infrastructure and satellite data and signals, as well as space-related administrative support and trainings. Between 2010 and 2023, the total committed amount reached USD 1.2 billion in constant 2023 dollars. EU institutions, France and Japan were the biggest donors over the period, financing projects notably in general environmental protection, disaster risk prevention, satellite-based media and information as well as in other sectors (Figure 1.8). Most geographically targeted assistance went to Africa and Asia, but a lot of the ODA was not tied to specific region. For example, the Norwegian satellite data programme covers tropical forest countries across the world.

Figure 1.8. Environmental protection and disaster risk reduction are key areas of space-related official development assistance

Flows between the biggest donors and sectors and between sectors and recipient regions in the 2010-23 period



Source: Calculations based on OECD (2025^[10]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

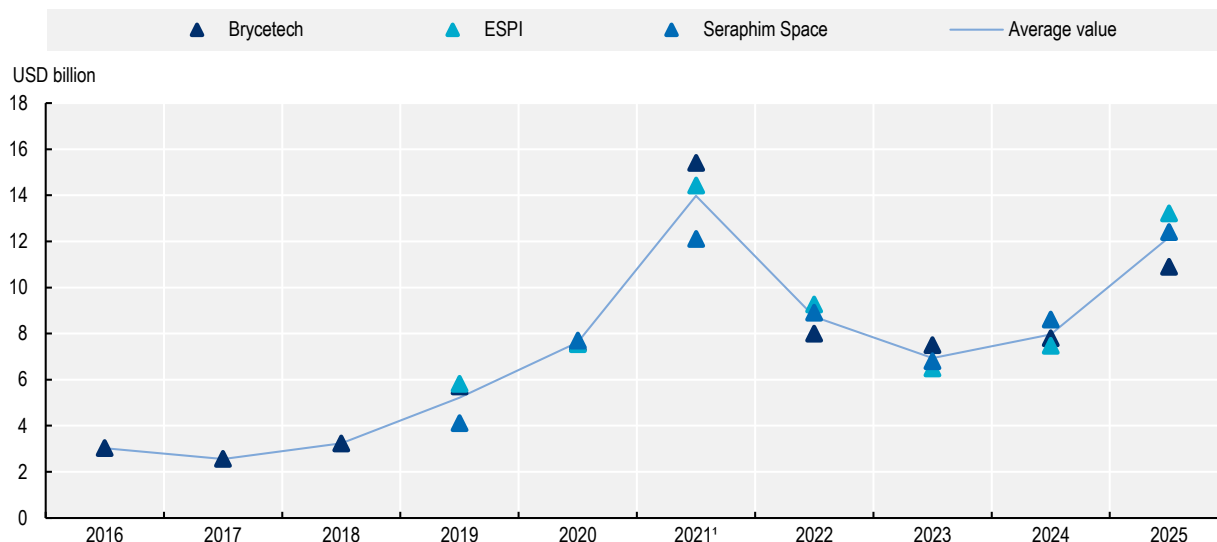
Private finance and firm-level R&D are reshaping investment conditions

Private financing mechanisms in the space sector have diversified markedly over the past decade. They now include venture capital and private equity for early-stage firms, corporate and institutional investors, debt instruments and public markets, including initial public offerings (IPOs) and special purpose acquisition companies. Complementary arrangements – such as public-private partnerships, anchor-customer contracts and other risk-sharing schemes – also play a key role in mobilising private capital, particularly in capital-intensive segments of the space economy.

In 2025, private capital flows to the space sector reached an estimated USD 11-13 billion, depending on scope and definition, the highest level since 2021 (Figure 1.9). Investment was concentrated mainly in mature firms and in capital-intensive segments such as space manufacturing and launch activities. Private investment has also diversified geographically. In 2025, 67% of recipients were based in the United States, 15.6% in China and 12.8% in Europe. The investor base has also widened: in 2025, 39.6% of investors were located in the United States, 27.2% in Europe and 26.5% in Asia (Brycetek, 2026^[11]).

Figure 1.9. Access to private capital is improving, 2016-25

Different industry estimates of space equity funding, acquisitions and debt finance



Notes: The data producers' definitions differ for the space sector, eligible companies and types of investment. 1. the year 2021 is an outlier, marked by low interest rates and multiple special purpose acquisition company listings.

Sources: Adapted from Brycetek (2026^[11]), "Start-up Space 2026", ESPI (2026^[12]), "Space venture 2025"; and Seraphim Space (2026^[13]), "Seraphim space index: 2025: Q4", and similar reports from previous years.

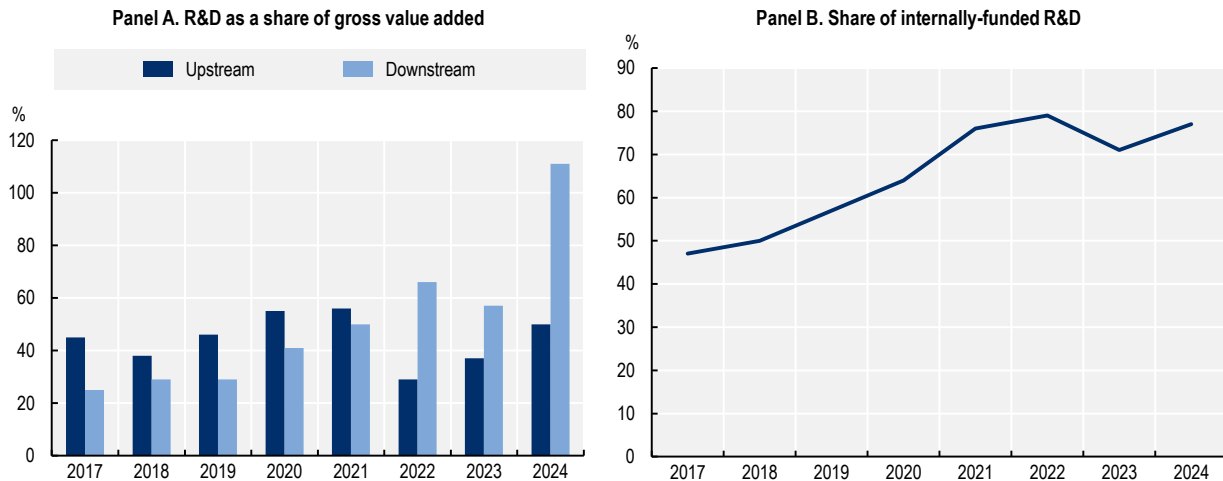
Access to private finance remains uneven. Space start-ups, particularly in manufacturing and launchers, often face long development cycles, high technological risk and uncertain revenue prospects, which may not align with typical venture capital investment horizons of around five years (OECD, 2024^[14]). Public-backed venture financing for the space economy has expanded across several countries.

- In 2018, Japan announced a JPY 100 billion, or around USD 940 million, funding pool to support space economy firms over five years, with backing from the Development Bank of Japan, the Industrial Innovation Organisation and other institutions. The same year, the French space agency CNES created CosmiCapital, a EUR 70 million fund dedicated to the space sector.
- In 2020, Italy launched Primo Space, Europe's first venture capital fund dedicated to the space economy, with EUR 85 million in committed capital and support from the European Investment Fund and CDP Venture Capital SGR. Luxembourg also supported the creation of Orbital Ventures, a EUR 70 million venture capital fund targeting space-related companies.
- The European Commission followed in 2022 with the launch of CASSINI, which brings together around EUR 1 billion in support measures for space start-ups and SMEs through 2027. In 2023, Italy further expanded its support with Italia Space Ventures, a venture capital and co-investment fund managed by CDP Venture Capital and financed through the PNRR Complementary Fund. This initiative supported the development of the Galaxia technology transfer hub and the Take-Off accelerator programme.
- Korea has also scaled up public-backed space investment. Since 2023, the Korea Aerospace Administration has operated the *New Space Fund*. As of March 2026, three funds totalling KRW 30.1 billion, or around USD 20 million, had been established, while a fourth fund was expanded in 2026 to KRW 200 billion, or around USD 135 million.

This relative improvement in access to funding is reflected at the firm level in some countries, with a growing shift towards internally financed R&D among space companies, supported by retained earnings

as well as debt and equity financing. In Canada, for example, business firms in the space sector increasingly rely on such internal resources in addition to government funding. This trend appears to have contributed to a rise in overall R&D intensity measured as a share of gross value added (Figure 1.10). From 2017 to 2024, the internally financed share of Canada's business expenditures on R&D (BERD) rose from 47% to 77% of total investment. Over the same period, the downstream sector's share of BERD increased from 44% to 71%. Together, these trends point to an expanding commercial ecosystem increasingly oriented toward downstream space activities.

Figure 1.10. R&D investment in Canadian space-related business firms, 2017-2024



Source: CSA (2026_[15]), 'State of the Canadian space sector report 2025', <https://www.asc-csa.gc.ca/eng/publications/2025-state-canadian-space-sector.asp>.

In addition, public listings have become an important but uneven channel of private financing for the space sector. Based on a conservative OECD sample of 40 listed space companies in the United States, Europe and Asia, primary proceeds raised over the past decade are estimated at no less than USD 100 billion. SpaceX should be treated separately: its 2026 IPO alone raised USD 85.7 billion, with much of its value connected to broader artificial intelligence developments.

The trend has broadened public-market exposure to space activities beyond traditional satellite operators. Earth observation and geospatial companies account for the largest number of listings in the sample, followed by launch, in-space infrastructure and lunar services, and satellite communications and connectivity. Investor demand appears strongest in segments linked to government, defence, intelligence, connectivity and infrastructure markets. Space exposure has long existed through diversified aerospace, defence and communications groups, where space is one activity among many. Newly listed space firms offer more direct exposure, but also higher concentration risk, weaker profitability in many cases, and greater sensitivity to technical milestones, launch cadence, public contracts and equity-market conditions.

Overall, public listings have expanded access to capital for a number of space companies, visibility and exit routes for investors, while increasing market discipline and volatility. The past decade therefore marks the emergence of a new financing regime rather than a simple wave of successful space IPOs. More space activities are now investable, but outcomes remain highly uneven, with recent cases such as Virgin Orbit's bankruptcy and Astra's Nasdaq exit highlighting the fragility of some listed space business models. This requires for investors a clear distinction between mature incumbents, credible growth firms and more speculative companies.

These developments do not reduce the importance of public investment. Rather, they change the way public and private roles interact. Government contracts, grants, co-funding arrangements and anchor-customer commitments can provide predictable revenue and signal technical credibility to investors. Private finance can in turn help scale technologies, accelerate commercial deployment and expand the range of market participants. The policy challenge is to ensure that public support crowds in high-quality private funding, while avoiding excessive dependence on short-term financial cycles or speculative investment.

Governments are using a wider set of policy tools to support space entrepreneurship and commercialisation.

Governments are increasingly using a broader mix of instruments to support private-sector innovation, entrepreneurship and commercialisation in the space economy. Traditional instruments such as R&D procurement and grants remain important, but policy approaches have shifted since 2010 towards more diverse support for technology transfer, public-sector data sharing, collaboration platforms, advisory services, incubators and access to equity finance.

Public procurement remains one of the main channels through which governments stimulate innovation and create demand. Across the OECD, government agencies are reshaping their relationships with the private sector to foster innovation and improve value for public money (Undseth, Jolly and Olivari, 2021^[16]; OECD, 2023^[7]). Traditional procurement models, based on government ownership, detailed specifications and cost-plus contracting, are increasingly being complemented by more flexible arrangements that emphasise service procurement, co-funding, firm fixed-price contracts and private-sector retention of intellectual property. These practices are intended to make public contracting more accessible to start-ups and privately financed firms, while using government demand to stimulate wider commercial activity.

The United States has been the pioneer of this shift (Figure 1.11). Since the mid-2000s, NASA has advanced commercial as-a-service models, beginning with the Commercial Orbital Transportation Services programme. This approach was later extended to cargo and crew transport to the International Space Station, commercial lunar payload delivery services and, since 2021, commercial low-earth orbit destinations. US agencies have also been encouraged to purchase commercial data to support public missions, and the FY2027 budget request foresees transitioning the civilian Landsat earth observation programme towards a commercial solution. Figure 1.11 shows the changes in NASA procurement practices from the mid-1990s to 2024. In 1996, firm fixed price and cost-plus award fees accounted for 10% and 72% of total procurement obligations, respectively. Firm fixed price instruments may work best for routine, low-risk projects. In 2024, firm fixed price was the most common procurement instrument, covering 31% of obligations, compared to 28% for cost plus award fees (NASA, 2025^[17]). For 2009-18, average cost growth for projects with firm fixed price was 61%, compared to 46% for project with cost-plus (Sobel and Tibor, 2022^[18]).

These models are influencing procurement approaches in Europe and Asia, although adoption has generally started later and from a lower base. ESA, the United Kingdom and Japan have procured debris removal services, while EUMETSAT has begun purchasing commercial meteorological data. The planned IRIS² secure connectivity constellation will be delivered by a private consortium, although it is not a service buy in the strict sense, given the involvement of public actors in its development.

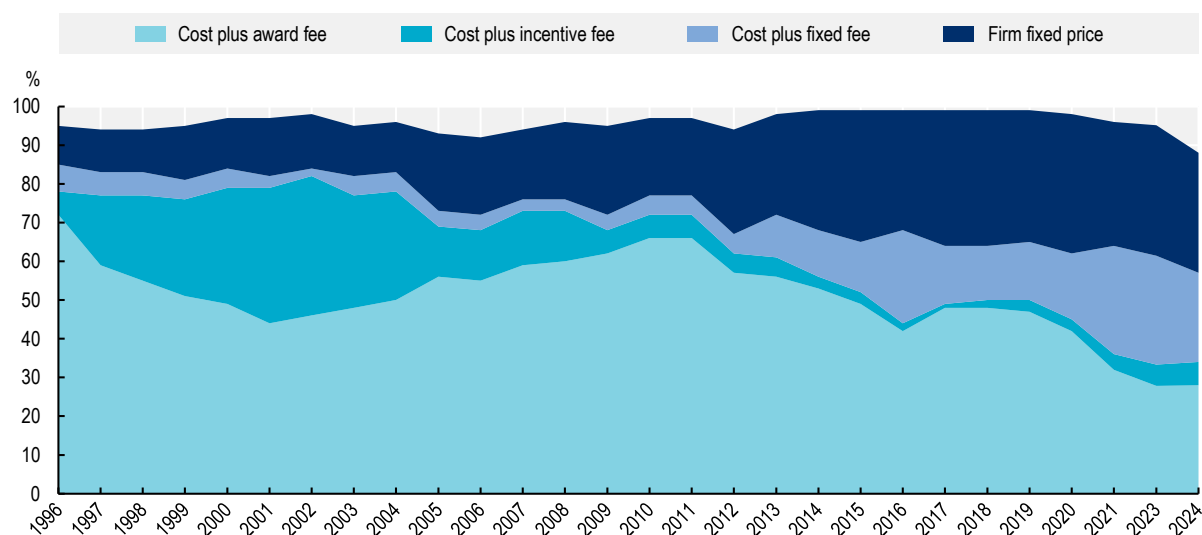
Contractual instruments have evolved in parallel. In the United States, Other Transaction agreements, including Space Act Agreements used under COTS, and Broad Agency Announcements provide greater flexibility than standard federal procurement rules. In Europe, ESA's 2025 Contracting Regulations introduced co-operative agreements, creating a more flexible framework than traditional procurement and placing greater emphasis on co-funding and firm fixed-price arrangements.

These changes can also help mobilise third-party finance. Government contracts provide predictable revenue and can act as a signal of technical credibility to investors. Intellectual property provisions are particularly important. While traditional procurement often entails transferring IP ownership to the public party, US Space Act Agreements and ESA co-operative agreements allow inventors to retain ownership. Evidence from start-ups beyond the space sector suggests that firms with proprietary IP have higher turnover and stronger growth prospects, making IP rights an important signal for investors (EPO/EUIPO, 2023^[19]).

At the same time, contractual arrangements need to be adapted to mission characteristics, risk profiles and market maturity. Firm fixed-price instruments may work best for routine or lower-risk projects, while more uncertain, complex or exploratory missions may require different risk-sharing arrangements (OECD, 2024^[14]). The growing use of procurement as an innovation tool therefore does not imply a single preferred model. Rather, it points to a more differentiated set of procurement practices designed to match public objectives, technological risk and commercial maturity.

Figure 1.11. Firm fixed price is now NASA's most common procurement instrument, 1996-2024

NASA procurement instruments as a percentage of total obligations

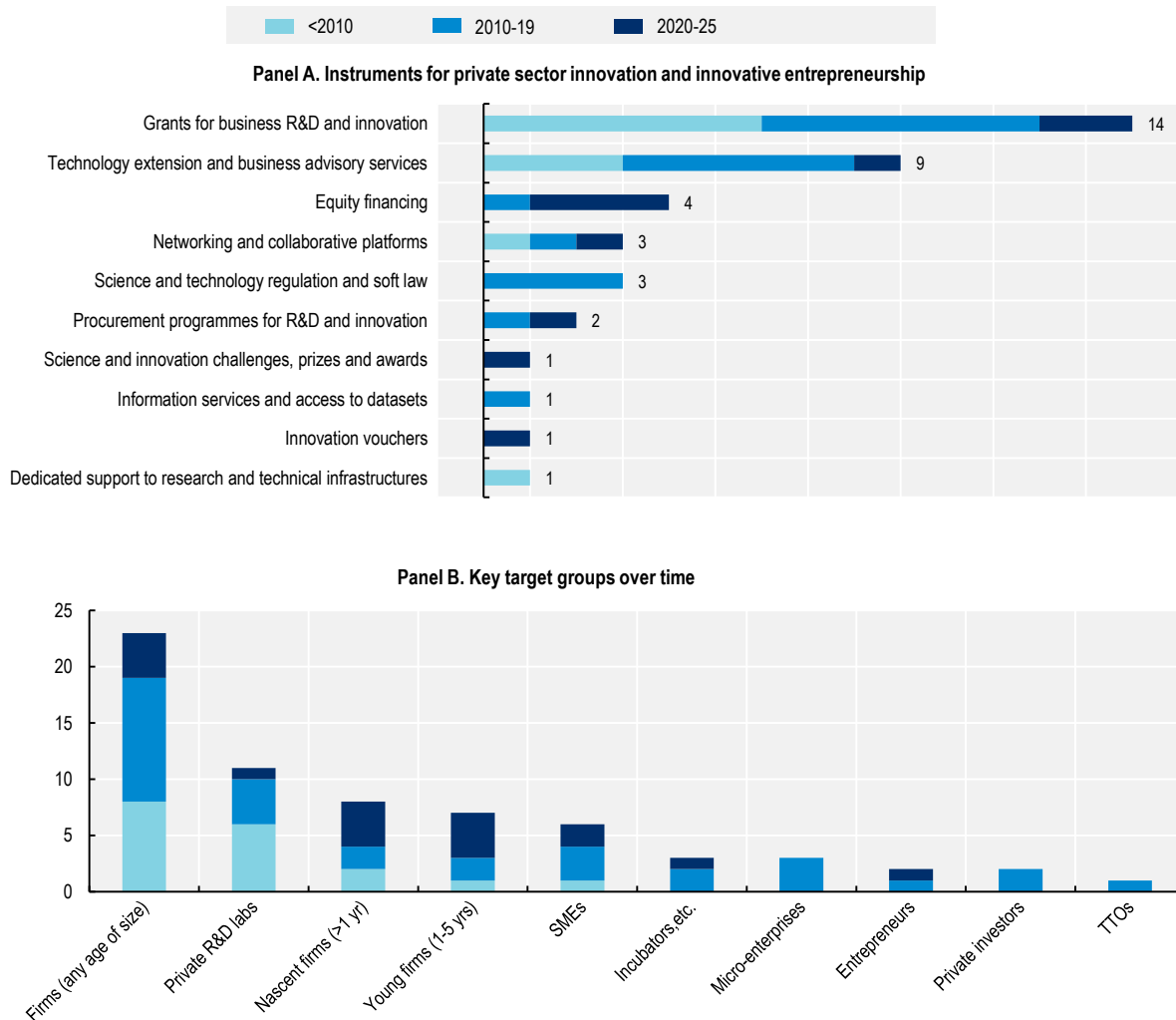


Source: NASA (2025^[17]), FY 2024: Agency Financial Report, <https://www.nasa.gov/wp-content/uploads/2023/11/nasa-fy-2024-afr.pdf> and NASA (2023^[20]), State of NASA Procurement Report: Fiscal Year 2022 in Review, <https://www.nasa.gov/wp-content/uploads/2023/09/state-of-nasa-procurement-2022.pdf?emrc=65cb8e6c1f3fa> and reports from previous years.

Beyond procurement, governments are broadening entrepreneurship-oriented support (Figure 1.12). The space economy thematic portal of the EC-OECD STIP Compass database identified 212 space-related policy initiatives and 289 policy instruments across 45 countries, the European Union and the European Space Agency in 2026 (EC-OECD, 2026^[21]).³ These initiatives show that innovation is increasingly viewed through an entrepreneurial lens, with greater emphasis on knowledge transfer, business advisory services, collaboration platforms, public data access, incubators and equity finance.

Figure 1.12. Governments are broadening policy instruments to support private sector innovation in the space sector, 2010-2025

Counts of policy instruments



Note: The space portal includes 212 policy initiatives. Most initiatives cover more than one policy instrument or target groups.

Source: [Space economy thematic portal](#) powered by EC-OECD (2026^[21]), *STIP Compass: International Database on Science, Technology and Innovation Policy (STIP)*, edition 06/2026, <https://stip.oecd.org>.

Several programmes illustrate this shift. Connect by CNES, launched in 2018, facilitates access to space data and signals for start-ups, companies and institutional users, while also offering technology extension and business advisory services. At European level, ESA Business Incubation Centres have become a major instrument for supporting early-stage firms, with around 36 centres operating in 23 countries by the end of 2025. Similar efforts can be observed in Japan, where the J-SPARC initiative, introduced in 2018, promotes collaborative projects between private companies and JAXA. In the United Kingdom, the UK Space Agency Accelerator programme has helped 289 founders raise more than GBP 100 million (USD 133 million) in the 2021-26 period (UKSA, 2026^[22]).

Initiatives may target specific activities or objectives. Norwegian and Netherlands government agencies fund platforms to facilitate domestic users' access to satellite data from the European Copernicus programme, in order to develop earth observation applications. In the Netherlands, the Satellite Data Portal

now also includes higher-resolution optical and radar imagery as a supplement to Copernicus data. In Switzerland, the European Space Deep-Tech Innovation Centre, established in 2025, seeks to accelerate the commercial adoption of “deep-tech”, comprising technologies such as quantum, data and materials.

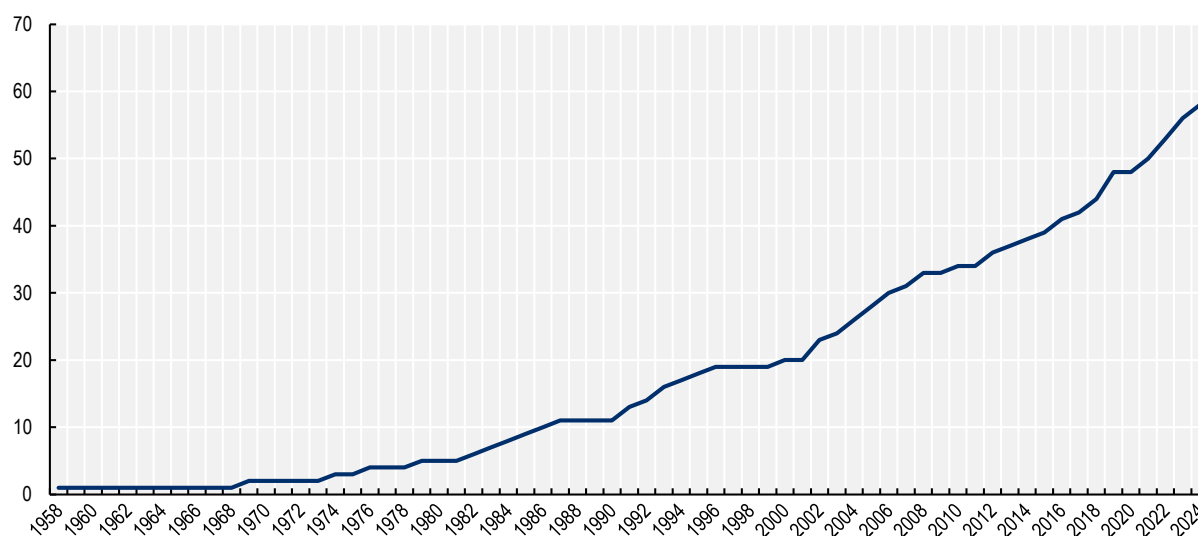
The diversification of policy instruments reflects a broader structural transformation in the space economy. As private firms take important roles in launch, satellite manufacturing, earth observation, communications and downstream services, governments increasingly need to combine direct funding with measures that support ecosystems, reduce barriers to entry, connect firms to public missions and enable the commercial use of space-based data and infrastructure.

Legal and regulatory frameworks are becoming more supportive of commercial space activities

Legal and regulatory frameworks are increasingly important for national space competitiveness. As more countries and private actors engage in space activities, governments need rules that enable market entry and innovation while managing risks related to safety, liability, security, sustainability and international obligations. The number of countries with space-related regulations has increased steadily, nearing 60 by 2025 (Figure 1.13).

Figure 1.13. Increasing number of countries with space regulations, 1958-2025

Cumulative count of countries with space regulatory instruments



Source: Adapted from UNOOSA (2026^[23]), “National space law”, webpage, <https://www.unoosa.org/oosa/en/ourwork/spacelaw/nationalspacelaw/index.html>.

Countries have accompanied their investments in space programmes by creating or updating legal frameworks. Examples include the 2017 *Outer Space and High-Altitude Activities Act* in New Zealand and the 2018 *Space Industry Act* in the United Kingdom, in order to regulate emerging domestic commercial space activities. National regulatory instruments are mainly associated with satellite registration and space activity licensing, but increasingly also address issues such as in-orbit insurance and liability, satellite imagery exports, orbital resource exploitation and debris mitigation.

However, keeping pace with accelerating technological development is a challenge. Regulators need to account for the risks of deploying new technologies while simultaneously enabling innovation. Earth observation imagery provides a useful example, where the availability of sophisticated commercial imagery has increased considerably over the past 5-10 years (see Chapter 2), yet only a few OECD countries have explicit data regulation in place in 2026, including Canada, France, Germany, Japan and the United States. These frameworks regulate the conditions for reporting and/or disseminating private sector data for national security purposes, typically addressing technical characteristics such as temporal, spatial and spectral resolution or frequency domains. To position their commercial sector to compete in the global market for earth observation data, the United States introduced a new tiered licensing approach for private earth observation systems in 2020, linking regulatory stringency to the existence and technological capabilities of foreign competition. The risk burden is transferred to the US government and focuses more on international comparisons than technological specifications. Overall, having these regulations in place is important, and ensuring their effectiveness requires ongoing review and periodic updates to requirements in order to reflect new technologies and changing market conditions.

Regulatory frameworks also play a role in managing externalities associated with growing space activity, including orbital debris, radio-frequency interference and the sustainability of orbital resources. While these issues are addressed in greater detail later in the report, they are increasingly part of the competitiveness agenda: firms and countries depend on safe, predictable and sustainable access to orbital environments, spectrum and launch opportunities.

Overall, the competitiveness of national space economies will increasingly depend on the ability of governments to combine sustained public investment with well-designed market-shaping policies. Public funding remains indispensable for maintaining strategic capabilities, supporting long-term R&D and addressing public-good objectives. Its impact will be amplified where it mobilises private capital, encourages firm-level innovation, opens opportunities for new entrants and provides clear regulatory conditions. Countries that can provide stable demand, effective innovation ecosystems and predictable rules will be better placed to capture the economic, scientific and strategic benefits of the expanding space economy.

Conclusion

Space competitiveness is increasingly shaped by the capacity of governments to combine strategic public investment with effective market-shaping policies. Public funding, R&D support and institutional demand remain indispensable for sustaining critical capabilities and addressing public-good objectives, but their impact increasingly depends on how well they mobilise private finance, support firm-level innovation and enable new entrants. As space-based services become more deeply embedded across economies, countries will also need stronger statistical evidence, predictable regulatory frameworks and policy instruments that connect public missions with commercial opportunities. Those able to provide stable demand, effective innovation ecosystems and proportionate rules will be better positioned to capture the economic, scientific and strategic benefits of the expanding space economy.

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Notes

¹ These estimates do not include public sector activities, unlike the US thematic account.

² The OECD Space Forum tracks non-confidential government civilian and dual-use space budgets, making it possible to analyse government space priorities over time and across countries. This indicator covers a broad range of activities, comprising for example R&D support, space system procurement and operations, provision of physical and data space-related infrastructure and data/signal utilisation. It also includes contributions to the European Organisation for the Exploitation of Meteorological Satellites, the European Space Agency and estimated contributions to the European Union Space Programme. To improve comparability, activities with purely military objectives have been excluded.

³ The EC-OECD STIP Compass collects data on national science, technology and innovation policies under the auspices of the OECD Committee for Scientific and Technological Policy and the European Union's European Research Area and Innovation Committee. In June 2026, the database comprised more than 8 000 initiatives and 12 000 policy instruments from 63 countries. The space thematic portal counted 212 policy initiatives and 289 policy instruments from 45 countries, the European Union and the European Space Agency.

2 Shifting market dynamics in the space economy: Concentration, diversification and value creation

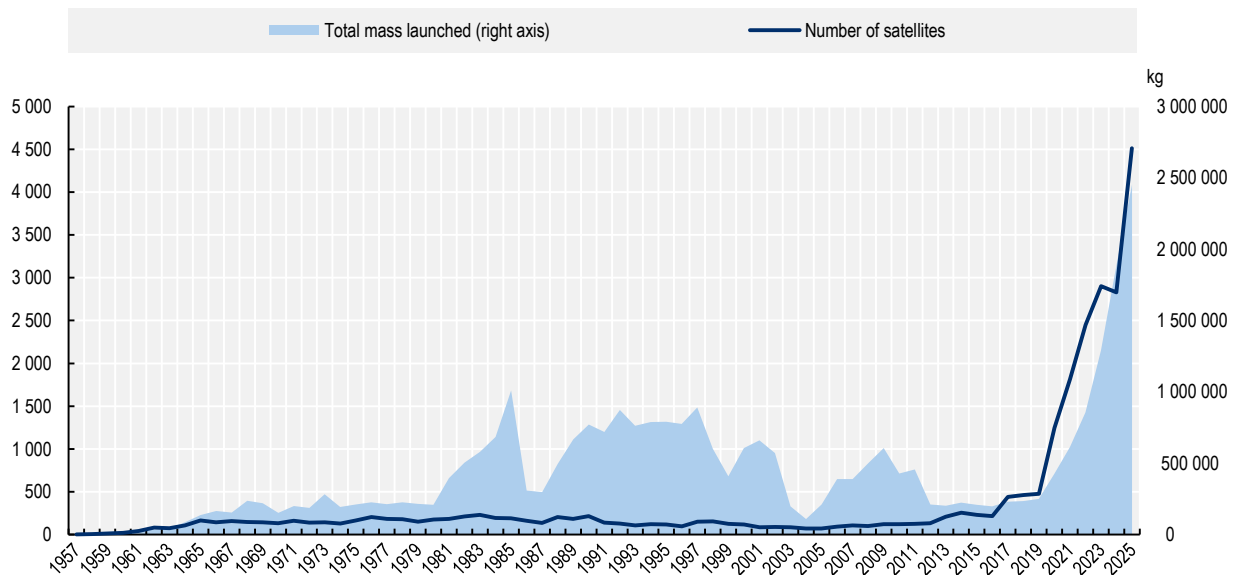
This chapter examines how the space economy is becoming more commercial, more concentrated and more constellation driven. It focuses on market transformation, orbital infrastructure and commercial value creation. The chapter shows that access to space has broadened, but strategic capabilities, launch activity, orbital infrastructure and several emerging commercial markets remain concentrated among a limited number of actors.

A transformation of orbital occupancy and ownership in the last ten years

The space economy has changed dramatically in the past decade. New market entrants and technological innovations have reduced the cost of access to space, supported by reusable launcher components, miniaturised satellites and commercial rideshare opportunities. Digital advances have also transformed the use of satellite signals and imagery by facilitating data sharing, processing and analysis. At the same time, governments have promoted private-sector participation through commercialisation programmes, procurement models and regulatory measures in selected market segments.

As a result, satellite ownership and orbital activity have expanded rapidly. In 2025 alone, 329 space launches were conducted, carrying around 4 900 objects into space. By the end of the year, more than 14 000 active, operational satellites were circling Earth in low, medium and geostationary orbits (Figure 2.1). In terms of operational satellites, US organisations led by large satellite broadband deployments operated 76% of active satellites by the end of 2025, followed by operators in Western Europe and China. By mid-2026, the number had reached almost 15 000. These developments are both recent and unprecedented, and they have shifted the scale, composition and geography of orbital activity.

Figure 2.1. The scale of launch activity is increasing, in object numbers and mass



Source: Adapted from McDowell (2026^[1]), *General Catalog of Artificial Space Objects (GCAT)*, <https://planet4589.org/space/gcat/>.

The transformation of orbital occupancy has important market implications. It creates opportunities for new commercially-offered services, including satellite broadband, earth observation, direct-to-device communications, mobility services and data-enabled applications. It also increases competition for launch capacity, spectrum, ground infrastructure and orbital resources (see for example OECD (2022^[2]; 2023^[3]; 2024^[4])). These developments have contributed to a space economy that is larger in some segments and increasingly commercially oriented.

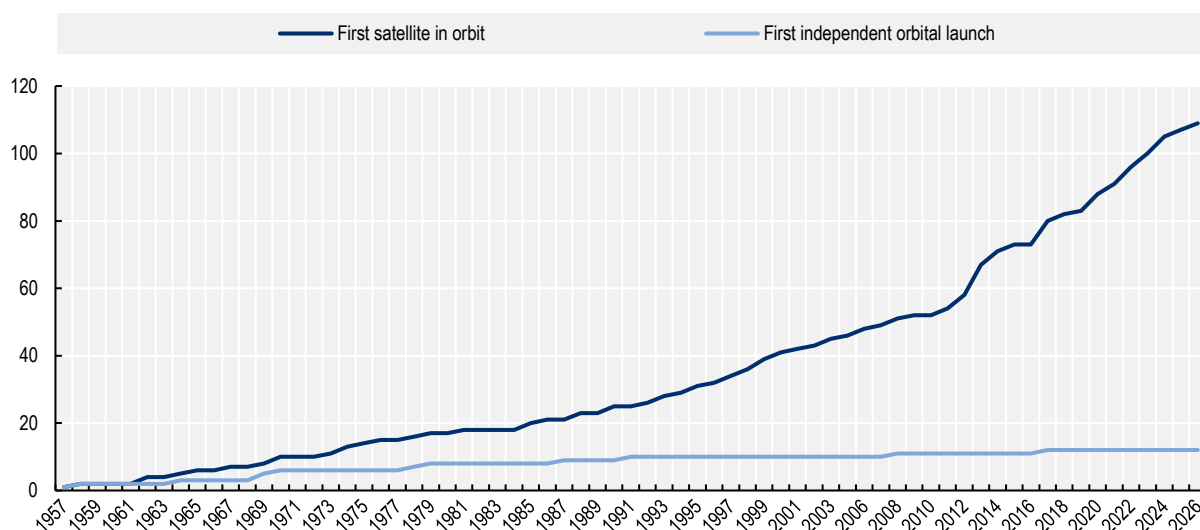
Participation in space activities has broadened, but launch capability remains concentrated

Access to space has broadened markedly. In fifteen years, as many countries launched their first satellite as in the previous five decades combined, bringing the total to 109 countries by the end of 2025 (Figure 2.2). This acceleration has been enabled to a large extent by CubeSats, commercial rideshare launches, international training programmes and increasingly available off-the-shelf satellite technologies.¹

Recent first satellite missions illustrate the diversity of pathways through which countries are entering space. In Africa, Kenya, Uganda, Zimbabwe and Botswana have launched small satellites through university-led projects, international training programmes and commercial small-satellite expertise. Similar approaches are visible in Asia and Latin America, where Bhutan, Costa Rica and Guatemala launched early missions focused on technology demonstration, environmental monitoring and natural resource management. These examples show that countries can enter space through academic capacity-building missions, commercial procurement and gradual development of domestic design, manufacturing and operational capabilities.

Figure 2.2. Over half of the countries in the world have successfully launched at least one satellite

Cumulative counts of “firsts” for satellite ownership and independent orbital launches, as of May 2026



Source: OECD analysis, 2026.

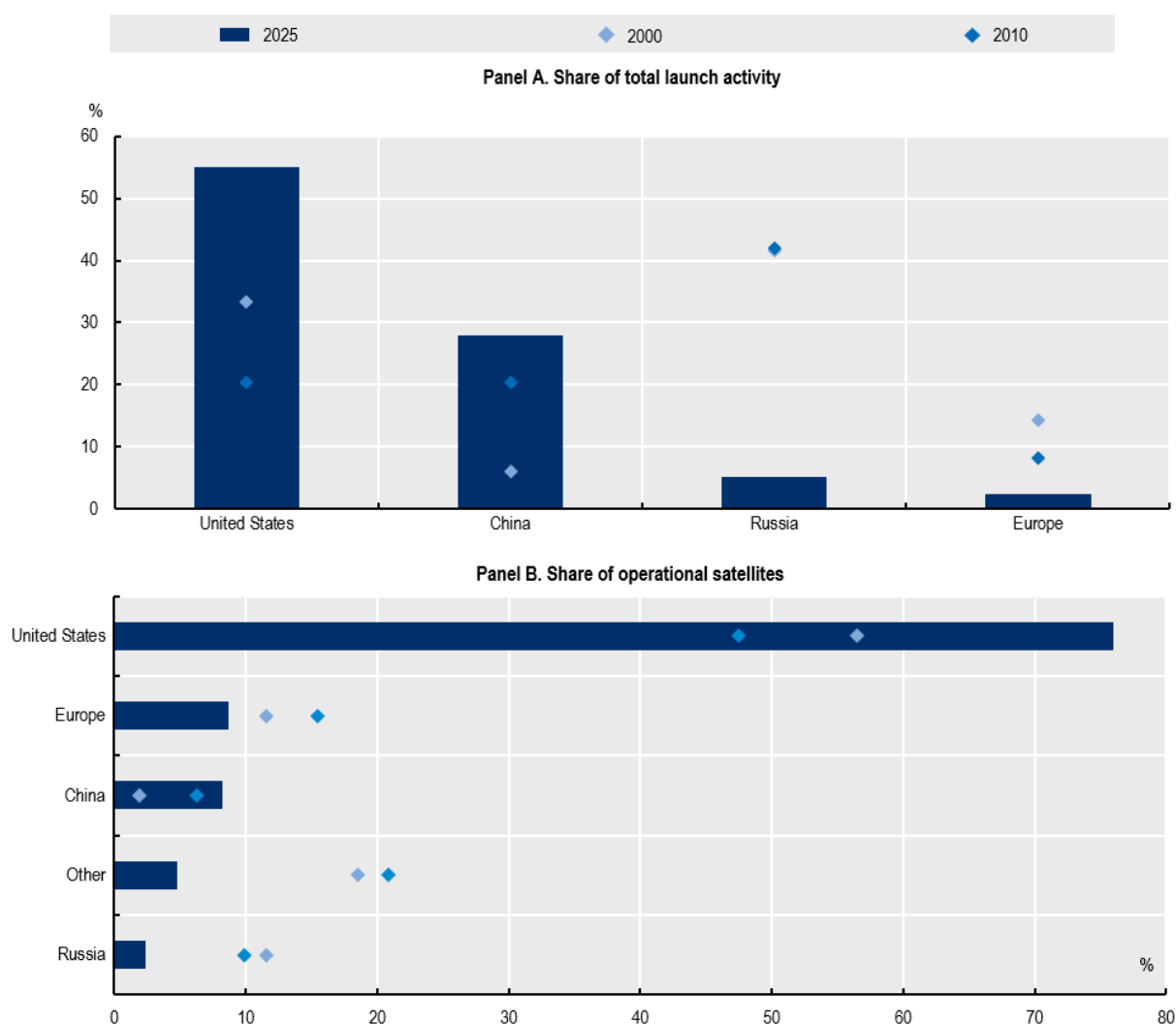
However, sovereign orbital launch capability remains restricted to a handful of economies. By mid-2026, only 12 countries had demonstrated the ability to place objects in orbit.² Several countries – including Australia, Canada, Germany, Norway, Portugal, Sweden and the United Kingdom – are developing new commercial or sovereign launch-site projects at varying levels of maturity, but by mid-2026 many had conducted only suborbital activity or unsuccessful orbital attempts.³

Launch activity is also highly concentrated in practice. In 2025, the United States accounted for 55% of all launches and 87% of launched objects, largely reflecting the high launch cadence of SpaceX, which conducted most US missions (Figure 2.3). China followed with 28% of launches and 8.4% of launched objects. Other active launch actors included Russia, Europe through the Ariane and Vega programmes, Japan, India, Iran and New Zealand, while Israel, the Republic of Korea and North Korea also conducted

launches or orbital launch attempts during 2024–2025. The United States and China are furthermore the only economies that have successfully recovered first stage rocket boosters so far. US operator SpaceX first achieved this in 2015 with its Falcon-9 rocket and then with its Starship/Super Heavy booster in 2024. SpaceX had by early 2026 conducted more than 600 Falcon-9 booster recoveries, more than 50 out of which have been reused at least a second time. US firm Blue Origin successfully landed its New Glenn booster in 2025. In 2026, China Rocket, the commercial launch subsidiary of the state-owned China Aerospace Science and Technology Corporation, successfully recovered the first stage of a Long March 10B rocket using a sea-based net capture system, marking China's first recovery of an orbital-class launch vehicle booster.

Figure 2.3. The United States accounts for a growing share of orbital launches and active satellites

Share of annual orbital launches and of operational satellites in Earth's orbits, 2000, 2010 and 2025



Source: Adapted from McDowell (2026^[1]), *General Catalog of Artificial Space Objects (GCAT)*, <https://planet4589.org/space/gcat/>.

While the global launch market is estimated at USD 12-16 billion in 2025, transparency remains low because most providers do not publish pricing or revenue information and large institutional segments continue to operate as captive markets. Much demand is effectively reserved for domestic launch

providers, as governments and institutional customers often require national satellites to be launched on national rockets for strategic, security, or industrial policy reasons.

Over the past decade, more than 30 small-launcher projects have been announced worldwide, including over a dozen in Europe, supported through national programmes, ESA initiatives and private investment. While only a limited number have reached orbit or entered commercial service, many remain in the demonstration, test-flight or early operational phase. In Europe, notable examples include Germany's Isar Aerospace and Rocket Factory Augsburg, Spain's PLD Space, the United Kingdom's Orbex and Skyrora, and France's MaiaSpace and Latitude. Similar developments are underway in Asia, where companies such as China's LandSpace, iSpace and Galactic Energy, Japan's Space One, South Korea's Innospace, and India's Skyroot Aerospace and Agnikul Cosmos, alongside Indian Space Research Organisation's Small Satellite Launch Vehicle, are pursuing dedicated launch solutions for small satellites. This proliferation reflects strong demand for responsive launch services but also raises questions about market consolidation, as not all projects are expected to achieve long-term commercial viability.

This combination of broader participation, new investments and concentrated strategic capability is central to current market dynamics. More countries and firms can access space-based services and aim to deploy small satellites in particular, but large-scale launch, constellation operation and orbital infrastructure still remain controlled by a limited number of actors. This raises policy questions about access conditions, resilience, interoperability and competitive market development, which become more important as an increasing number of downstream users depend on space-based infrastructure.

Private operators now dominate activity in orbit

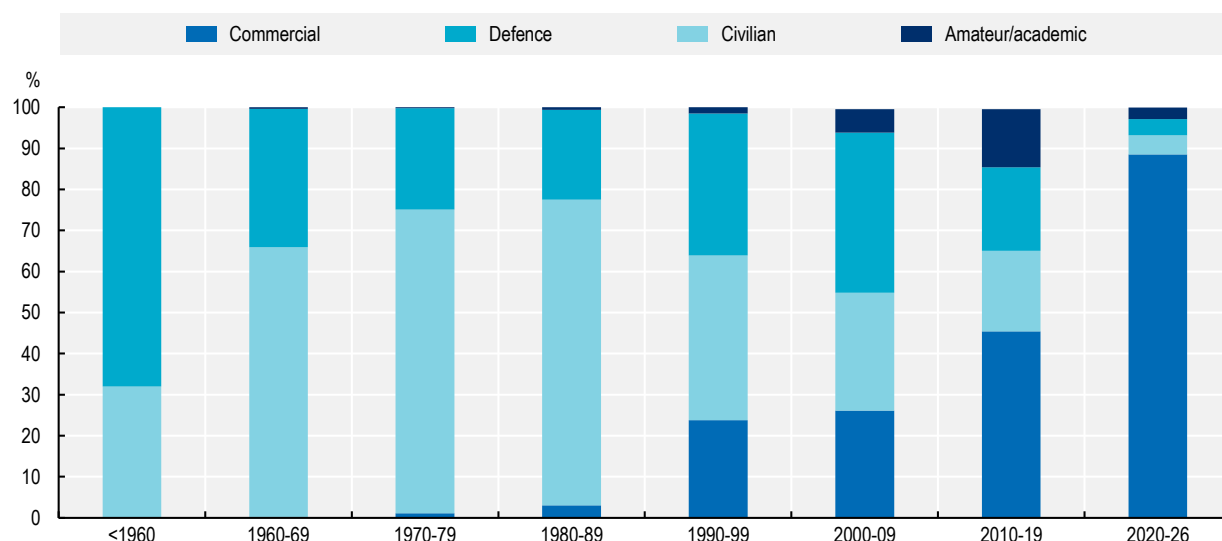
Private operators now account for the large majority of active satellites. Their share of satellites launched into orbit increased from 23% in 2010 to 88% in 2025, marking a significant shift from a space environment historically dominated by governments and public agencies towards one increasingly shaped by commercial actors (Figure 2.4).

The nature of private-sector activity has also changed. During the 2010s, in addition to large commercial geostationary satellites providing telecommunication and television broadcasting, many private missions relied on small, standardised and stackable satellites, including CubeSats, which lowered costs and enabled more frequent launches. These satellites increased the number of objects launched, but because they were relatively light, their effect on total launch mass was more limited.

Since 2019, the deployment of large private satellite broadband constellations has become the main driver of launch activity and orbital occupancy. One operator in particular, through the rapid deployment of the Starlink constellation, accounted for around 60% of all active satellites in orbit in 2025, with the constellation continuing to expand rapidly in 2026. This illustrates the growing concentration of orbital occupancy among a small number of large commercial operators.

Figure 2.4. Private sector increasingly dominant in orbit

Owner/operator' share of launched objects, data as of 9 May 2026



Source: Adapted from McDowell (2026^[1]), *General Catalog of Artificial Space Objects (GCAT)*, <https://planet4589.org/space/gcat/>.

This shift has important implications for market structure. Commercial operators increasingly control assets that are central to connectivity, orbital infrastructure, data flows and service delivery. At the same time, public agencies often remain key customers, regulators and mission partners. The boundary between public and private activity is therefore evolving, with commercial operators playing a larger operational role while governments continue to shape markets through demand, regulation and public missions.

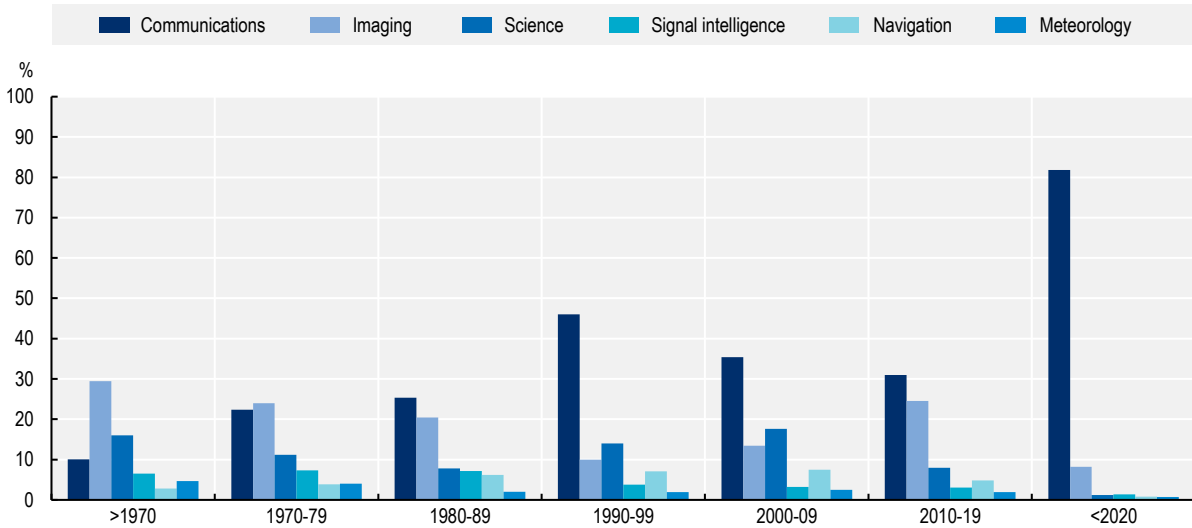
Telecommunications and LEO constellations are reshaping orbital demand

Telecommunications have become the main driver of recent satellite deployment. International satellite communications were liberalised in the 1980s, with subsequent commercial growth initially driven by direct-to-home television from geostationary orbit. Mobile satellite services emerged in the 1990s, while satellite broadband solutions from geostationary orbit appeared in the early 2000s but faced cost and latency constraints.

Since the mid-2010s, the bulk of orbital activity has shifted from geostationary orbit to low-earth orbit, located roughly between 160 km and 2 000 km altitude. Telecommunications satellites have accounted for a growing share of global satellite launches since 2020, representing 82% of all satellites launched in 2025, compared with 31% over the 2010-19 period (Figure 2.5).

Figure 2.5. Telecommunications accounted for more than 80% of satellites launched in 2025

Percentage of satellites launched in main application groups

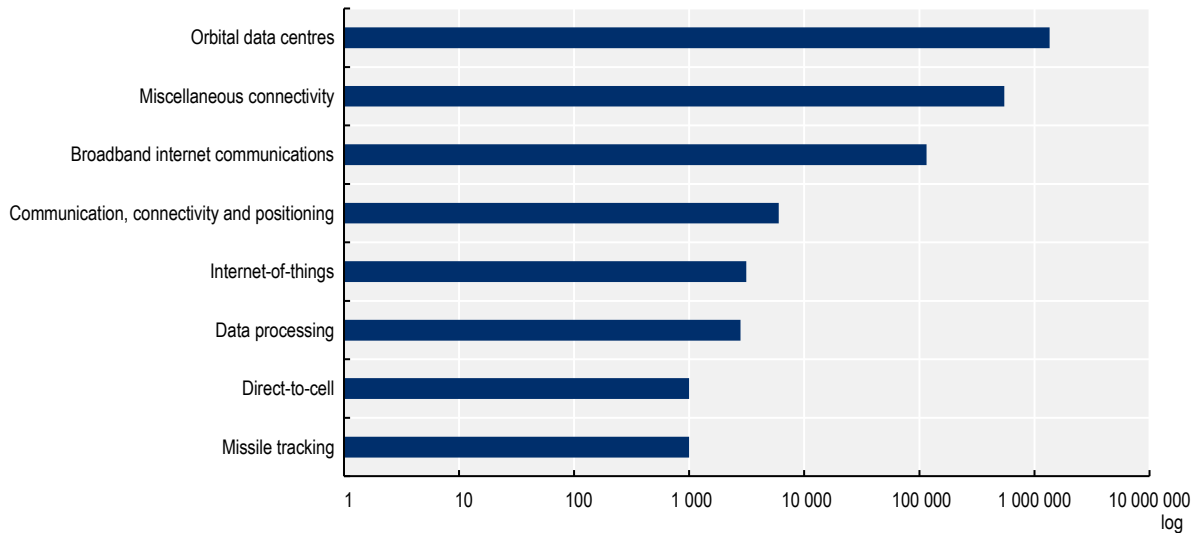


Note: Key applications have been grouped according to the primary mission

Source: Adapted from McDowell (2026^[1]), *General Catalog of Artificial Space Objects (GCAT)*, <https://planet4589.org/space/gcat/>.

This marked shift is largely driven by the deployment of large-scale satellite broadband constellations and rising demand for space-based connectivity services.

Figure 2.6. Announced future mega constellations comprise more than 2 million satellites over the next decade



Note: Regulatory satellite filings can be speculative, e.g. used to reserve frequencies and slots or to signal commercial ambition.

Source: Adapting and expanding on McDowell, J. (2026^[5]), *Jonathan's Space Report*, <https://planet4589.org/> and Kulu, E. (2026^[6]), *NewSpace Index*, <https://www.newspace.im/>.

The ongoing and planned deployment of mega-constellations is now a major driver of current launch activity and projected demand for orbital capacity. By mid-2026, more than 12 000 satellites had been

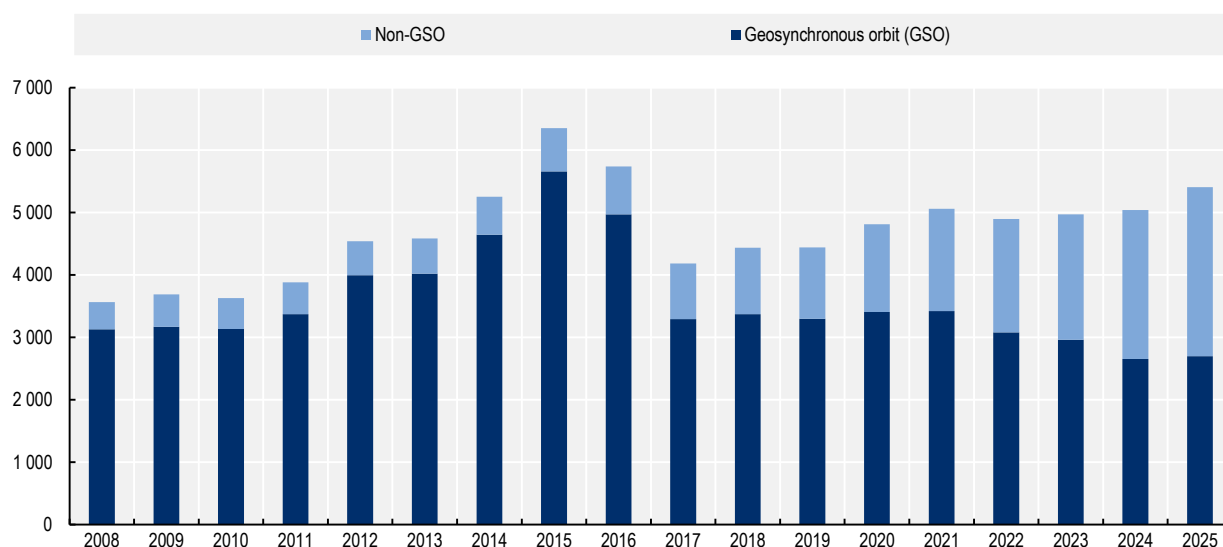
launched by 13 constellation operators based in the United States, China and the United Kingdom, with Starlink representing the largest deployed constellation. Announced projects and regulatory filings refer to more than 2 million possible future satellites over the next decade, mainly for satellite broadband, direct-to-device communications, orbital data centres and other connectivity services (Figure 2.6).⁴ In 2026, several unanticipated filings proposed very large space-based computing infrastructures, including proposals involving up to one million satellites. However, the underlying technologies remain largely unproven at scale, and commercial viability would depend on launch and in-orbit operations costs falling substantially.

International Telecommunication Union satellite filing patterns also point to this shift. Filings for geostationary orbit systems declined between 2015 and 2025, while non-geostationary filings rose sharply and overtook geostationary filings (Figure 2.7). This matters because the ITU plays a central role in co-ordinating the international use of radio-frequency spectrum and orbital resources, helping to avoid harmful interference between satellite systems.

All satellite filings should be interpreted cautiously. Many are non-binding, speculative or unlikely to be implemented in full. Some filings are used to reserve frequencies and orbital resources or to signal commercial ambition. Nevertheless, they provide an indication of the scale of commercial interest in low-earth orbit and the growing pressure on spectrum, launch capacity and orbital management.

Figure 2.7. Satellite filings shift from geostationary to non-geostationary systems, as low-earth orbit constellations expand

Total number of ITU filings, 2008-2025

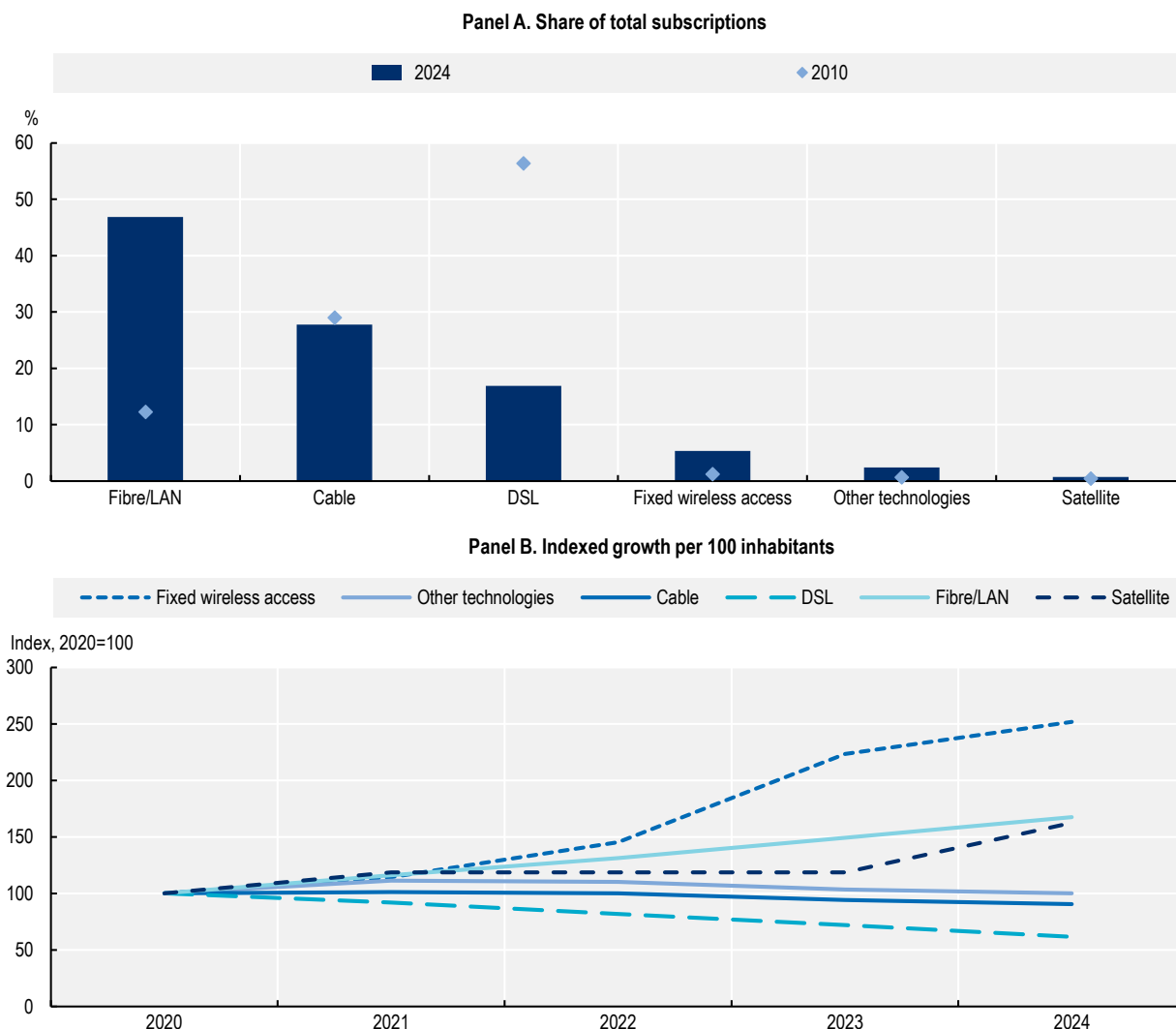


Source: ITU (2026^[7]), "ITU Radiocommunications Bureau (BR), 2025 Annual Space Report to the STSC 2025", <https://www.itu.int/en/ITU-R/space/snl/pages/reportsts.aspx>.

Satellite broadband and emerging direct-to-cell communications are moving from niche activities to growth markets. Satellite connections represented around 0.7% of fixed broadband subscriptions in OECD countries in 2023, but they are growing steadily and play an important role in extending coverage to rural and remote areas where alternative infrastructures may be limited (Figure 2.8) (OECD, 2026^[8]; forthcoming^[9]). Low-earth orbit constellations are reshaping the market by enabling higher-speed, lower-latency services and by supporting new use cases in mobility, resilience and direct-to-device connectivity.

The pace at which satellite connectivity businesses can scale is illustrated by Starlink, whose revenues reportedly nearly tripled between 2023 and 2025, reaching approximately USD 11.4 billion, driven by an expanding subscriber base and service portfolio (SpaceX, 2026^[10]). Further growth is expected as more firms diversify into maritime and aviation connectivity, direct-to-cell capabilities and hybrid satellite-terrestrial services (OECD, forthcoming^[11]). These developments may strengthen network resilience and extend coverage, but they also reinforce the importance of monitoring market concentration, interoperability and access conditions.

Figure 2.8. Satellite takes a small but growing share of OECD fixed broadband subscriptions



Source: OECD (2026^[8]), *Broadband and telecom databases*, data last updated 20 March. <https://data-explorer.oecd.org/>.

The rapid expansion of low-earth orbit constellations is also increasing pressure on orbital resources. This chapter treats congestion only as a market-structure and resource-management issue; the broader sustainability and debris implications are addressed in Chapter 4. From a market perspective, the concentration of satellites in attractive orbital bands can affect entry conditions, collision-avoidance costs, spectrum co-ordination and the long-term availability of orbital capacity. This will require national and

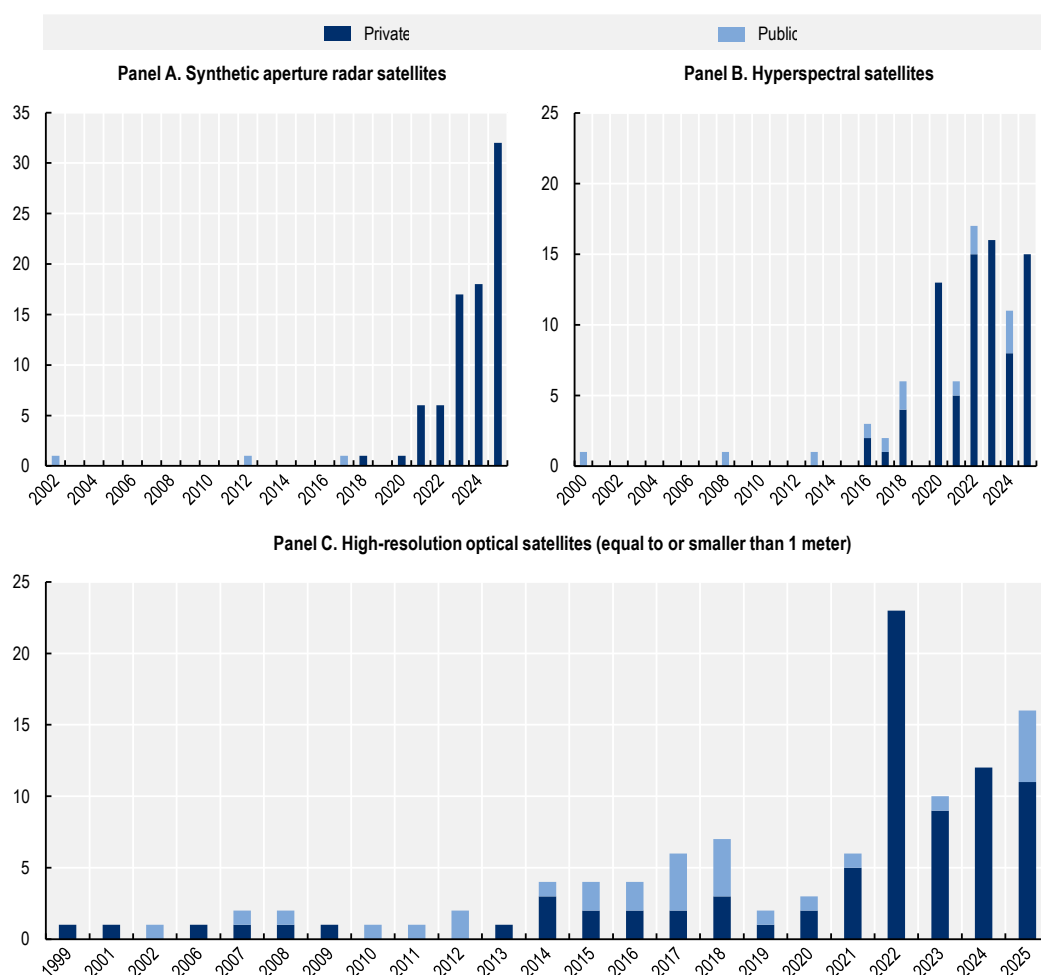
international co-ordination efforts. The management of spectrum is a case in point, where regulators need to consider both existing and future needs while making room for innovation (OECD, 2022^[2]).

Commercial earth observation services have grown in scale and sophistication

Commercial earth observation markets are evolving rapidly. Technological progress, standardisation and miniaturisation on the launcher and satellite hardware side have further lowered the barriers of entry and made commercial products more affordable and performant (OECD, 2024^[12]). Benchmarking exercises put start-ups are at the forefront of technological excellence in commercial imagery (Bingen, Gauthier and Chang, 2024^[13]). These developments are expanding the range of commercially available observations, including higher-resolution optical imagery, synthetic aperture radar and hyperspectral data.

Figure 2.9. Rapid growth in the launch of advanced sensors

Number of satellites, 2002-24



Sources: Clauson. et al. (2024^[14]), *Earth Observing Sensing Satellites Online Compendium: U.S. Geological Survey digital data*, <https://calval.cr.usgs.gov/apps/compendium>, data accessed 31 May 2026.

There has been notable growth in the launch of advanced earth observation technologies, including synthetic aperture radar, hyperspectral sensors and optical imagery with sub-metre resolution (Figure 2.9).

These capabilities can improve observational and analytical performance, particularly where time series are sustained and data access conditions support public and commercial use. They can also create new markets for monitoring infrastructure, agriculture, maritime activity, natural resources, insurance, climate risks and security-related applications.

As the cost and complexity of earth observation missions decline, demand is also changing. Some countries increasingly seek technology transfers and domestic control over earth observation systems, rather than relying only on external data providers. In parallel, large utilities, transport operators and infrastructure companies are beginning to assess whether small, dedicated constellations could serve their own monitoring needs, for instance for pipelines, railways, ports, energy assets, land-use impacts or climate-related risks. This broadening of potential users reflects the growing operational relevance of earth observation, but it may also fragment demand across highly specialised use cases.

In their in-depth review of satellite imagery/earth observation technology in official statistics, the United Nations Economic Commission for Europe highlighted the need to calibrate earth observation data against other datasets as well as investments in infrastructure to fully support data processing, interpretation and analysis (UNECE, 2019^[15]). Private sector actors have expressed similar types of concern about high investment needs with uncertain returns (OECD, 2024^[4]). The introduction of AI models could in some cases deepen rather than alleviate distrust in these technologies, as potential users understand even less the underlying technology. Indeed, there seems to be widespread use in the earth observation field of AI methods that are not explainable (Gevaert, 2022^[16]). Building new commercial applications also depends on the constant and reliable supply of data. Meanwhile, high-resolution imagery, both in terms of pixel size and temporal revisits, can be subject to government “shutter control”, to avoid revealing sensitive information, e.g. in times of conflict. Commercial GNSS-enabled applications only developed after the US government decided to discontinue its use of “selective ability”, an intentional degradation of public GPS signals in 2000.

This fragmentation creates challenges for commercial earth observation providers. Many markets remain specialised, project-based and characterised by limited repeat demand. Customers often need tailored products, sector-specific analytics, calibration with other data sources and assurance on continuity, quality and latency. As a result, revenue streams can be uneven, and some private operators face bottlenecks in converting technical performance into scalable and recurring commercial revenues (Cerbaro et al., 2020^[17]; Burke et al., 2021^[18]; CEOS, 2019^[19]). This helps explain why government demand remains important, even as commercial offerings become more capable.

Commercial earth observation is also increasingly shaped by the interaction between public missions and private provision. Governments have opened access to public satellite catalogues while also supporting private providers through deregulation, procurement and service buys. This dual role has expanded the availability of satellite data while creating demand for commercial services. The US National Oceanic and Atmospheric Administration (NOAA) has been purchasing commercial radio occultation data since 2016 for integration into the agency’s numerical weather prediction models (NOAA, 2023^[20]). NASA has been using the Commercial Smallsat Data Acquisition programme since 2017 to augment and complement its own data or that of other partners (NASA, 2023^[21]). In 2022, the National Reconnaissance Office made its largest commercial contracts yet to three data providers, worth several USD billion until 2032, as part of its Electro-Optical Commercial Layer (EOCL) programme (NRO, 2022^[22]). Other countries and organisations followed suit. Eumetsat launched a commercial radio occultation third-party pilot data service in 2022 (Eumetsat, 2022^[23]).

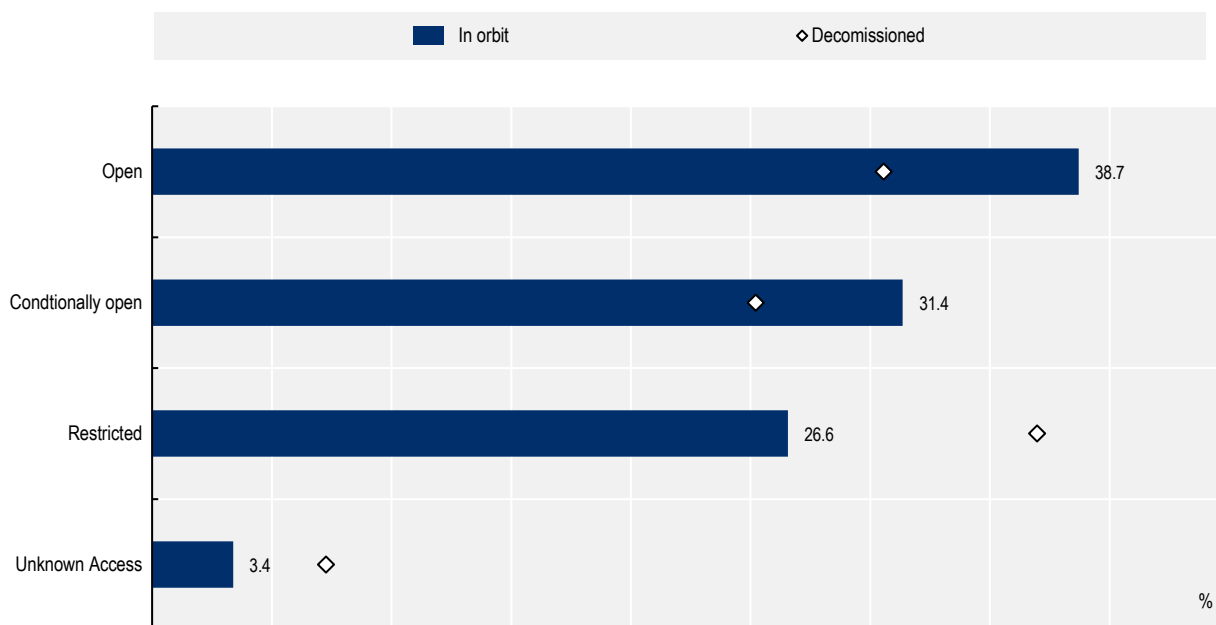
In the case of weather information, seven-day weather forecasts today are about as accurate as five-day forecasts were around 2000, reflecting major advances in modelling and, above all, the rapid expansion of earth observation systems. More than 90% of the observations assimilated into modern weather forecasting systems comes from satellites, making space infrastructure the foundation of both traditional and new AI-based forecasting. Traditional numerical weather prediction (NWP) combines these

observations with previous forecasts through data assimilation before solving the physical equations governing the atmosphere, while AI models learn atmospheric behaviour from decades of reanalysis data and can generate forecasts in minutes. The two approaches are increasingly complementary, with operational centres such as the European Centre for Medium-Range Weather Forecasts (ECMWF) already using both in parallel. However, neither AI nor NWP can compensate for missing observations. Continued investment in public meteorological satellite programmes (e.g. NOAA, EUMETSAT and JAXA), alongside the growing ecosystem of commercial weather data providers (e.g. Spire Global, Tomorrow.io and PlanetIQ) remain essential to further improve forecast accuracy.

Open public earth observation data are therefore a major source of public value and a foundation for commercial innovation. Data from government missions are increasingly openly available. The Committee on Earth Observation Satellites reports that some 70% of data from active missions (and 55% from decommissioned missions) are open access, if including those requiring registration (Figure 2.10). The Landsat and Copernicus programmes provide free and open access to long time series of earth observation data, supporting science, public policy and private-sector applications. These datasets provide de facto standards for geometric, spectral and radiometric calibration, enabling reliable detection of change across time and locations. They also provide training data and reference datasets for digitally enabled analysis, including machine-learning applications and geospatial foundation models.

Figure 2.10. Accessibility of earth observation data from CEOS missions

Share (%) of instruments, data as of 1 June 2026



Note: Conditionally open data may require registration or a formal application for data access.

Source: CEOS (2026^[24]), CEOS Virtualization Environment (COVE), <http://www.ceos-cove.org/en/>.

Open public earth observation data and commercial markets are closely interconnected. Open datasets can lower barriers to entry, support value-added analytics, enable benchmarking and strengthen user confidence in commercial products. At the same time, governments need to balance the benefits of open access with conditions that allow commercial providers to develop sustainable business models.

GNSS-enabled digital services are growing fast

Global navigation satellite system (GNSS) signals are among the most widely diffused space-enabled inputs in the economy. Since GNSS chips were first introduced in mobile phones, wearables and cars in the early 2000s, satellite positioning, navigation and timing services have become embedded in a broad range of digital devices, platforms and services. Unlike upstream space activities, much of this value is generated downstream, where satellite signals are combined with software, maps, sensors, connectivity, data analytics and user-facing applications.⁵

This makes GNSS a useful example of how space infrastructure supports commercial value creation beyond the space sector itself. Location-based services now underpin consumer navigation, personal and asset tracking, emergency applications, health and fitness services, mapping, logistics, tourism, agriculture, construction and other digital markets. In many of these applications, the space component is largely invisible to end users, but it remains an essential enabling input.

These developments point to expanding commercial opportunities, although market development remains uneven. The European Union Agency for the Space Programme forecasts accumulated revenues of around EUR 2.8 trillion over 2024-34 for GNSS-enabled consumer solutions, tourism and health applications (EUSPA, 2026^[25]). These estimates, although covering data markets beyond the core space economy, point to new potential commercial opportunities associated with satellite signals, particularly as GNSS capabilities are integrated into smartphones, vehicles, wearables, connected devices and sector-specific digital services.

For policy makers, GNSS-enabled markets illustrate why the economic value of space cannot be assessed only through launch, satellite manufacturing or satellite operations. Much of the value emerges through broader applications, business models and user adoption. This reinforces the importance of reliable and resilient positioning, navigation and timing signals, as well as interoperability, standards, cybersecurity and user capabilities across the wider digital economy.

Table 2.1. Examples of GNSS-enabled applications

Application area	Examples of GNSS-enabled services	Main value proposition
Consumer and lifestyle services	Smartphone navigation, geo-tagging, sport and fitness applications, social networks, tourism services	Enables everyday location-aware digital services and personalised user experiences
Health and safety	Emergency location, mHealth applications, air-quality alerts, support for visually impaired users	Improves safety, accessibility and responsiveness of public and private services
Navigation and tracking	Personal tracking, asset tracking, vehicle navigation, logistics and fleet management	Supports routing, monitoring, efficiency and traceability across transport and supply chains
Corporate and mapping services	Mapping and GIS, location-based billing, geo-advertising, business analytics	Enables location-based business models and spatial decision-making
Agriculture, construction and robotics	Precision steering, machine guidance, consumer robotics, site management	Supports automation, productivity gains and more efficient use of inputs

Source: OECD analysis.

Profitability remains uneven across space economy segments

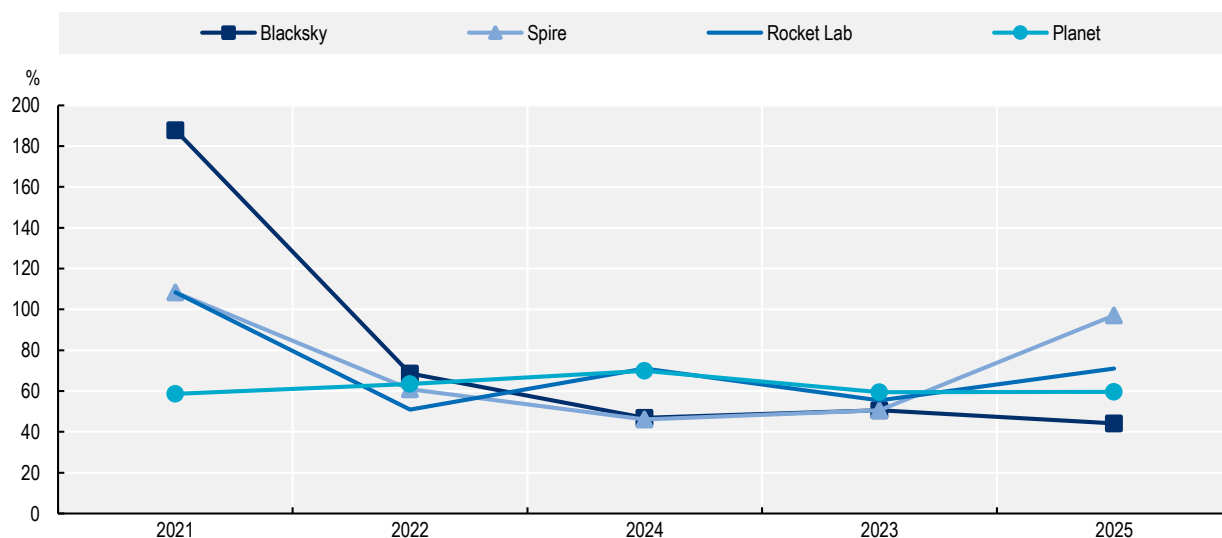
Despite the increasing number of commercial actors and activities, profitability remains uneven across space economy segments. This is particularly true in space manufacturing and launch, where firms often face high capital expenditure, high R&D costs, long development cycles and regulatory barriers (OECD, 2024^[12]).

Despite being leaders in different space market segments, several recent entrants that went public around 2021 have yet to achieve sustained profitability.⁶ For selected listed space manufacturers, combined R&D and capital expenditure represented a large share of revenues, sometimes exceeding 60% of revenues, over the 2021-25 period. This reflects the challenges of developing, manufacturing and launching space systems, as well as the reaching the scale needed to become profitable in such capital-intensive industries (Figure 2.11).

In Europe for example, commercial activity and investment are expanding rapidly across the space economy, but profitability remains uneven as many firms continue to balance strong growth ambitions against high R&D, manufacturing and capital expenditure requirements. According to the European industry association Eurospace, the sector's overall financial performance weakened significantly between 2019 and 2024, with aggregate operating margins declining from a small profit (+2% of sales) to a sizeable loss (-16%). However, this deterioration was largely driven by substantial losses incurred by a limited number of companies. Profitability challenges remained concentrated among a few large actors rather than affecting the industry as a whole (Eurospace, 2026^[26]).

Figure 2.11. High capital and R&D expenses for emerging space manufacturers

Combined capital and R&D expenditure as a share of revenues for selected listed space manufacturers



Source: OECD calculations based on public company financial statements from StockAnalysis, 2026.

Downstream segments linked to the delivery and exploitation of satellite data and signals show more heterogeneous performance than upstream activities. Satellite operators face periodic spikes in capital expenditure, for example when procuring and launching satellites, but their R&D intensity tends to be lower than in manufacturing and launch. For listed operators such as Iridium and Viasat, R&D expenditure averaged around 3.4% of revenues over 2021-25 (Stock Analysis, 2026^[27]). Closer to end-user markets, space-related activities increasingly follow broader digital economy trends, where value derives from data, software, devices and integrated services.

Market composition is also changing. Satellite broadcasting, once the dominant privately generated space economy segment in some countries, is declining as streaming services gain traction: its share in value added in the United States fell from 40% in 2016 to 25% in 2023, as recorded in the US space economy thematic account, with further decreases observed more recently (Georgi and Surfield, 2025^[6]). By contrast, the manufacturing of consumer equipment linked to global navigation satellite systems has

become a major industry segment, reflecting the growing integration of satellite-based data into everyday digital applications.

Against this backdrop, horizontal consolidation and vertical integration are becoming increasingly important business strategies in the space sector. Proposed consolidation among European space-systems manufacturers and digital space services providers, including Airbus, Leonardo and Thales, alongside the recent SES-Intelsat and Eutelsat-OneWeb mergers in satellite communications, reflect efforts to build scale across space services. Some companies are as well expanding beyond their core activities to control a larger share of the value chain, combining infrastructure, manufacturing and services under a single business model. Such strategies may improve profitability by securing demand, reducing transaction costs, capturing revenues across multiple segments and providing greater control over critical technologies and supply chains. Similar trends have been observed in the wider aerospace and defence industries, where major incumbents such as Boeing, Airbus and Lockheed Martin have often integrated upstream and downstream activities to improve margins, reduce dependence on external suppliers and strengthen resilience.

Recent developments suggest that this trend is accelerating. Beyond SpaceX's combination of launch services, satellite manufacturing and communications through Starlink, several firms expanded their activities in 2025-26. In 2026, Rocket Lab announced the acquisition of satellite communications operator Iridium, extending its activities from launch and spacecraft manufacturing into satellite network operations and connectivity services. MDA Space expanded into both manufacturing and analytics through the acquisitions of spacecraft manufacturer Blue Canyon Technologies and a majority stake in earth observation analytics company CLS. York Space acquired satcom terminal provider All.Space, while Mitsubishi Electric acquired ground-segment company Infostellar. While these strategies may increase efficiency and support investment at scale, they could also reinforce market concentration and raise barriers to entry for smaller firms. These patterns show that commercialisation does not automatically translate into profitability across all space economy segments. Some activities may remain dependent on public demand, anchor contracts or patient capital, while others may scale through consumer, enterprise or mobility markets. For policy makers, this underlines the importance of distinguishing between market growth, strategic capability and sustainable business models when designing support measures and procurement strategies.

Conclusion

This chapter has shown that the space economy is becoming more commercial, more constellation-driven and more concentrated. Access to orbit has broadened, with more countries and firms able to deploy satellites, but launch capability, orbital infrastructure and large-scale satellite operations remain concentrated among a small number of actors. Telecommunications and low-earth orbit constellations are now major drivers of launch activity and demand for orbital resources, while satellite broadband, direct-to-cell services and commercial earth observation are creating new market opportunities.

These developments are reshaping the conditions for competition and value creation. Public procurement continues to play an important role in forming markets, particularly in earth observation, while profitability remains uneven across space economy segments. For policy makers, the challenge is to support commercial growth and innovation while monitoring concentration, access conditions, interoperability and the sustainability of orbital resources. The broader public value, resilience and environmental implications of these trends are addressed in Chapter 4.

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Notes

¹ CubeSats are miniaturised, standardised satellites (typically built in 10 × 10 × 10 cm units) designed for low-cost space missions, research, and commercial applications. Commercial rideshare launches are launch services that allow multiple satellites from different customers to share a single rocket, significantly reducing launch costs for each participant.

² 12 countries have demonstrated the ability to place objects in orbit by mid-2026. In chronological order, these are: the Soviet Union/Russian Federation (1957), the United States (1958), France (1965), Japan (1970), the People's Republic of China (1970), the United Kingdom (1971), India (1980), Israel (1988), the Islamic Republic of Iran (2009), the Democratic People's Republic of Korea (2012), New Zealand (2018) and the Republic of Korea (2022).

³ The United Kingdom carried out one successful orbital launch in 1971 from an air force base in Australia. The domestic launcher programme was terminated the same year.

⁴ ITU filings should not be interpreted as equivalent to operational authorisation. They are formally submitted to the ITU by national administrations, often on behalf of satellite operators, as part of the international process for co-ordinating radio-frequency spectrum and orbital resources. Operators still generally require national authorisation and licensing from the relevant state before launch and operation. The ITU filing and co-ordination process may begin earlier and proceed in parallel with national licensing, but it does not by itself confer permission to operate. As a result, many filings may remain non-binding, speculative or only partially implemented.

⁵ Upstream comprises the manufacture and launch of satellites and their supporting ground segment; downstream begins once a satellite is operational, encompassing the services, applications and analytics built on the signals and data it provides.

⁶ Planet is also a satellite operator but has been included in space manufacturing because it manufactures its own satellites.

3

Innovation capabilities and emerging technologies in the space economy

Innovation is a key determinant of the future trajectory of the space economy, shaping the development of new technologies, applications and capabilities. Measuring space innovation, however, requires a multi-dimensional approach, as no single indicator captures its full scope. This chapter combines complementary evidence from scientific publications, patent data and selected technology developments to explore the changing geography, intensity and direction of innovation in the space economy. It provides granular insights into research capabilities, collaboration networks and emerging scientific fields, as well as inventive activity and technological development. It further seeks to identify early signals of emerging technologies advances that may not yet be visible in bibliometric or patent statistics.

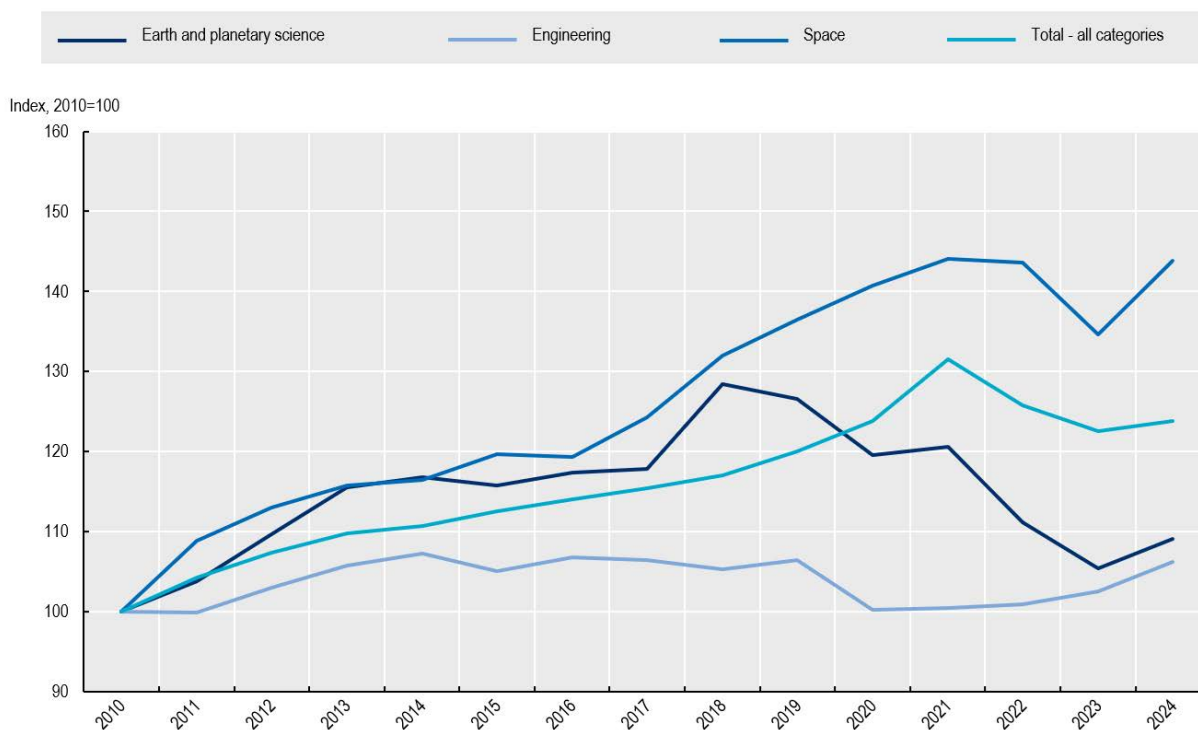
Space-related scientific publication activities are growing

Scientific publications provide a valuable window into the research base underpinning space innovation. They offer insights into the scale and direction of scientific activity, the development of research capabilities, patterns of international collaboration, and the emergence of new scientific fields and technological opportunities. While publications do not directly measure innovation outcomes, they can provide early signals of evolving knowledge frontiers and help identify areas where future technological advances may emerge. For policy makers, they provide an early view of emerging fields that may shape future technological development and competitiveness in the space economy.

Space-related publication has become an increasingly dynamic field of scientific production. Since 2009, space-related scientific output of publications in journals and conference proceedings has more than doubled, growing more quickly than broader related scientific fields such as engineering and “earth and planetary science” and total scientific publication activity, pointing to growing interest and capabilities in this area. This expansion reflects the growing strategic and economic relevance of space, the wider availability of satellite data and signals, and the development of new missions and applications across earth observation, communications, navigation, space science, exploration and downstream data services (Figure 3.1).

Figure 3.1 Scientific production in space-related fields is outpacing broader related research areas, 2010-24

Indexed growth in fractional counts for space literature and selected ASJC codes, 2010=100



Note: the number of fractional counts will be affected by its associated number of international author affiliations and Scopus All Science Journal Classification (ASJC) codes. A paper with only one country affiliation and ASJC code equals 1, whereas multiple ASJC codes and international affiliations reduces a paper's relative weight in the total count.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

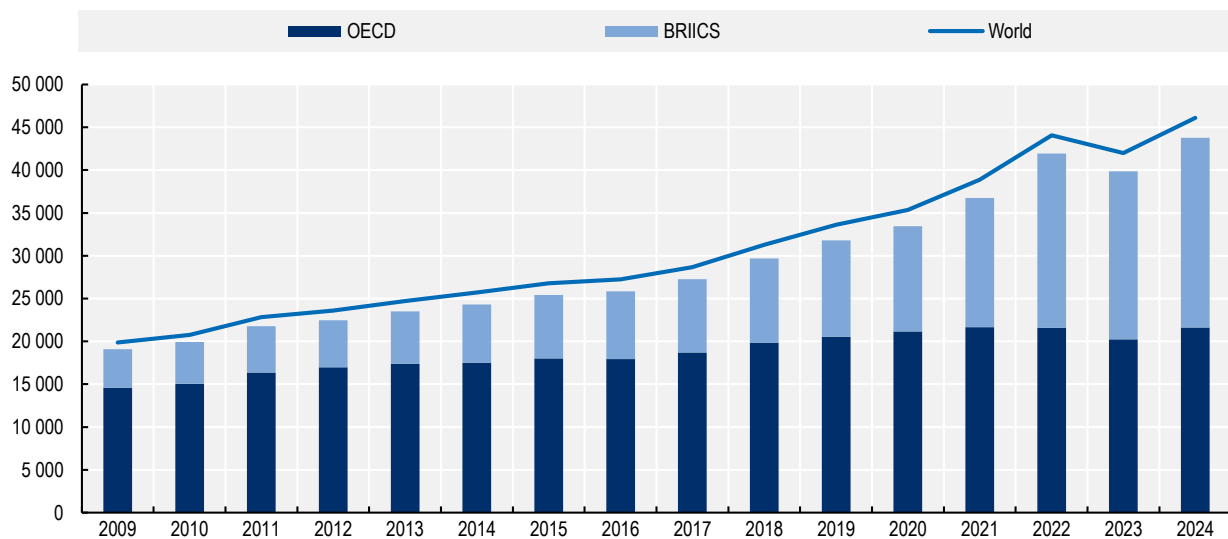
Space remains a specialised but growing field within the broader scientific landscape. In 2024, space-related publications accounted for 1.3% of total fractional publication counts in the OECD area. This modest share, alongside sustained growth, points to a niche yet expanding area of scientific activity. Growth is driven by the expansion of national space programmes, wider use of space-based data, and a broader set of actors involved in missions and downstream services. It also reflects the cross-disciplinary nature of space research, spanning astrophysics, aerospace engineering, Earth and planetary sciences, remote sensing, data science, materials, robotics and communications.

Scientific production is becoming more multipolar, with emerging economies accounting for much of recent growth

Global space-related scientific output has increased over the past 15 years, alongside a marked shift in the geography of research (Figure 3.2). Emerging economies now play a larger role, with China and India contributing significantly to growth in space-related publications and to a more multipolar innovation landscape.

Figure 3.2. Scientific production in space-related fields has become more geographically diversified, 2009-2024

Annual counts of journal papers and conference proceedings fractionalised by ASJC category and country affiliation



Notes: BRIICS refers to Brazil, Russia, India, Indonesia, China and South Africa; Fractional counts depend on the number of international author affiliations and Scopus All Science Journal Classification (ASJC) codes associated with a publication. A paper with one country affiliation and one ASJC code receives a weight of 1, while papers with multiple country affiliations or ASJC codes receive a lower relative weight in the total count.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

In 2024, China accounted for the largest share of space-related scientific literature, with 40.8% of global fractional counts. Space has been a recurring priority in China's medium-term planning, with successive space white papers outlining objectives spanning launch capabilities, satellites, human spaceflight, lunar and deep-space exploration, earth observation, navigation, communications and space science. These ambitions have created sustained demand for scientific and engineering expertise across a wide range of disciplines, including aerospace engineering, planetary science, remote sensing, telecommunications, robotics and materials science. The rapid growth of China's space-related scientific literature therefore

reflects more than an increase in publication volumes. It also points to the expansion and diversification of national capabilities associated with major space missions and the development of supporting technologies.

The OECD area continues to represent a substantial share of global space-related publications, but its relative contribution has declined. Between 2009 and 2024, the OECD area's share of global space-related fractional counts decreased from 73.3% to 46.9%.

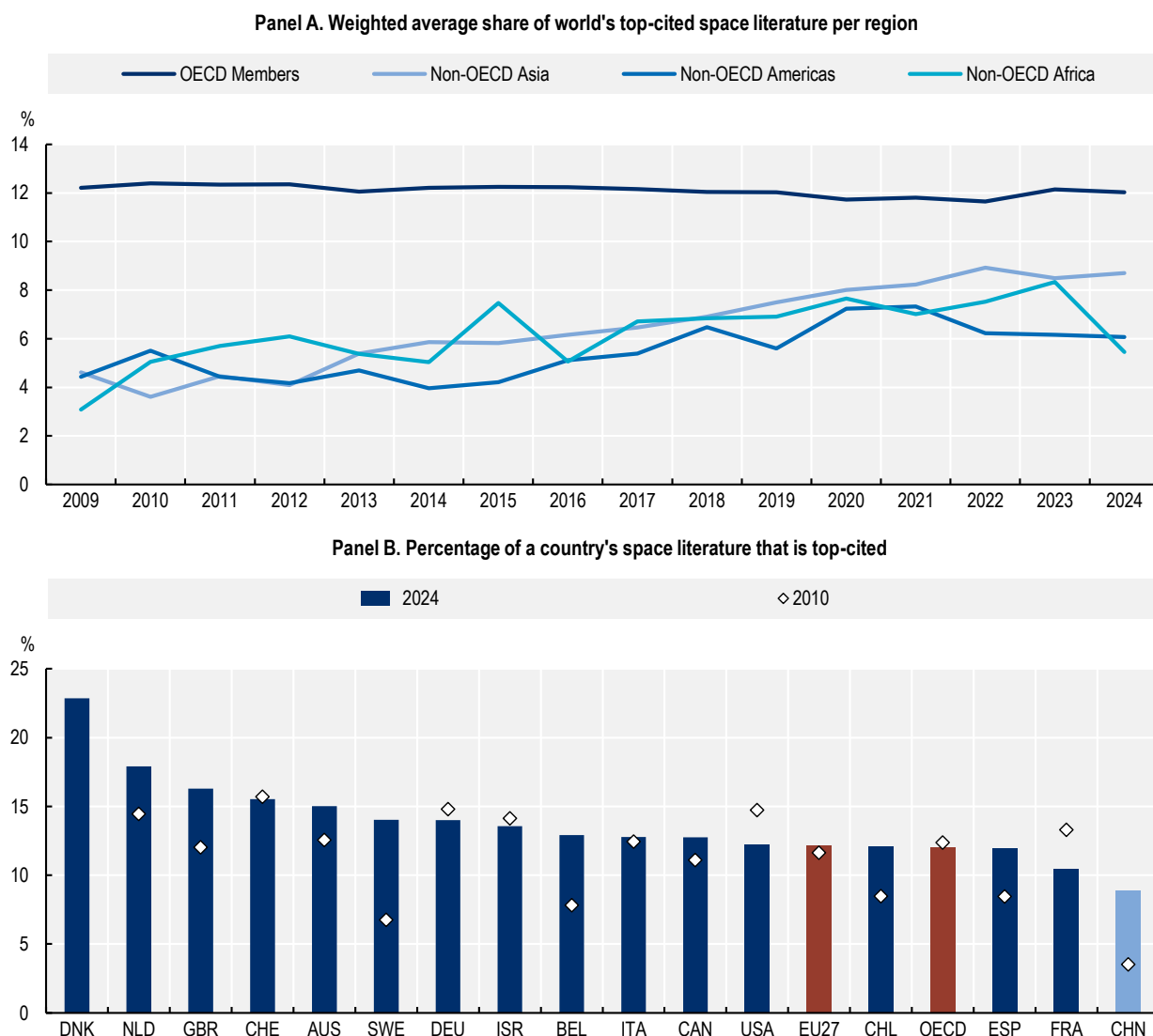
The OECD area continues to lead in scientific excellence

Citation-based indicators provide a complementary perspective on scientific impact and research excellence. The OECD area remains a leading contributor to highly influential space research, accounting on average for 12% of the world's top-cited space-related publications (Figure 3.3). Several OECD countries also demonstrate particularly research excellence relative to their overall scientific output, producing results with high international visibility and impact. In 2024, Denmark recorded the highest share of space publications among the world's top 10% most-cited publications (22.9%), followed by the Netherlands (17.9%), United Kingdom (16.3%) and Switzerland (15.5%).

These indicators point to two complementary dimensions of scientific performance. Large producers such as China and the United States account for a high share of the world's most-cited space literature in absolute terms. Smaller and highly internationalised research systems, many of them in the OECD area, such as the Netherlands and Switzerland, often show stronger performance when excellence is measured relative to their own publication output.

Figure 3.3. OECD countries remain at the forefront of high-impact space research

Scientific excellence measured as the share of documents that are among the top-10 most highly cited



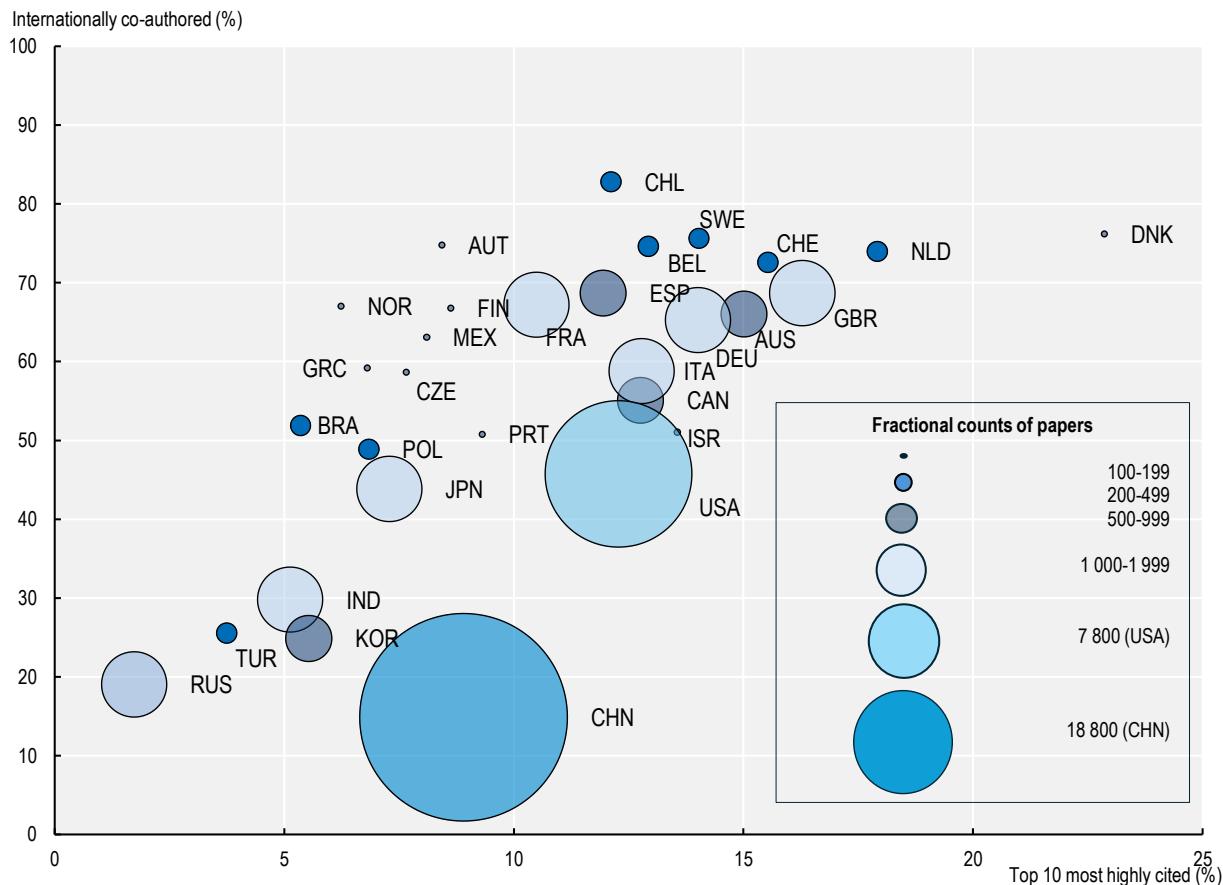
Note: Panel A shows the publication-weighted average share of each regional group's space publications (fractional counts) that rank among the world's top 10% most-cited papers. Countries with larger publication outputs carry greater weight in the average. Higher values indicate a larger share of highly cited space research. In both panels, the analysis covers economies with at least 100 fractional-count publications.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

International collaboration appears to play an important role in scientific excellence (Figure 3.4). Countries with higher levels of international co-authorship tend to perform well in citation-based indicators, reflecting the wider visibility, dissemination and potential impact of collaborative research. Citation indicators should nevertheless be interpreted with care, as citation patterns vary across fields, publication types and over time, and ratio-based indicators can favour countries with smaller publication volumes.¹

Figure 3.4. High citation rates often coincide with high shares of international co-authorship

Share of an economy's papers in top-10 percentile for citations (x-axis) and internationally co-authored (y-axis)



Note: The analysis includes economies that have produced 100 or more fractional document counts.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

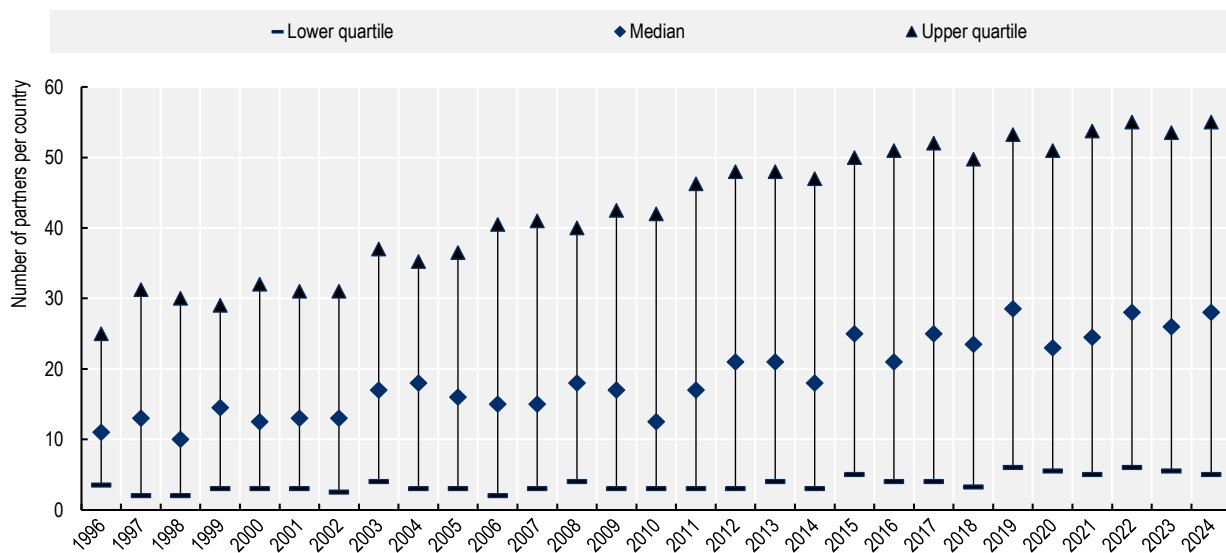
International collaboration in space research is expanding and deepening

International co-authorship indicates scientific collaboration and is often linked to higher research quality and visibility. In space-related literature, the number of participating countries rose from 71 in 1996 to 119 in 2024, while active bilateral links more than tripled, from 566 to 1 958. Over the past 28 years, authors from countries engaged in space research worked, on average, with three times as many international partners (Figure 3.5).

At country-level, the United States consistently recorded the largest number of international partners in space research, rising from 52 unique partners in 1996 to 101 in 2024. In an aggregate measure of research intensity between bilateral country partners, the United States maintains the highest ranking in every year across the time period. This suggests the United States is engaged in broader and more intense international collaboration in space research than all other countries. Other notable countries in this regard include France, Germany, and the United Kingdom which all appear in the top-5 countries for total number of partners and total research intensity in all years between 1996 and 2024.

Figure 3.5 Countries engaged in space research worked, on average, with three times as many international partners in 2024 as in 1996

Lower quartile, median, and upper quartile values from count of distinct bilateral partners per country per year



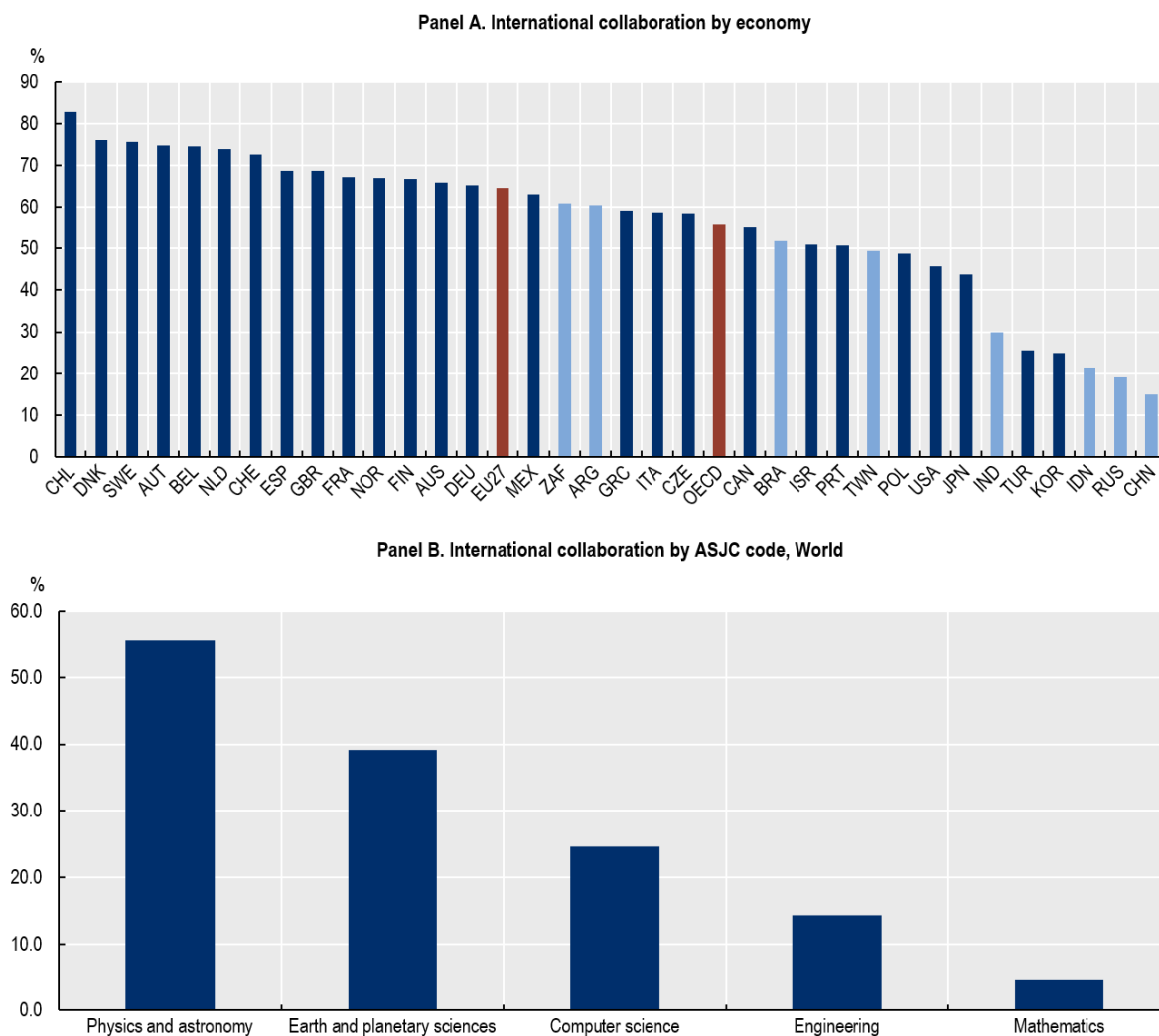
Notes: The chart shows, for each year, the distribution across countries of the number of distinct international partner countries in space-related publications, using the lower quartile, median and upper quartile. The dataset covers eligible peer-reviewed articles, reviews and conference papers from a predefined set of “pure space” journals. Publications are assigned to country pairs based on author affiliation countries recorded in citations. Only country pairs with at least 10 co-authored publications over the period are included, and affiliation countries are counted only for authors appearing in at least two publications in the dataset.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

The importance of international co-authorships in a country's total publications of space-related literature however varies widely across countries. In 2024, among countries producing more than 100 space-related fractional counts of papers, the share of publications involving international co-authorship ranged from 82.8% in Chile to 14.8% in China (Figure 3.6). These differences reflect several factors, including the size of national research communities, countries' areas of specialisation and their integration into international research networks. Smaller or more specialised research systems may rely more heavily on international collaboration, particularly in fields that require access to specific infrastructure, data or locations. Chile is a case in point: its strong specialisation in astronomical observations is supported by favourable geographic and climatic conditions, as well as by the presence of major international observatories.

Figure 3.6. International co-authorship in space publications varies by economy

Share of fractional counts with international co-authorship, 2024



Notes: International co-authorships are based on the affiliation of the authors. The analysis includes economies that have produced 100 or more fractional document counts. Multiple All Science Journal Classification (ASJC) codes can be assigned to each journal.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

These collaboration patterns are important for policy makers because space research relies on shared infrastructure, international missions, data access, distributed expertise and long-term institutional partnerships. A key policy challenge will be to preserve openness while managing security risks, resilience needs and strategic dependencies in space research and innovation.

Box 3.1. Measuring space-related scientific output through bibliometric indicators

Bibliometric indicators based on academic journal articles and conference proceedings provide useful insights into countries' scientific capabilities, collaboration patterns and emerging research fields. The analysis draws on 254 space-related journals and conference proceedings indexed in Elsevier's Scopus database, selected using journal names, Scopus All Science Journal Classification codes and the SCImago Journal Rank. The dataset covers 491 496 fractional counts of scientific papers published between 2009 and 2024, referred to here as "space literature". Fractional counts are used to attribute documents across fields and country affiliations. This approach improves comparability, but results should be interpreted with care, particularly for multidisciplinary journals and internationally co-authored publications. Publication volumes, co-authorship, citations and research themes can help identify international collaboration networks, areas of research excellence and emerging topics of interest to policy makers. However, publication behaviour, citation practices, journal coverage and the growing use of generative artificial intelligence may affect cross-country and comparisons over time.

Scientific publications point to emerging space innovation fields

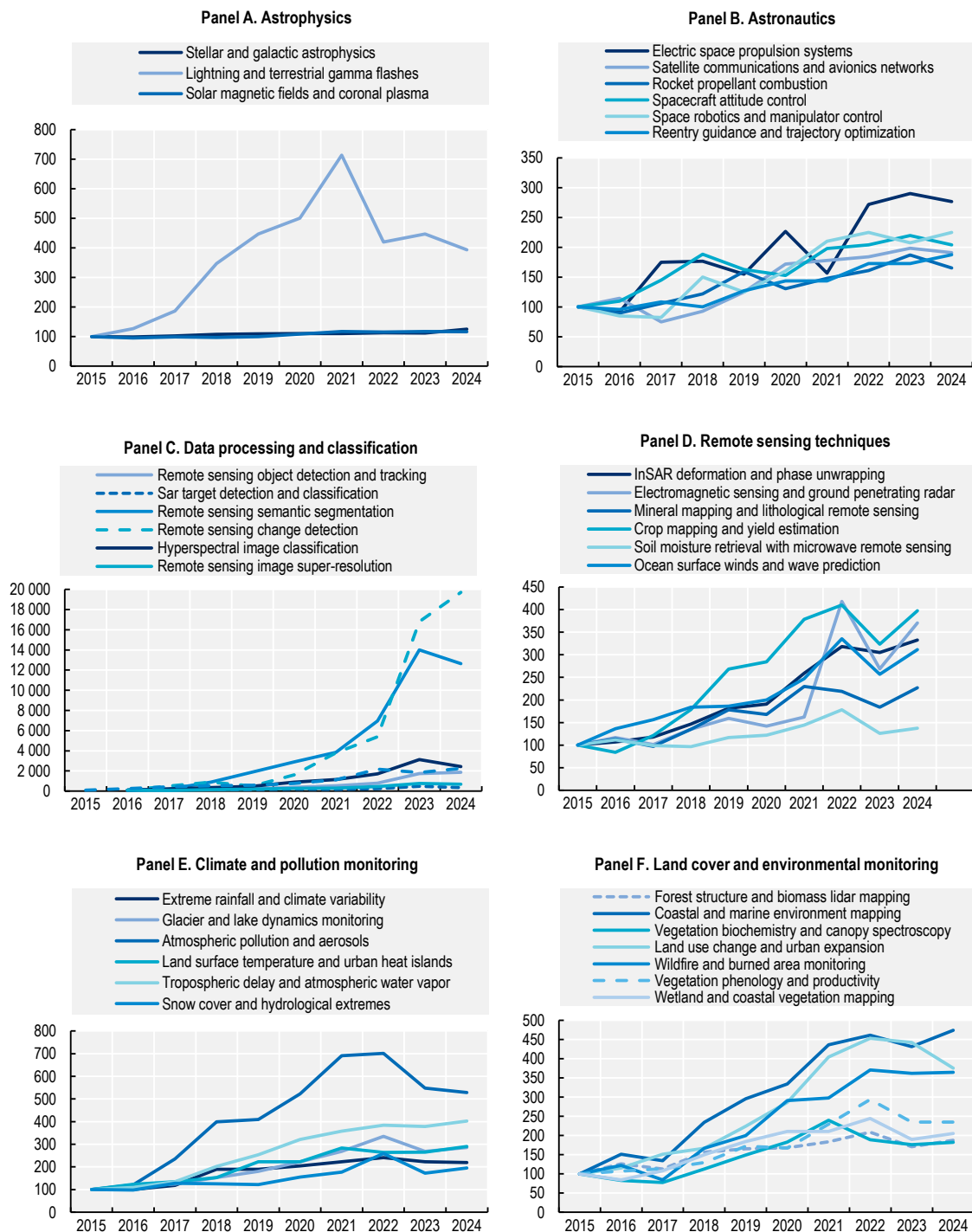
Scientific publications can provide early signals of emerging innovation fields. Multiple factors can influence thematic trends in scientific literature, including technological developments, access to new data, government funding strategies and broader societal trends.

When looking at trending topics in space literature since 2015, remote sensing change detection and remote sensing semantic segmentation² stand out in terms of novelty and growth in the number of associated papers (Figure 3.7). This coincides with the increasing use of digital technologies to process and analyse satellite imagery. Other significant topics include atmospheric pollution and aerosols, lightning and terrestrial gamma flashes, and coastal and marine environment mapping, corresponding with improved space-based sensors and the broader deployment of earth observation satellites.

These trends point to the growing importance of data-intensive space innovation. Advances in sensing technology, satellite revisit times, cloud computing, artificial intelligence and access to large open datasets are expanding the analytical uses of satellite imagery and signals. This creates opportunities for environmental monitoring, climate science, disaster management, agriculture, security and commercial data services.

Figure 3.7. Trending topics in space publications highlight the growing role of digital technologies in satellite imagery

Indexed growth in papers, 2015=100



Note: Topics identified using a BERTopic influenced topic model (cluster size 750).

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

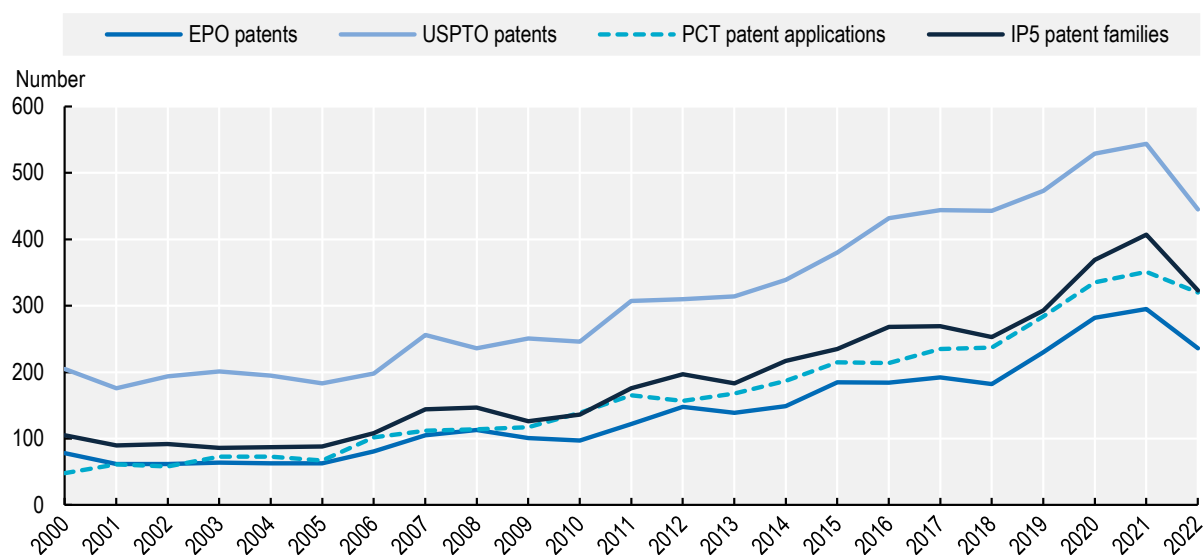
Space-related patenting points to new technological development

Patent data provide a useful, albeit partial, proxy for measuring innovation in the space economy. They capture codified technological advances and make it possible to track trends in inventions across countries, organisations and technology domains. However, not all forms of space innovation are covered by patents, particularly where knowledge is kept confidential or where advances are embedded in system integration, software, operational processes or mission design.

Space-related patenting has increased steadily since 2000, pointing to sustained growth in technological development across the sector (Figure 3.8). Patenting activity has risen across six major technological fields: propulsion, spacecraft electrical power, space system control, life support, space debris and quantum technologies for space applications. A slight decline has been observed since 2019 in most fields.

Figure 3.8. Space-related patenting has increased steadily since 2000

Number of patents, by patent offices and priority date

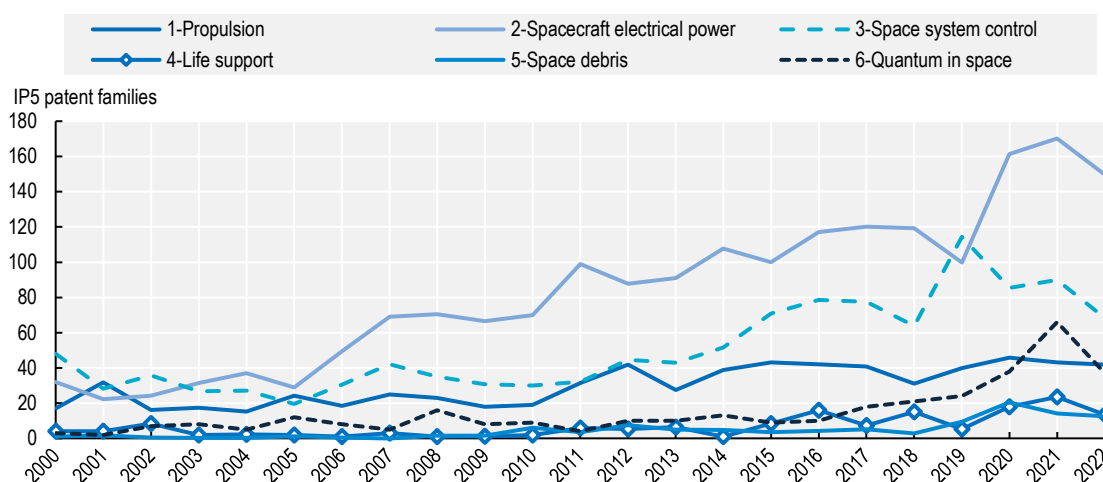


Note: Partial information for patent applications from priority year 2022. IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD (2026^[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

The growth in space-related patenting has been driven primarily by advances in spacecraft electrical power and space system control technologies (Figure 3.9). These two fields account for a large share of patent applications in the sample, reflecting the importance of energy generation, storage and management, as well as guidance, navigation and control capabilities, for increasingly complex space missions. In 2023, spacecraft electrical power alone represented 46% of space-related patent applications, underscoring its central role in contemporary space innovation. The period also saw the emergence of newer technology areas, including quantum technologies for space applications and space debris-related technologies. Although still smaller in volume than more established fields, these areas point to evolving innovation priorities, linked respectively to next-generation sensing, communications and timing applications, and to the growing need to monitor, mitigate and manage orbital debris.

Figure 3.9. Evolution of space-related patents by field, driven mainly by advances in spacecraft electrical power and space system control technologies



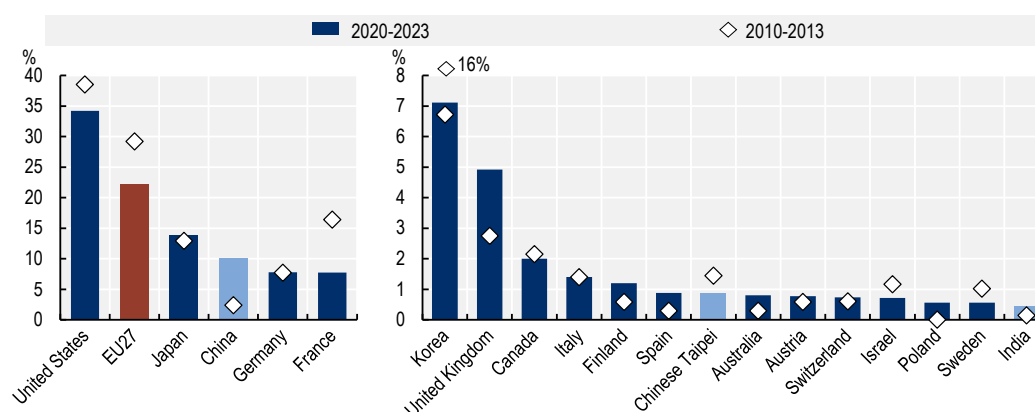
Note: Partial information for patent applications from priority year 2022. IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD (2026_[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

The United States remained the leading source of space-related patenting activity in 2020-23, accounting for 34.2% of applications. It was followed by the EU27, Japan and China (Figure 3.10).

Figure 3.10. Space-related invention remains led by economies with mature space programmes

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: Partial information for patent applications from priority year 2022. IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

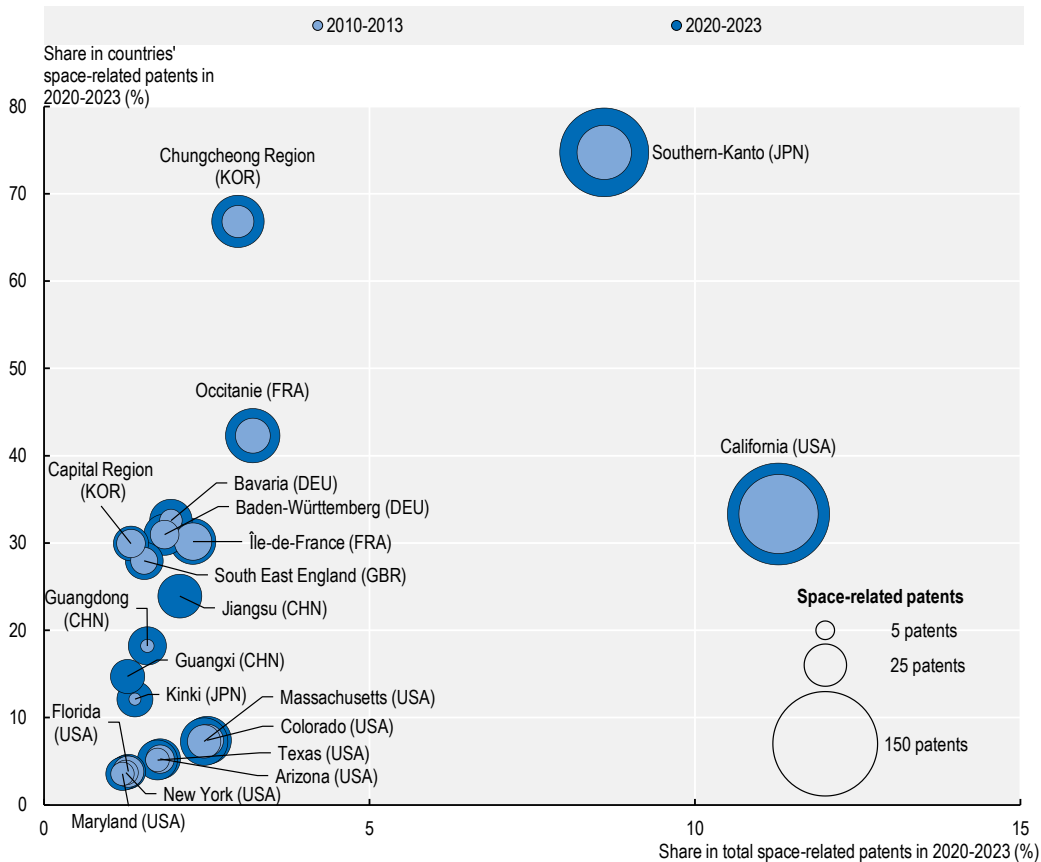
Source: OECD (2026_[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

This distribution points to the continued concentration of space-related inventive activities in economies with long-standing space programmes, large public research and defence capabilities and well-established industrial ecosystems.

At the same time, the geography of space-related innovation is gradually broadening. Several economies considerably increased their share of total space-related patenting activity over the period, notably Australia, China, Finland, India, Poland, Spain and the United Kingdom. These developments reflect a combination of national space strategies, participation in international programmes, specialised firms and research centres, and rising demand for space-based applications in connectivity, earth observation, navigation and security. Space-related patenting is also concentrated in a limited number of leading innovation regions, although their importance within national patenting activity has increased over time. California recorded the highest number of space-related patents over 2020-23, reflecting the region's strong concentration of space firms, research organisations and technology capabilities (Figure 3.11). It was followed by Southern Kanto, Occitanie, Chungcheong and Colorado, which combine established aerospace industrial bases, specialised research ecosystems and proximity to major public space or defence institutions.

Figure 3.11. Space innovation remains concentrated in leading regions, 2010–13 and 2020–23

Patent applications filed under the Patent Co-operation Treaty (PCT) by inventor's location and priority date



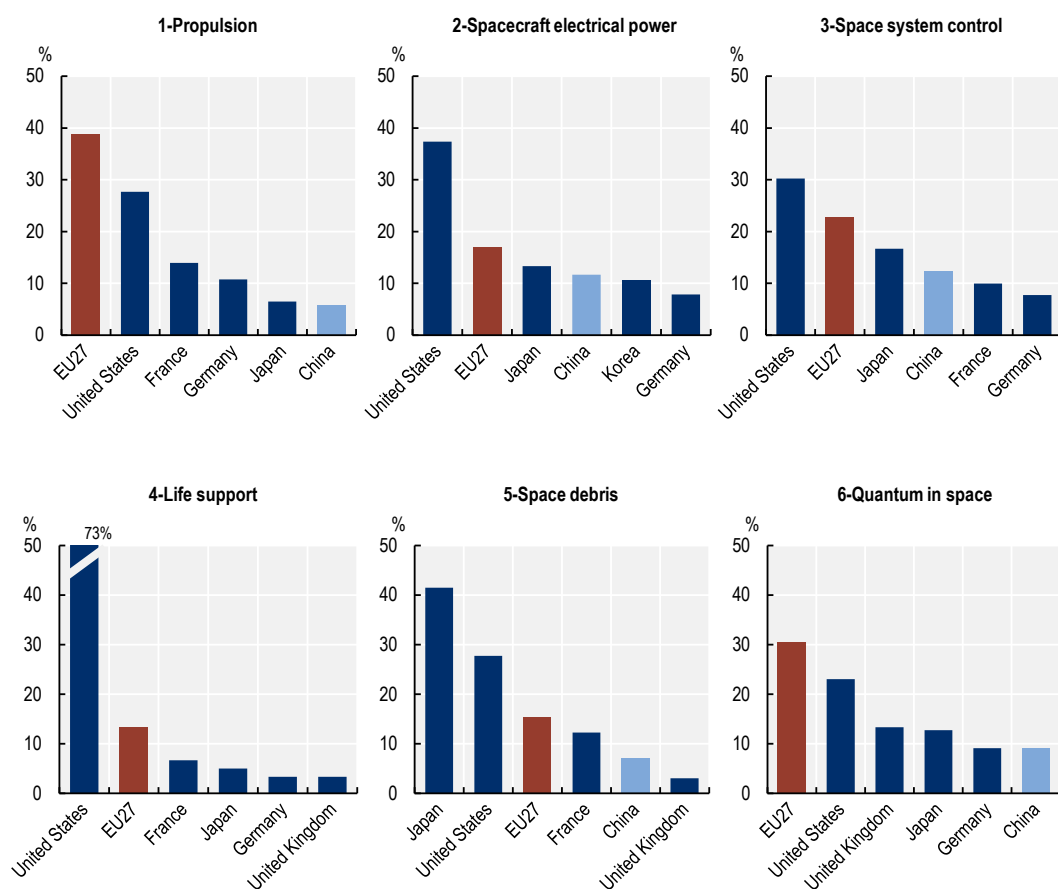
Note: Data refers to patent applications filed under the Patent Cooperation Treaty (PCT), by the inventor's region at Territory Level 2 (TL2) and priority date. Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD (2026^[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

Patent data for 2020-23 suggest distinct areas of space innovation specialisation across major economies (Figure 3.12). European organisations lead in propulsion and quantum-related patents. The United States leads in life support systems, spacecraft electrical power and space systems control. Japan stands out in space debris innovation, illustrated by mission activity related to active debris removal and inspection of non-co-operative objects. These patterns should be interpreted with caution, as relevant innovations may also be taking place in other economies or remain unpublished, unpatented or protected through secrecy for commercial, strategic or security reasons (Olivari, Jolly and Undseth, 2021^[2]).

Figure 3.12. Space-related patents indicate specialised innovation strengths across major economies, 2020-23

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD (2026^[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

AI-enabled data processing is becoming a key space innovation frontier

The convergence of technologies can be an important driver of innovation and new applications, as discussed in the *OECD Science, Technology and Innovation Outlook 2025* (OECD, 2025^[3]). In the space sector, convergence with digital technologies has been particularly impactful over the last decade.

Artificial intelligence (AI) has revolutionised data analysis (see for example OECD (2019^[4]; 2023^[5]), and it has become increasingly important across the space data value chain. Initially associated mainly with downstream data analysis, AI is now being integrated at earlier stages of data production, including onboard processing, data triage and autonomous prioritisation of downlinks. This is particularly relevant for earth observation missions, where satellite downlinks face bandwidth constraints and growing data volumes. The growing number and sophistication of earth observation sensors, discussed in Chapter 3, is increasing the volume and diversity of satellite data available for AI-enabled processing and analysis.

An emerging application of AI is onboard data processing and triage, allowing spacecraft to autonomously prioritise which data to downlink. Examples include segmenting and masking cloudy pixels, rapidly detecting natural disasters, classifying objects and compressing imagery to reduce downlink volume (Goodwill, Wilson and MacKinnon, 2023^[6]). In 2024, the European Space Agency launched the PhiSat-2 technology demonstration satellite, equipped with a multispectral camera and an AI computer able to turn imagery into maps, classify objects, detect wildfires and compress information before transmission.

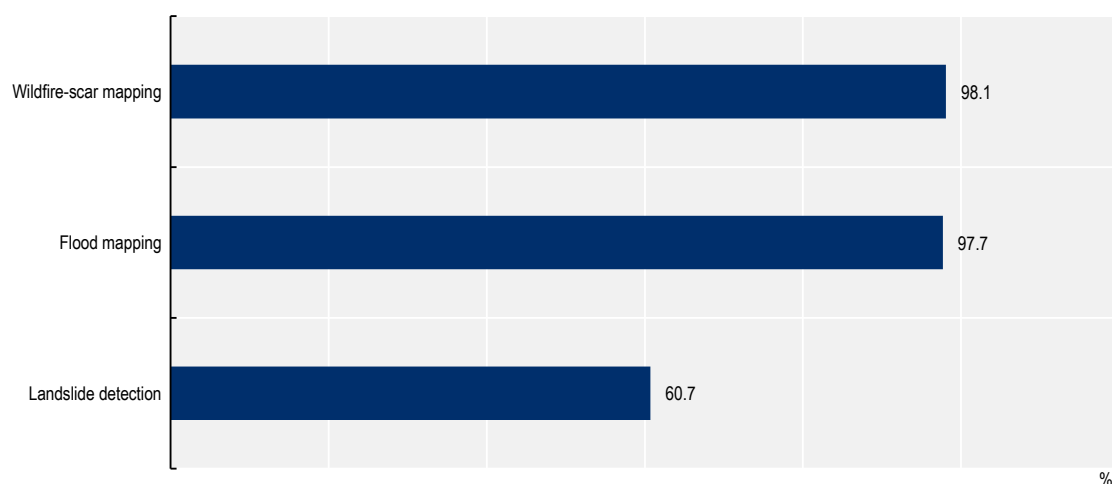
Machine learning can reduce the need for manual analysis, provide near-real-time access to information and identify patterns for predictive insights. However, many applications still require geospatial expertise, adequate ICT infrastructure and access to high-quality training datasets. These requirements can create barriers to broader uptake, particularly for smaller firms, public administrations and users in countries with limited digital and geospatial capacity.

The emergence of geospatial foundation models could represent a breakthrough for future satellite data analysis. These models may lower barriers to access while strengthening analytical capability. The NASA and IBM-led Prithvi models for earth observation, weather and climate illustrate this trend. The Prithvi-EO 2.0 model is pre-trained on millions of data points from harmonised Landsat and Sentinel-2 datasets and supports applications such as carbon flux estimation, landslide detection, burn intensity estimation, crop pattern identification and flood mapping. Figure 3.13 illustrates the model's performance across three disaster-response applications, using F1 scores ranging from 0 to 100% to capture the balance between recall, the ability to detect all relevant targets, and precision, the ability to avoid false positives. Prithvi-EO-2.0 performs nearly perfectly in wildfire-scar and flood mapping. Its performance is lower for landslide detection, where it is more effective at identifying landslide areas than at excluding false positives (Szwarcman et al., 2026^[7]).

However, there are still challenges associated with automated interpretation of aerial and satellite imagery. Figure 3.14 presents the results of a benchmark study assessing the accuracy of 13 vision-language models, which differ fundamentally from geospatial foundation models, across tasks including classification, counting and spatial reasoning (Danish et al., 2025^[8]). For the multiple-choice tasks shown, each question had five answer options: one correct answer, one semantically similar alternative and three other plausible alternatives. Random guessing would therefore yield an expected accuracy of 20%. The results show considerable dispersion in model performance, which may partly reflect differences between general-purpose and geospatially specialised models, as well as differences in architecture, training data and task difficulty. Although some models performed well on scene and land-use classification, performance remained close to random guessing for several granular counting and interpretation tasks, such as tree health assessment, general vehicle counting and crop classification.

Figure 3.13. Prithvi earth observation foundation model performs nearly perfectly on certain disaster response applications

F1 scores from separate application-specific evaluations

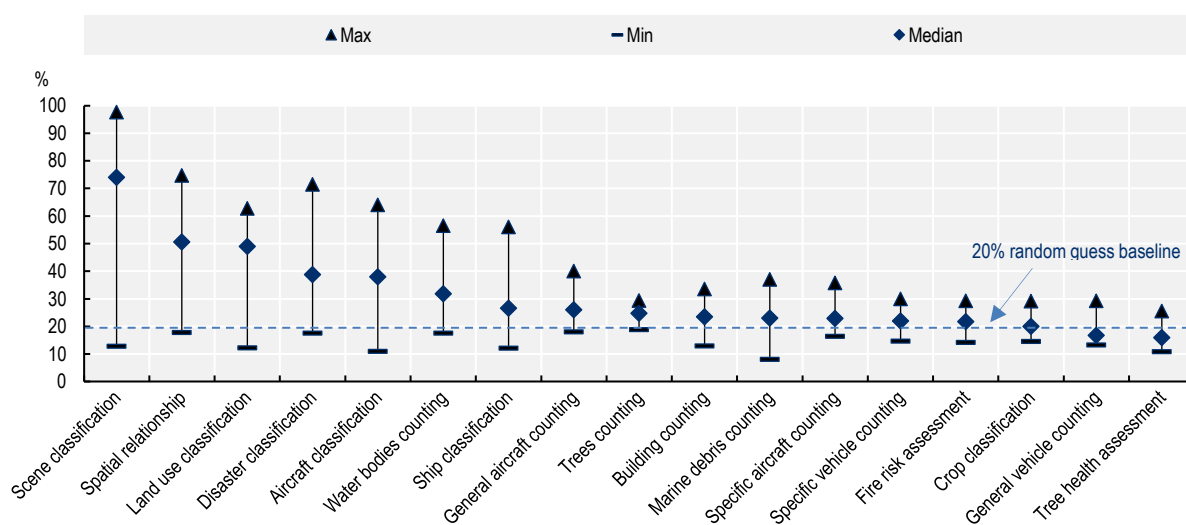


Notes: This represents the best reported results from separate application-specific evaluations; datasets, sample sizes and performance metrics differ across tasks. The F1 score balances false positives and missed detections (precision and recall). A score of 100% indicates that the model detected all relevant targets without producing false positives.

Source: Based on Szwarcman et al. (2026^[7]), “Prithvi-EO-2.0: A Versatile Multitemporal Foundation Model for Earth Observation Applications”, Tables 4-13, <https://doi.org/10.1109/TGRS.2025.3642610>.

Figure 3.14. Vision-language models display varying strengths across geospatial tasks

Highest, lowest and median shares of correct answers to multiple-choice questions.



Notes: The benchmark study assessed 13 vision-language models, featuring over 10 000 manually verified instructions and spanning diverse visual conditions, object types, and scales. Each question had five response options and one correct answer, yielding a random response baseline of 20%.

Source: Danish et al. (2025^[8]), “GEOBench-VLM: Benchmarking Vision-Language Models for Geospatial Tasks”, <https://doi.org/10.1109/ICCV51701.2025.00670>.

AI-enabled processing therefore links space innovation to broader digital transformation. It can increase the value of satellite data, support new commercial and public-service applications, and help users manage growing volumes of observations. At the same time, it raises questions about data quality, model transparency, validation, access to training datasets and the governance of automated geospatial analysis.

Emerging technologies shaping future space exploration

Space exploration is increasingly aimed at enabling sustained robotic and human operations in orbit and on planetary surfaces. This requires advances across a broad set of technologies. The International Space Exploration Coordination Group (ISECG), including some 27 organisations, helps co-ordinate global exploration planning by publishing regularly updated roadmaps of shared objectives, mission scenarios and technology needs (ISECG, 2013^[9]; 2018^[10]; 2024^[11]). Over time, there has been a gradual shift in strategic objectives, both in the destinations and the types of missions envisaged, which also affects technology needs. The Moon is no longer viewed mainly as a stepping stone to Mars, but as a destination for long-term presence. There is also greater emphasis on partnerships with industry and academia, including the use of commercial services such as lunar transport.

The list of critical technologies to mature or develop has been expanded from transportation and life support systems to technologies for long-duration habitation, power generation and in-space communication and navigation (ISECG, 2024^[11]). The 2024 Global Exploration Roadmap lists 48 critical technologies distributed across eight exploration areas: propulsion, landing and return; autonomous systems; life support and habitability; crew health and performance; communication and positioning, navigation and timing; power; transversal technologies; and extra-vehicular activity mobility and robotics, as shown in Table 3.1. They vary considerably in terms of technological maturity (for some, adequate testing facilities do not yet exist), reliance on third-party developments (autonomous systems and communications) and regulatory barriers (international trade and exchanges in nuclear fission material and technology are strictly controlled).

Table 3.1. Critical technologies needed for space exploration

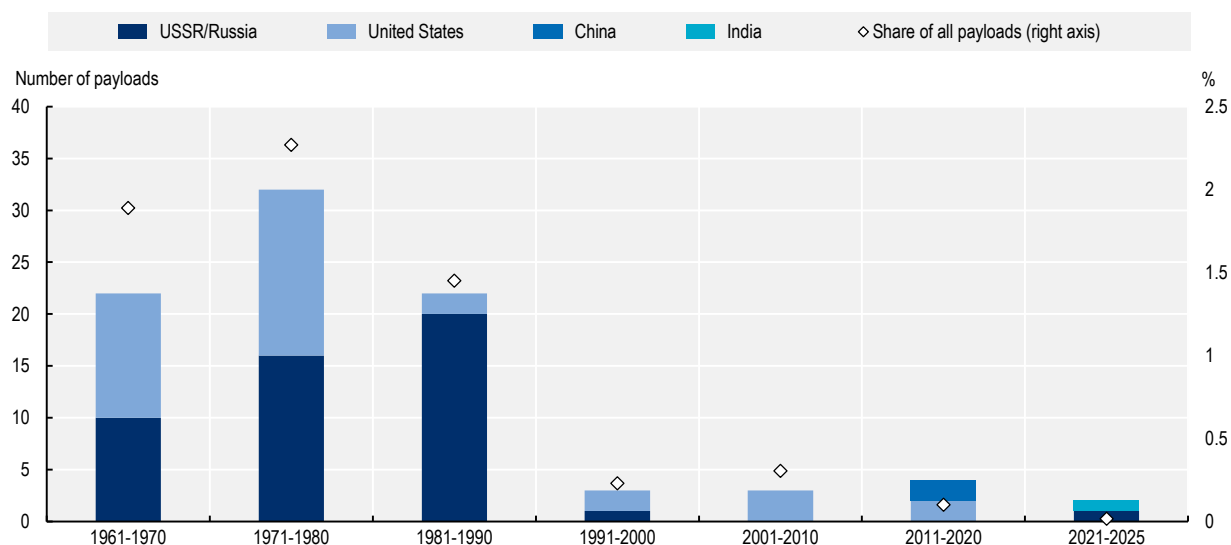
Strategic technology shift	What it means	Examples of critical technologies
From short missions to sustained presence	Future exploration is moving beyond one-off missions toward long-duration operations on the Moon, Mars and beyond. This requires systems that can keep crews alive, healthy and operational for extended periods.	Closed-loop life support, enhanced reliability life-support systems, long-duration medical care, crew autonomy beyond low-earth orbit, nutrition and storage of perishable goods, surface habitats.
From remote control to autonomy	Greater distance from Earth makes real-time control harder, so spacecraft, habitats, vehicles and robots will need to operate with more autonomy.	Autonomous rendezvous and docking, autonomous vehicle management, in-space timing and navigation, mission-control automation beyond LEO, autonomous surface habitation systems, robots working alongside suited crew.
From Earth-supplied missions to local resource use	Future missions will need to reduce dependence on supplies launched from Earth by using local resources and adapting to harsh planetary environments.	In-situ resource utilisation, dust mitigation, inflatable structures, low-temperature mechanisms, thermal management, surface mobility systems.
From solar-limited operations to resilient power systems	Sustained lunar, Mars and deep-space operations will require more reliable, high-density and long-duration power systems, especially where sunlight is limited or unavailable.	Deployable surface solar arrays, high-power in-space arrays, nuclear power for surface missions, multi-megawatt nuclear power for electric propulsion, high-energy-density fuel cells, long-life batteries, radiation-tolerant power management.

Source: Adapted from ISECG (2024^[11]), *The Global Exploration Roadmap: 2024*, <https://www.globalspaceexploration.org/wp-content/isecg/GER2024.pdf>.

Nuclear power in space could enable future exploration missions

Future space activities may require more resilient and increasingly autonomous power systems, including for operations in orbit, on planetary surfaces and in deep space. Space nuclear power could become an important enabler of such activities. Nuclear technologies have been used in space for more than six decades, mainly through radioisotope power systems providing electricity or heat for missions where solar power is unavailable, insufficient or unreliable (Figure 3.15).

Figure 3.15. Nuclear power systems launched into space



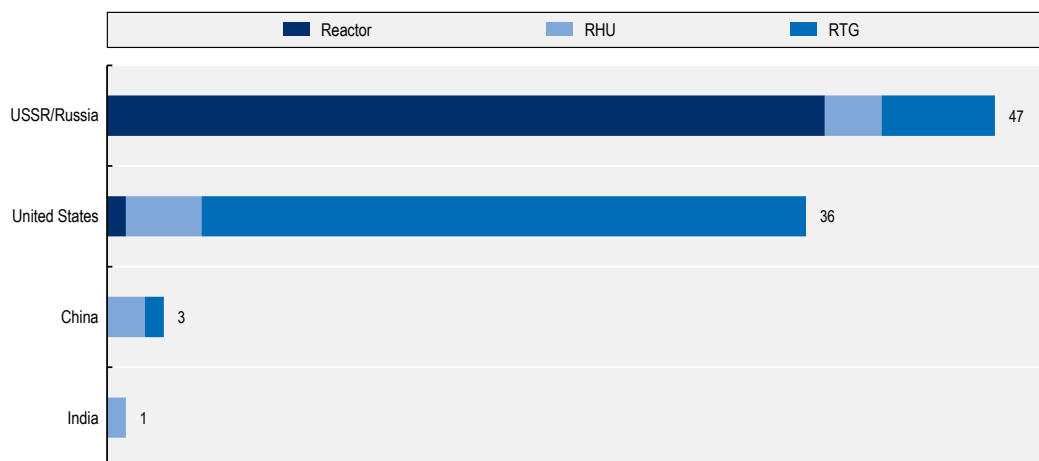
Notes: The payload of a space launch is normally its most valuable part, responsible for achieving the primary objectives of a mission (e.g. satellite, lunar lander, rover, etc.). The data includes nine failed missions: 4 in 1961-70, 2 in 1971-80, 1 in 1981-90, 1 in 1991-2000 and 1 in 2021-23.

Source: Adapted from Krebs (2025^[12]). "Nuclear Powered Payloads", *Gunter's Space Page*, https://space.skyrocket.de/doc_sat/nuclear.htm, data retrieved 13 October.

Four economies have launched nuclear power systems into space (Figure 3.16). The USSR and later the Russian Federation, and the United States, account for most historical missions, while China and India have more recently used nuclear power systems in lunar exploration missions. Most space nuclear applications have involved radioisotope heater units or radioisotope thermoelectric generators, which use natural heat radiation from selected radioisotopes to generate heat or electricity.

Figure 3.16. Types of nuclear power sources used in space, 1961-2025

Number of missions with nuclear power sources



Note: Reactor: Nuclear fission reactor; RHU: Radioisotope heater unit; RTG: Radioisotope thermoelectric generator. The data includes nine failed missions involving re-entries: 4 in 1961-70, 2 in 1971-80, 1 in 1981-90, 1 in 1991-2000 and 1 in 2021-23.

Source: Adapted from Krebs (2025^[12]). "Nuclear Powered Payloads", *Gunter's Space Page*, https://space.skyrocket.de/doc_sat/nuclear.htm, data retrieved 13 October.

Radioisotope power systems, fission surface power and nuclear propulsion offer high-density, reliable and long-duration energy solutions. These systems could support deep-space exploration, life support, in-situ resource utilisation, surface operations and transport. They may be particularly relevant for missions beyond Mars, lunar night operations, permanently shadowed regions of the Moon, high-power surface systems and future in-space propulsion applications. Nuclear fission reactors are a different technology. They rely on a controlled chain reaction that releases heat, which is subsequently converted into energy. Compared with radioisotope power systems, fission systems can generate higher power levels and may become relevant for more energy-intensive activities, including sustained surface operations and propulsion concepts. Opportunities also exist to exploit synergies with terrestrial small modular reactor development, including knowledge transfer on safety, materials, power conversion and systems engineering.

Key challenges vary by technology and include technology readiness, testing infrastructure, mission design, fuel form, power conversion, system reliability and integration with spacecraft or surface architectures. The wider safety, environmental, supply and governance implications of nuclear power in space are important but are addressed in the chapter on vulnerabilities and resilience.

Conclusion

Space innovation is expanding rapidly and becoming more geographically diverse. China now accounts for much of the recent growth in scientific output, while OECD countries continue to perform strongly in research excellence and international collaboration. Patent indicators show sustained technological development, particularly in spacecraft power systems, control systems and emerging fields such as quantum technologies. For policy makers, the key challenge will be to maintain strong research and industrial capabilities while supporting collaboration, skills development and access to specialised infrastructure and data. Emerging technologies such as AI-enabled data processing and nuclear power systems could open new mission possibilities but will require forward-looking policy frameworks. Issues related to safety, supply constraints and resilience are addressed in later chapters.

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Notes

¹The top 10% most cited documents is an indicator of “excellence”. This rate indicates the amount (in percentages) of a country’s scientific output that is included into the group of the 10% of the most cited papers in their respective scientific fields. The indicator of scientific excellence is calculated at the document level using fractional counts. Documents – organised by document type, ASJC field and year – are sorted in descending order based on the number of citations received. A threshold of 10% most cited documents is calculated for each category. Only documents with a fixed number of citations above the threshold are included. Documents with the same number of citations as the threshold are sorted according to the Scimago Scientific Journal Rankings (SJR) value of the journal in which they were published; those with the highest scores are selected. It is likely that this step of the procedure – used to select the citation tiebreak documents – favours countries publishing in higher prestige journals versus a more random approach. The resulting documents are included until 10% is attained. Only after this selection has been made can the identification by country and field be made to determine their share in total output. The citation window is based on the whole period, so top 10% most cited documents for the reference year uses whole period citations. No citation window is imposed as citation-based indicators are calculated on the basis of comparisons with documents published in the same year. The world average is 10% for the period (OECD, 2026^[13]).

² Change detection identifies changes in specific land features by sensing them at different times. Semantic segmentation means classifying every pixel within images in specific categories, such as vegetation, urban infrastructure, bare soil, etc. Satellite image calibration is essential for reliable change detection and semantic segmentation because it ensures that variations in pixel values reflect genuine changes on the ground rather than differences in sensors, atmospheric conditions, illumination, or acquisition settings. US Landsat imagery serves as a reference benchmark for other satellite sensors, due to their long and consistent time series, pre-launch testing, continuous comparison with ground reference sites and onboard calibration (US National Geospatial Advisory Committee, 2020^[14]).

4

Managing vulnerabilities, dependencies and sustainability risks of the space economy

This chapter closes the report by examining how the growth and diffusion of space activities create systemic vulnerabilities that require policy attention. It looks more closely at risks related to orbital sustainability, market concentration, supply chains and critical materials, the resilience of satellite signals and the environmental externalities of intensified space activity.

Growing dependence on space systems creates new vulnerabilities

Space systems have become critical to modern economies, public services and key infrastructures. They support science, communications, positioning, navigation and timing, weather forecasting, disaster response, transport, agriculture, finance and defence. As reliance on these systems grows, disruptions to space infrastructure can have wider economic and societal impacts.

The vulnerabilities examined in this chapter are diverse and should not be treated as a single category of risk. They differ in their causes, time horizons, affected actors and policy responses. Five broad types can be distinguished:

- Safety and sustainability risks, including orbital congestion, collision risks, space debris and the long-term usability of Earth's orbits.
- Market risks, including concentration in launch capacity, constellation-scale infrastructure and key space-enabled services, which may create strategic dependencies for governments, firms and users.
- Industrial supply-chain risks, including dependence on specialised components, materials, qualified suppliers and critical inputs such as radioisotopes.
- Security and resilience risks, including cyber vulnerabilities, jamming and spoofing of satellite signals, interference with critical services and the need for continuity arrangements in downstream sectors.
- Externalities of space activity, including impacts on astronomy, dark and quiet skies, radio-frequency environments, hazardous substances and the upper atmosphere.

Several of these risks are becoming more prominent as space activity expands. The space environment is increasingly contested, with jamming and spoofing of satellite navigation signals raising concerns in aviation, maritime transport and telecommunications. Although this chapter does not examine security and defence issues in depth, malicious interference, dual-use capabilities and geopolitical tensions form part of the resilience context for civil and commercial space systems.

Other pressures relate to the wider effects of orbital activity. Large constellations can affect optical and radio astronomy through satellite trails and radiofrequency interference and contribute to the loss of dark and quiet skies needed for science and biodiversity. These issues are distinct from, but connected to, orbital congestion and space traffic management. Atmospheric impacts are also emerging, as satellite and rocket re-entries can inject metals and other materials into the upper atmosphere (OECD, 2023^[1]). While the scale and long-term effects remain uncertain, rising launch and re-entry rates point to the need for stronger evidence and responsible mitigation.

These vulnerabilities are interrelated but require different policy responses. Orbital debris calls for collective rules, operational co-ordination and responsible behaviour by operators. Market concentration requires monitoring of dependencies, procurement choices, interoperability and contestability. Supply-chain risks call for granular mapping, diversification and access to specialised inputs. Signal-resilience risks involve cybersecurity, redundancy and preparedness among users. Distinguishing between these vulnerabilities helps clarify where national policy action is needed, where international co-operation is indispensable and where better evidence is still required.

Each of these vulnerabilities is addressed in turn in this chapter.

Orbital congestion is intensifying in attractive low-earth orbits

The first set of vulnerabilities relates to the safety and long-term viability of orbital environments, an issue covered in depth in OECD work on the sustainability of Earth's orbits (OECD, 2022^[2]; 2024^[3]). Earth's

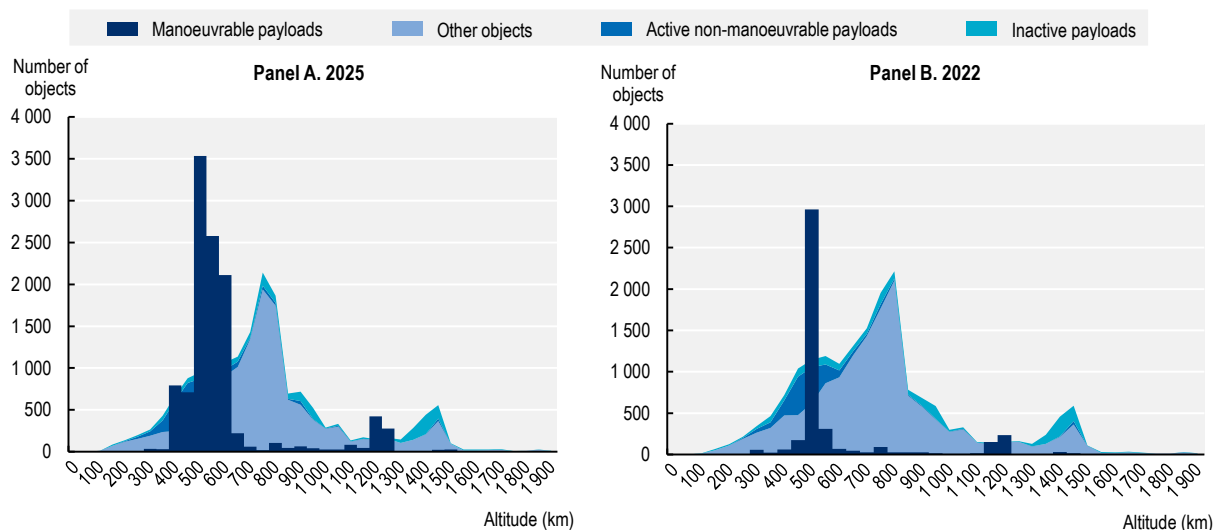
lower orbits are becoming increasingly crowded, but congestion is not evenly distributed. Active satellites are concentrated in a limited number of orbital bands, especially where latency, coverage, mission requirements and atmospheric drag conditions are favourable. As discussed in Chapter 3, low-earth orbit constellations and non-geostationary filings are major drivers of current and prospective demand for orbital capacity. The concentration is particularly visible between 400 and 600 km altitude, where communications satellites benefit from low latency and where the existing debris population is comparatively lower than in some higher orbital bands (Figure 4.1).

The challenge is therefore not only the total number of satellites, but their clustering at specific altitudes and orbital shells. Communications satellites, earth observation systems and other constellation-based services often seek similar operational conditions. This can increase collision-avoidance requirements, spectrum co-ordination needs and operational complexity, particularly as multiple large constellations scale up simultaneously (OECD, 2022^[2]; 2024^[3]).

This concentration of active satellites also interacts with the debris environment. Space debris refers to non-functional, human-made objects in orbit or re-entering the atmosphere, including fragments.¹ It has accumulated since the first orbital launch in 1957 and results from routine operations, accidents, explosions and collisions. The persistence of debris varies significantly by altitude. As shown in Figure 4.1, the population of debris objects above 700 km changed little between 2023 and 2025. At these higher altitudes, natural orbital decay – whereby objects gradually lose altitude and eventually burn up in the atmosphere – can take centuries or even thousands of years. At lower altitudes, decay is faster and may take decades, months or even days, depending on orbital conditions.

Figure 4.1. Satellites are concentrated in a small number of orbits

Distribution of satellites and debris objects across low-earth orbits, 2022 and 2025



Notes: Payloads in this figure refer to space objects (e.g. satellites, space probes) designed to perform a specific function in space, excluding launch functionality. Manoeuvrable payloads typically have an orbit control system (i.e. propulsion system).

Source: ESA (2026^[4]) and (2023^[5]), *Annual Space Environment Report*

https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf.

Manoeuvrable satellites can reduce some operational risks by conducting collision-avoidance manoeuvres and clearing their orbit at end of life. However, this depends on satellite design, operator practices, access to timely tracking information and effective co-ordination. Even in lower orbits where natural decay is faster,

rising satellite density can increase collision-avoidance burdens and create new operational and governance challenges.

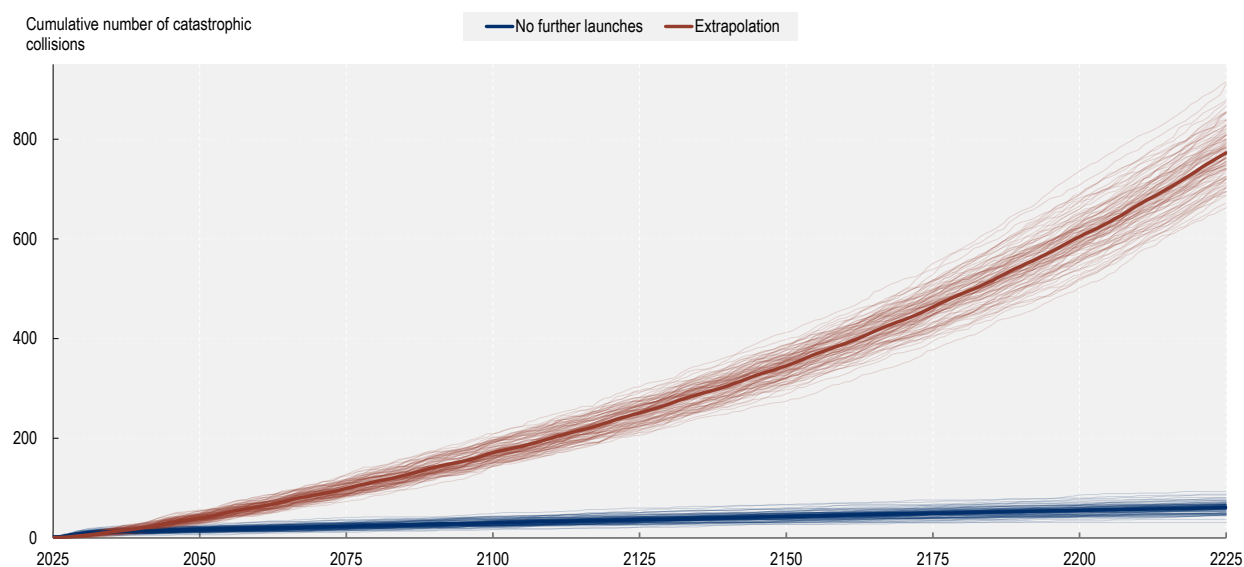
The space debris population is becoming self-generating

The debris population is large and only partly observable. Around 45 860 objects are regularly tracked, while models estimate about 54 000 objects larger than 10 cm, 1.2 million objects between 1 and 10 cm, and 140 million objects between 1 mm and 1 cm (ESA, 2026^[4]). Even small debris objects below 1 cm can damage spacecraft at orbital velocities, and most objects are too small to track operationally.

According to simulations by the European Space Agency, the debris population may become self-generating in some orbital regions: collisions among existing objects can create further debris even without additional launches (ESA, 2026^[4]). This is the so-called Kessler syndrome, first described in the late 1970s. In the worst case, debris accumulation in certain orbital regions could reduce their usability or increase operating costs substantially (Figure 4.2).

Figure 4.2. Long-term simulations point to self-generating space debris dynamics

Number of cumulative collisions in the low-earth orbit in the simulated scenarios of long-term evolution of the space environment



Source: ESA (2026^[4]), *Annual Space Environment Report 2026*,
https://www.sdo.esoc.esa.int/environment_report/Space_Environment_Report_latest.pdf.

Space debris already affects operations. The International Space Station has conducted more than 40 debris avoidance manoeuvres with crew on board since 1999. Large constellation operators also report rapidly growing numbers of collision-avoidance manoeuvres. SpaceX reported that its Starlink satellites performed more than 300 000 collision-avoidance manoeuvres in 2025, up from around 200 000 in 2024, a rise of roughly 50% in one year. Comparable public reporting across operators remains limited, making it difficult to assess system-wide operational burdens and trends.

Debris mitigation therefore requires more than post-mission disposal rules. It depends on satellite reliability, end-of-life disposal, passivation of rocket bodies, collision-avoidance co-ordination, data sharing,

active debris removal, and internationally aligned norms for responsible behaviour. The growing diversity of operators makes transparency and common reporting practices increasingly important.

Market concentration can create strategic dependencies

Market concentration can become a vulnerability when governments, firms or users depend on a small number of providers for services that cannot be readily substituted. Access to space provides one example. As shown in Chapter 2, participation in space activities has broadened significantly, with 109 countries having placed at least one satellite in orbit by end-2025, supported by CubeSats, commercial rideshare services and more accessible satellite technologies. Demand for launch services is also expected to remain strong over the coming decade, driven by both institutional and commercial users.

However, the expansion of space participation has not been matched by an equivalent diversification of launch supply. Global launch activity increased sharply between 2020 and 2025, although estimates vary depending on the treatment of failed attempts and test flights. Based on one commonly used metric, orbital launch activity rose from 114 attempts in 2020 to around 325 launches in 2025. The number of active launch providers also increased to more than 20 organisations (Table 4.1).

Yet effective launch supply, particularly among high-cadence providers, remained highly concentrated. In 2025, SpaceX, China Aerospace Science and Technology Corporation (CASC), Rocket Lab and Roscosmos together accounted for more than four-fifths of orbital launches, with SpaceX alone representing around half of the global total.²

Table 4.1. The number of space launch providers is growing

	2020	2025
Orbital launch attempts	114	325
Active launch providers	16	25
Providers with at least 3 launches	9	13
Share of launches by the 4 largest providers	~71%	~82%

Note: Active launch providers are organisations that conducted at least one orbital launch attempt during the year, including unsuccessful attempts. Providers are grouped at the launch-service or national launch-system level where appropriate; organisations developing launch vehicles but not conducting an orbital attempt during the year are excluded.

Source: OECD analysis.

The gap between emerging launch capabilities and launch services available at scale has implications for competition, resilience and assured access to space. Policy approaches should therefore consider not only the emergence of new launch systems or maiden flights, but also providers' ability to sustain reliable operations at scale, the availability of effective alternatives for satellite operators, and potential vulnerabilities associated with reliance on a limited number of high-cadence providers.

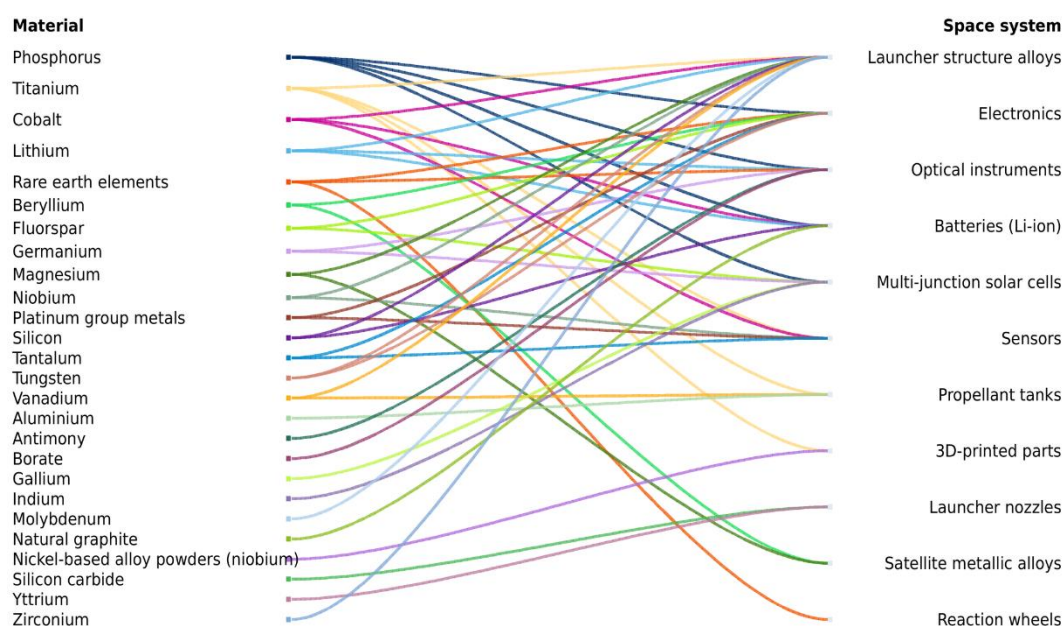
Space manufacturing is increasingly exposed to critical supply-chain risks

Space manufacturing supply chains have traditionally been characterised by low production volumes, high levels of specialisation and demanding qualification requirements. Components must often withstand launch vibration, extreme temperature changes, vacuum and radiation. The list of qualified suppliers can therefore be limited, and substitution can be slow because of safety, reliability and certification requirements (OECD, 2014^[6]; Altana, 2026^[7]).

Recent developments may increase exposure to supply-chain risks. Higher launch and production volumes increase demand for equipment and raw materials. At the same time, spacecraft and launch technologies are renewed more frequently and increasingly use technologies similar to those found in terrestrial high-tech markets. The growing use of commercial off-the-shelf components can reduce costs but may also expose the sector to competition for inputs from larger industries.

Critical and strategic materials are a particular concern. Space systems require specialised metals, minerals, gases, electronics, batteries, optical instruments, solar cells and structural alloys. For several materials used in space systems, production is highly concentrated, increasing exposure to disruptions, export restrictions or bottlenecks affecting specific grades, components or processing stages (Figure 4.3). Space systems rely on a broad range of critical minerals across launchers, spacecraft and ground-linked subsystems. Launcher and spacecraft structures, propulsion and thermal-control systems use materials such as titanium, aluminium, nickel, niobium, molybdenum, rhenium and platinum group metals, reflecting requirements for strength, heat resistance and performance under extreme operating conditions. Power subsystems depend on gallium, germanium, indium and silver for high-efficiency solar arrays, as well as lithium, nickel, cobalt and graphite for batteries. Electronics require gallium and germanium compounds, tantalum capacitors, tungsten, hafnium and noble gases. Sensors, optics and communications systems, including detectors, infrared optics, lasers and antennas, rely on germanium, indium, tellurium and rare earth elements. Overall, these materials are central to the performance, reliability and resilience of space systems, but they also expose the sector to supply-chain concentration, price volatility and strategic dependency risks.

Figure 4.3. European space systems' needs of strategic and critical materials



Source: Expanding on Maury-Micolier et al. (2022^[8]), "Criticality assessment of the EU space systems' supply chains", [https://indico.esa.int/event/416/contributions/7279/attachments/4854/7549/20221011_CSID_criticality%20assessment%20space%20supply%20chain_vf%20\(1\).pdf](https://indico.esa.int/event/416/contributions/7279/attachments/4854/7549/20221011_CSID_criticality%20assessment%20space%20supply%20chain_vf%20(1).pdf).

Determining dependencies in the space sector requires granular supply-chain mapping. A raw material may be refined and processed in multiple ways, and shortages may concern specific grades, by-products or components rather than the raw material alone (Carrara et al., 2023^[9]). The small size of the space sector can make stockpiling easier in some cases, but it can also reduce purchasing power and visibility compared with larger industrial sectors. Shortages can exist for very specific grades (e.g. nickel and

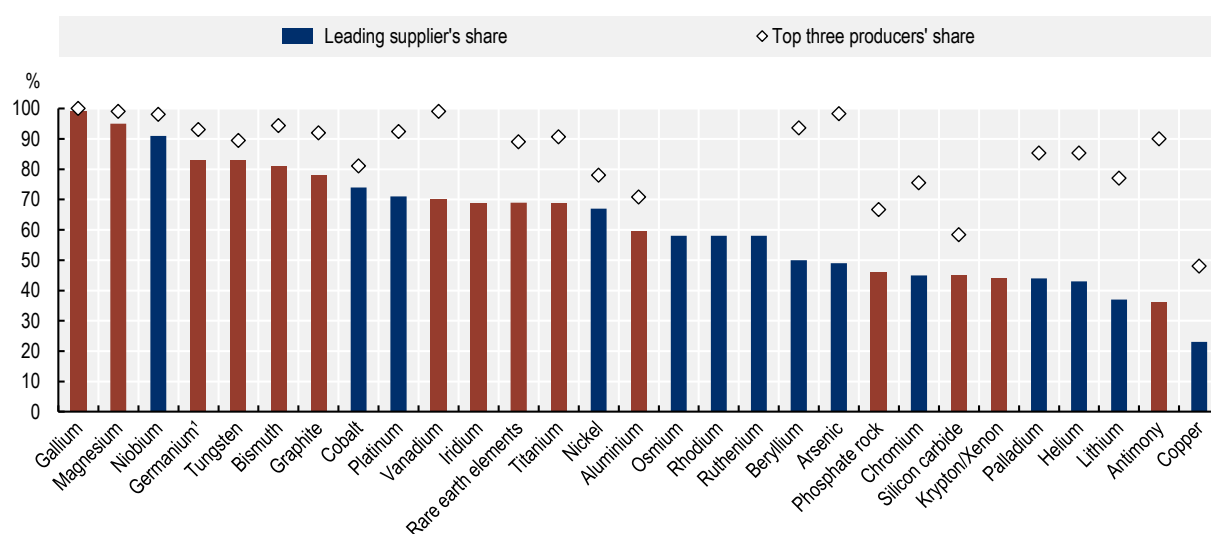
titanium) or for by-products of other production streams (germanium, gallium). The supply chain is complex and the substitution of products can be long and challenging due to safety and certification procedures.

Figure 4.3 shows a simplified overview of the critical and strategic materials needed in specific space systems – each column ranked from top to bottom in terms of the number of inputs. On the left side, phosphorus, titanium, cobalt, lithium and rare earth elements are used in the highest number of systems. On the right side, structural alloys, electronics and batteries, optical instruments and solar cells require the highest variety of critical material inputs. It is worth noting that this list does not account for input intensity or potential substitute materials.

For 14 out of the 29 products on the list of selected critical materials used in space manufacturing, metals, minerals and gases, one economy (mostly China) accounts for at least two-thirds of global production in 2025 (Figure 4.4). For 16 products, three producers accounted for 85% of production or more (USGS, 2026^[10]). At first glance, the 2035 outlook for mineral supply shows that supply gaps are narrowing, with copper as a major exception (IEA, 2025^[11]). However, in terms of diversification, the growth in refined critical minerals has been driven almost entirely by top producers, meaning that the geographic concentration of refined products has increased across nearly all critical minerals.

Figure 4.4. Leading producers of critical and strategic materials for space systems in 2025

Share of world production



Notes: For columns in red, China is the lead supplier. 1. Refers to refined production.

Source: Based on data from USGS (2026^[10]), *Mineral Commodity Summaries 2026*, <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026.pdf>.

Radioisotopes for space exploration are also in short supply. Radioisotope power systems can provide reliable heat or electricity for missions where solar power is insufficient or unavailable, but out of the more than 3 000 known radioisotopes, only a very limited number of radioisotopes are adapted for use in space, considering their half-life, radiation emission, power density, fuel form, cost and availability (National Research Council, 2009^[12]; Dustin and Borrelli, 2021^[13]). Only 22 radioisotopes have half-lives between 15 and 100 years, preferred for NASA missions (National Research Council, 2009^[12]). Plutonium-238 has been widely used by the United States, but supply declined after Cold War production facilities closed and has only recently been rebuilt. Americium-241 is being explored as an alternative by Europe and some commercial actors, while other candidates have limitations related to half-life, shielding, safety or availability (Oak Ridge National Laboratory, 2023^[14]).

In their comparative study of radioisotopes for space, Dustin and Borelli (2021^[13]) identified americium-241 as the best alternative to plutonium-238. India used americium-241 in its Chandrayaan-3 lunar mission. The European Space Agency is also opting for this radioisotope in the development of their own radioisotope thermoelectric generators. The US startup Zeno is using americium-241 to develop commercial radioisotope power systems for space. Compared with plutonium-238, americium-241 is cheaper and more readily available, as it is recovered from civil plutonium waste. It has a considerably longer half-life (432 years compared to 88 years) which could be a drawback or an advantage, depending on the mission duration. Finally, it has higher gamma radiation, requiring more shielding, but emits less neutrons. Strontium-90 and Curium-244 are other possible candidates in terms of technological readiness, availability, safety and thermal output, but their short half-life (28 and 18 years, respectively), limits their applicability (Dustin and Borelli, 2021^[13]).

Space infrastructure, signals and data are increasingly subject to malicious attacks

Space-based systems are designed to resist the multiple stresses of launch as well as the extreme temperature fluctuations and radiation of the space environment, and are, to a significantly lesser extent, shielded against minor collisions with debris. However, they are generally less protected against *malicious acts*. Civilian spacecraft follow predictable, publicly available, orbital paths and can be destroyed or blinded by physical anti-satellite weapons (Froehlich, 2021^[15]). Several economies have demonstrated anti-satellite capabilities in recent years, including China, India, the United States and Russia.

Furthermore, electronic attacks such as jamming and spoofing can interfere with the signals to and from a satellite, and in this way disrupt operations or send fake signals. Finally, ground systems, satellites or end-user equipment can all be the targets of cyberattacks.

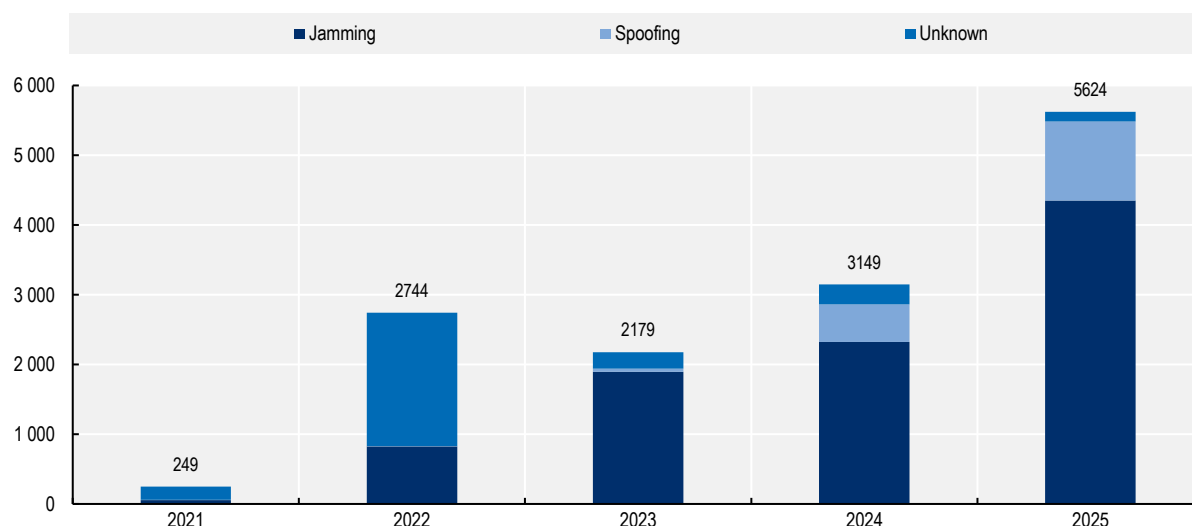
As the importance of space systems grow, so do the number of incidents and their potential impact, as illustrated by different examples.

- In Russia's war of aggression against Ukraine, a cyberattack targeting Viasat's KA-SAT fixed broadband network led to widespread network outages in Central and Eastern Europe on the day of the invasion, as the attack knocked out thousands of modems communicating with the geostationary satellite. Government users were not affected (Viasat, 2022^[16]).
- The European Organisation for the Safety of Air Navigation (EUROCONTROL) reports that GNSS radio-frequency interference reports affecting European air transport have increased from 249 in 2021 to 5 624 in 2025, as shown in Figure 4.5.
- Recent evidence suggests that GNSS interference is also conducted from satellites, not only from terrestrial sources (Clements, 2026^[17]),

The integrity of the satellite imagery supply chain is also vulnerable to malicious attacks. Several earth observation satellites critical for both national and global disaster management send unauthenticated or decryptable signals, making images susceptible to spoofing attacks for "malicious misdirection" (Salkield et al., 2023^[18]). Data can be intentionally intercepted or tampered with when communicating with ground stations at various geographic locations, subject to growing scrutiny and importance. As part of the growing trend of digital disinformation, the trust in satellite data can be eroded by fake, misinterpreted or intentionally misrepresented imagery. For example, in 2025 there were several examples of deepfake satellite imagery used to exaggerate the effect of military strikes in ongoing conflicts.

Figure 4.5. European air traffic faces growing threat of jammed and spoofed GNSS signals

Number of EUROCONTROL Voluntary ATM Information Reports on GNSS radio-interference cases



Source: EUROCONTROL (2026^[19]), "Performance review report 2025: an assessment of air traffic management in Europe", <https://www.eurocontrol.int/sites/default/files/2026-03/eurocontrol-performance-review-report-2025.pdf>.

A final example concerns terrestrial infrastructures vulnerabilities. In 2022, one of the two undersea fibre-optic cables linking the SvalSat ground station in Svalbard to mainland Norway was damaged, leaving one of the world's largest commercial satellite ground stations without full redundancy for 11 days. SvalSat hosts more than 150 antennas used by public and private space organisations and supports major Earth observation and weather missions, including Landsat and Copernicus. The incident was not attributed to a malicious act; the cause remains undetermined, and damage to undersea cables can also result from natural causes or accidental interference, including from fishing activities.

Conclusion and policy implications

The expansion of the space economy creates important opportunities for innovation, connectivity, scientific discovery and public services, but it also makes resilience, sustainability and responsible use of orbital resources central policy concerns. The vulnerabilities discussed in this chapter are mutually reinforcing. Expanding constellations increase demand for orbital capacity and traffic co-ordination; concentration in launch and constellation services can create strategic dependencies; supply-chain risks can affect the development and maintenance of critical capabilities; and environmental, security and signal-resilience risks can undermine trust in space-enabled services.

Because these vulnerabilities differ in nature, policy responses also need to be differentiated. Orbital congestion and debris require collective rules, operational co-ordination, data sharing and responsible behaviour by operators. Market concentration calls for monitoring of dependencies, procurement choices that preserve contestability and resilience, and interoperability or open standards where appropriate. Supply-chain risks require granular mapping, diversification of suppliers and secure access to specialised inputs, including critical materials and radioisotopes. Signal-resilience and environmental risks require stronger evidence, preparedness, cybersecurity, mitigation practices and continuity arrangements for critical services.

Resilience also needs to be embedded beyond space policy itself, in the many downstream sectors that depend on space-based infrastructure. This includes contingency planning for GNSS and communications

disruptions, cybersecurity and signal-resilience measures, alternative capabilities where necessary, and stronger user awareness of both the benefits and limits of space-enabled services. Countries will continue to compete in space, but they also share a common interest in preserving safe, sustainable and reliable access to space for economic, scientific and societal benefit.

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Notes

¹ The Inter-Agency Space Debris Coordination Committee defines space debris as "all man-made objects including fragments and elements thereof, in Earth orbit or re-entering the atmosphere, that are non-functional" (IADC, 2007^[20]).

² Active launch providers are defined here as organisations that conducted at least one orbital launch attempt during the year, including unsuccessful attempts. The classification includes commercial firms, government agencies, and state-owned enterprises. Under this definition, the 25 orbital launch providers identified in 2025 were: Arianespace, Blue Origin, China Academy of Science (CAS) Space Technology Co., Ltd., China Aerospace Science and Technology Corporation (CASC), China Rocket, ExPace, Firefly Aerospace, Galactic Energy, Gilmour Space Technologies, INNOSPACE, the Iranian Aerospace Force, iSpace (Interstellar Glory), Isar Aerospace, Israel Aerospace Industries, Indian Space Research Organisation (ISRO), Korea Aerospace Research Institute (KARI), LandSpace, Mitsubishi Heavy Industries, Northrop Grumman, Orienspace, Rocket Lab, Roscosmos, Space Pioneer, SpaceX, and United Launch Alliance (ULA).

5

The Space Economy at a Glance: Country Profiles

Using a common framework to present information, country profiles provide facts and indicators for members of the OECD Space Forum. Each profile provides information on the state of the country's space sector; space-related government budgets, recent policy developments, as well as key commercial activities. These findings are supported by a selection of internationally comparable indicators, subject to data availability: "fast facts" indicators, space budget trends and main programmes, top applicants of space-related patents, production and excellence in space-related scientific journal categories and space-related official development assistance commitments

Canada

Canada is among the world's longest-standing space-faring nations. Its first satellite, Alouette-1, launched in 1962, made Canada the third country to design and build a successful satellite, and it operated far beyond its planned lifetime, supporting ionospheric research for a decade. Since then, Canada has developed a strong record in human spaceflight, with ten Canadians having participated in space missions and having the first non-American take part in a lunar flyby during the Artemis II mission (April 2026).

Canada at a glance

Indicator	Canada	OECD average
First satellite in orbit	Alouette 1(1962)	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	60 (73%)	
Number of satellites launched since 2019 (June 2026)	69	
Number of spaceports	1 (under development)	
Space-related workforce	14 622 full-time equivalents (2024)	
Space-related commercial revenues	CAD 5 billion (2024)	
Institutional civil space budget as a share of gross domestic product	0.043%	0.065%
Institutional civil space budget per capita	23.3	33.6
Space literature per 100 000 inhabitants	1.9	2.1

Canada remains closely integrated into international exploration activities. It also maintains a long-standing and unique cooperation agreement with the European Space Agency, enabling Canadian firms and researchers to participate in selected ESA programmes and tenders. Canadian technological strengths include robotics, earth observation, satellite communications and downstream services, with flagship capabilities in synthetic-aperture radar through the RADARSAT programme and commercial satellite communications. These capabilities also contribute to broader national priorities such as Arctic monitoring, emergency management and remote connectivity.

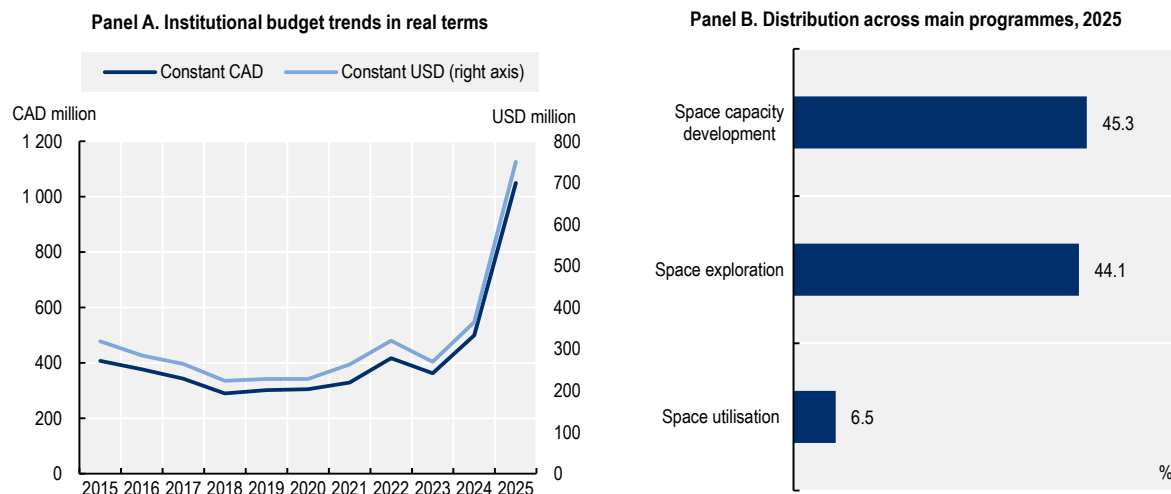
As of June 2026, Canada ranked 11th globally in satellite ownership and 10th for satellites launched since 2019, largely reflecting strong commercial activity in the sector.

In 2025, Canada's civilian institutional space budget amounted to USD 973.8 million (CAD 1 361 million), representing 0.043% of Canada's gross domestic product. The amount may not capture all space-related funding and represents already a steep increase compared with the years 2015-24, mainly associated with increased investment in European Space Agency programmes and space exploration activities (Figure 5.1). In 2026, Canada introduced the *Canadian Space Launch Act*, establishing a regulatory framework for space launch and re-entry activities. Several Canadian space companies are developing domestic launch capabilities, and commercial spaceports in Newfoundland and Nova Scotia are currently under development.

The Government of Canada is supporting the growth of this sovereign dual-use space capability through a range of funding programs and initiatives. Canada's civil space budget is also benefiting from growing interest in dual-use capabilities within the defence sector. In 2026, the government released Canada's Defense Industrial Strategy, which identifies key sovereign capabilities in the space domain. This includes Space-Based Intelligence; Surveillance and Reconnaissance; Space Domain Awareness; Satellite Communications; and Space Launch. Strategic investments in these areas have already begun, primarily in the launch sector, where the government announced a 10-year CAD 200 million agreement to lease a dedicated space launch pad as well as a CAD 105 million contest to launch Canadian payloads from Canadian soil.

Evolution of Canada's civil government space budget

In constant CAD and USD million, base year 2015



Source: OECD analysis based on institutional sources.

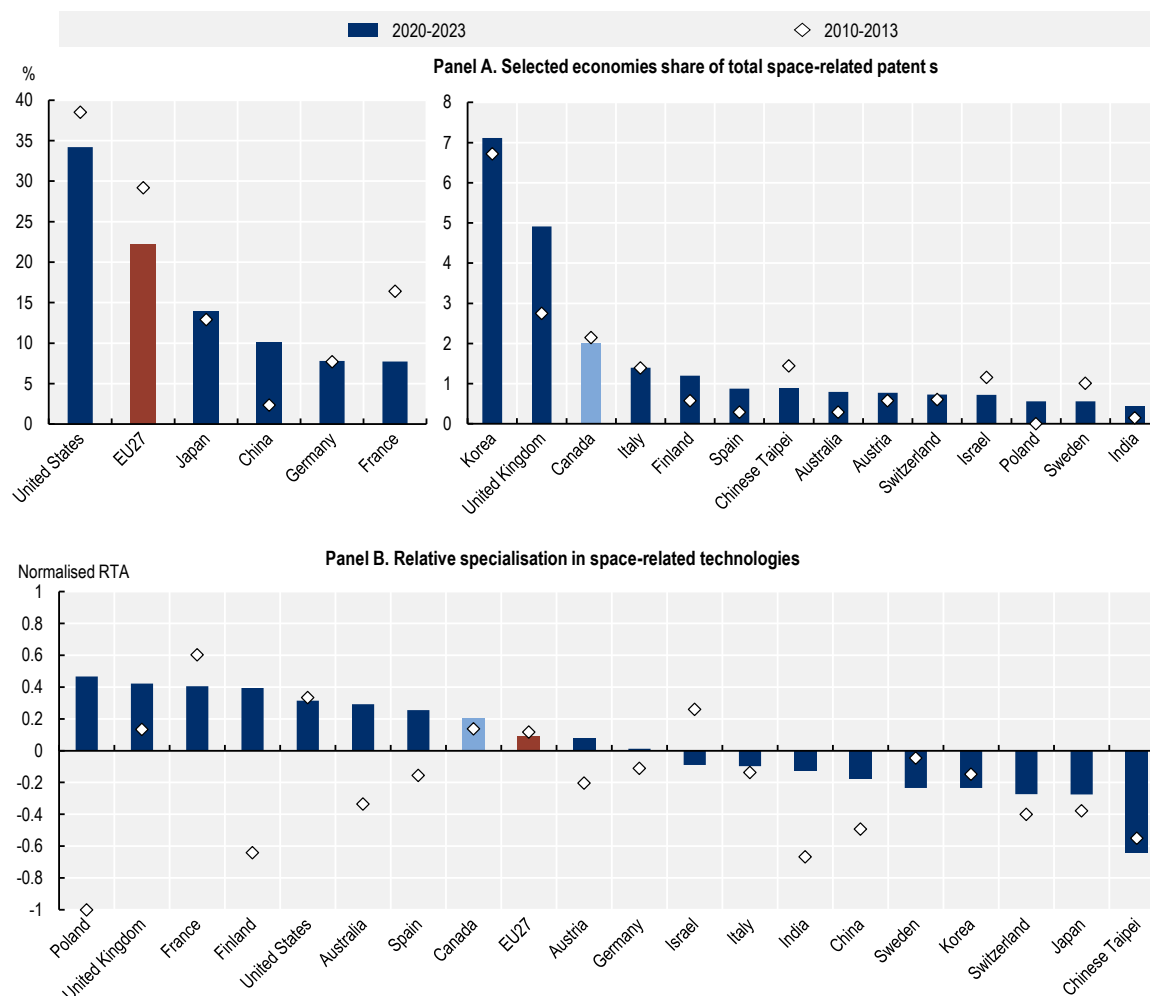
Note: In Panel B, the values sum to 100%, with the remaining budget (4.1%) allocated towards corporate services.

Canada accounted for about 2% of global space-related patent applications in the 2020-23 period (Figure 5.2), which is about the same as in 2010-13. The great majority of applications were filed by private firms. Canada has a positive revealed technological advantage (RTA), meaning that its share in space-related patenting is higher than its share in overall patenting activities. This relative specialisation has increased since 2010-13.

The Canadian space industry covers the entire value chain, from manufacturing to the delivery of satellite services. The sector generated USD 3.6 billion (CAD 5 billion) in revenues in 2024, down from previous years due to the continued decline in satellite broadcasting (which is a general global phenomenon not reserved to Canada). However, non-broadcasting revenues reached a record CAD 3.6 billion in 2024, with space systems manufacturing accounting for a growing share of total revenues. There are several other signs of growth in the Canadian space economy. In the same year, the sector contributed CAD 3.8 billion to Canada's GDP, a 15.4% increase in real terms compared to 2019. The space sector also employed over 14 600 direct full-time-equivalents in 2024, a 24% increase since 2019.

Canada: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



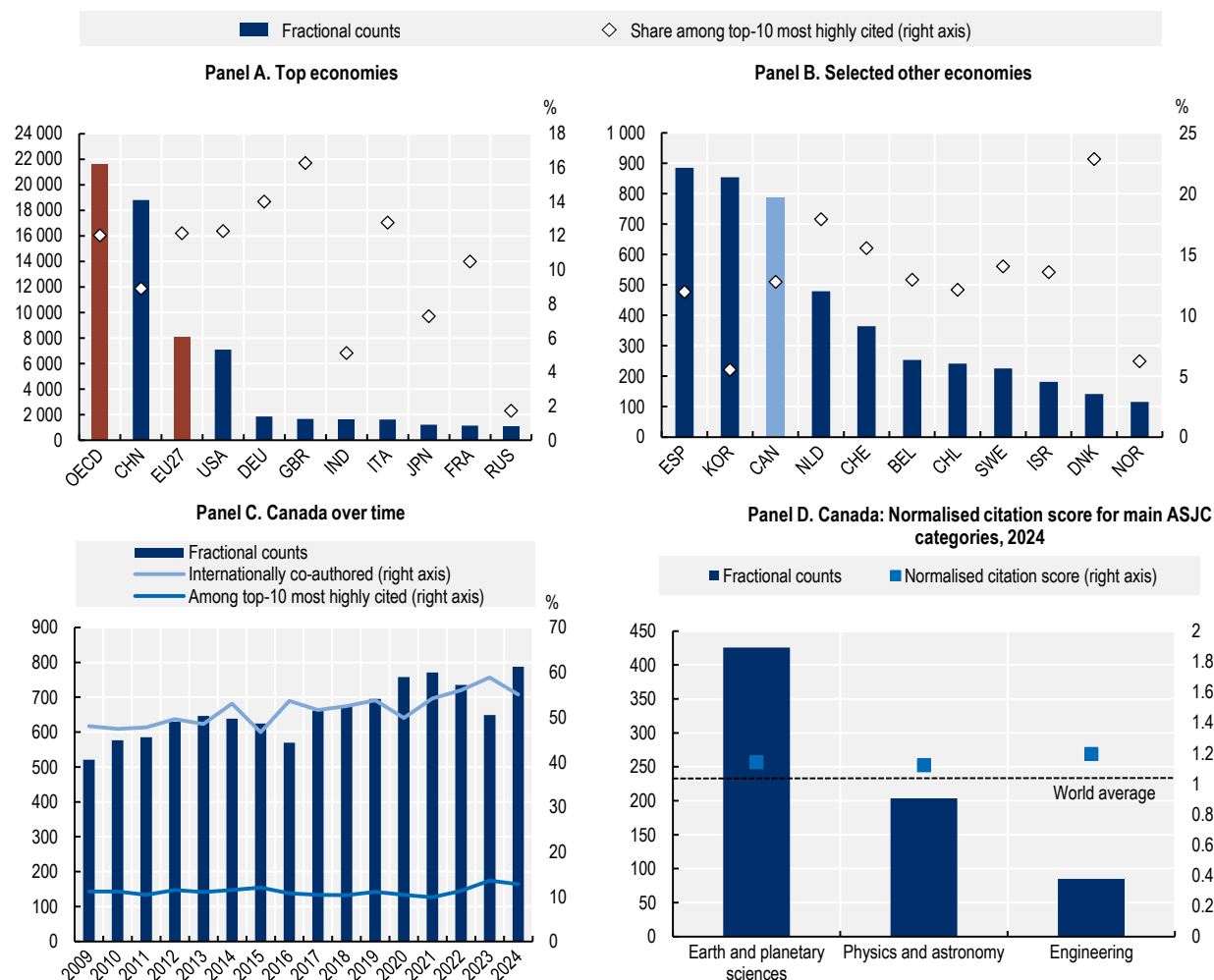
Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD (2026_[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

In terms of space-related scientific production and excellence, Canada's space literature (measured in fractional counts) has increased since 2009, as has its share of internationally co-authored papers (Figure 5.3). When looking at citations, a common proxy for scientific excellence, 13.1% of Canada's space literature was in the top-10 percentile for citations in 2024. This is above the OECD average of 12%. Canada's normalised citation score, comparing citation patterns within specific scientific fields, is above the world average in the All-Science Journal Classification (ASJC) categories "earth and planetary sciences", "physics and astronomy", and "engineering".

Canada: Space-related scientific output and excellence indicators in 2024

Fractional counts



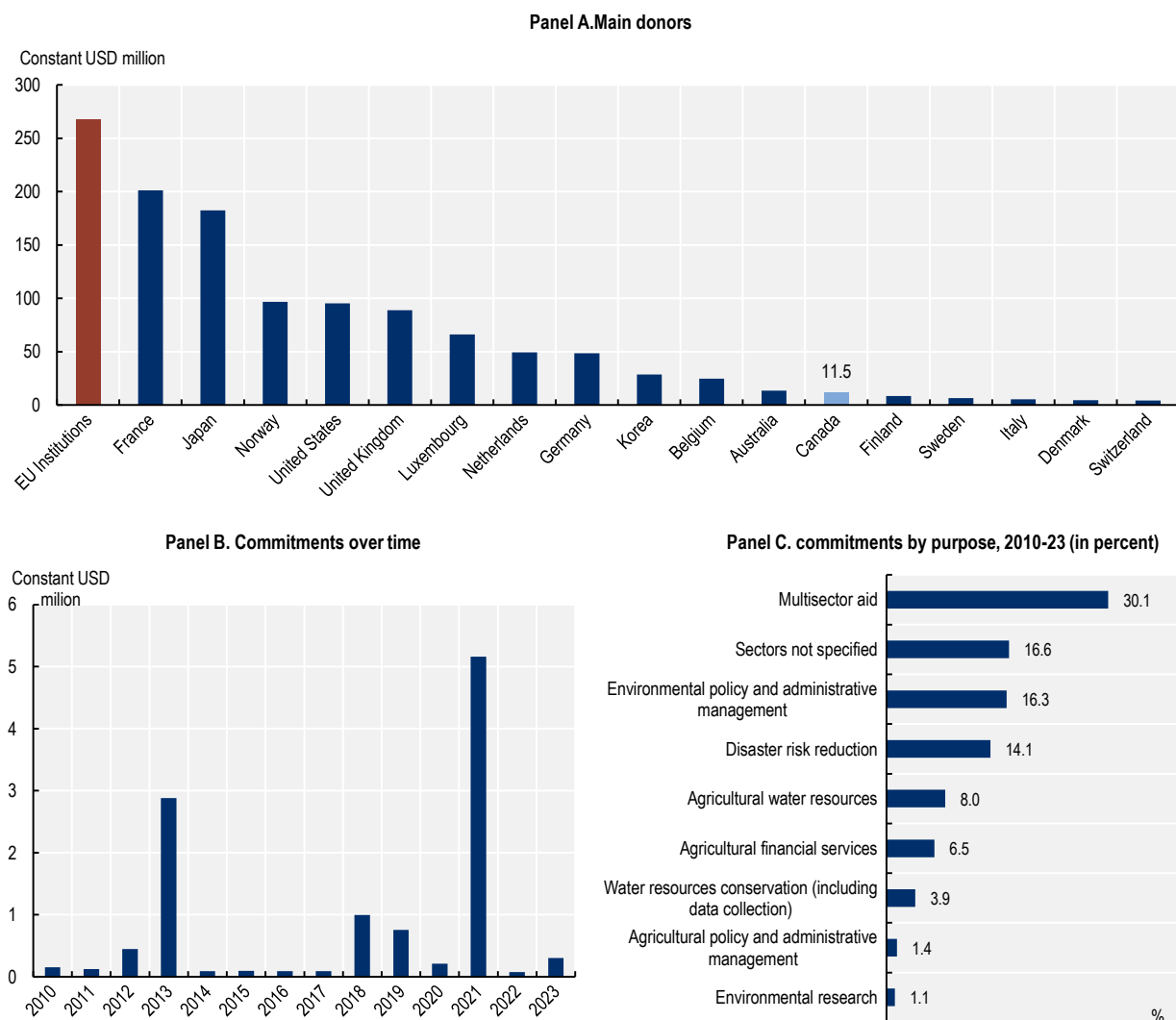
Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

Based on data in the OECD Development Assistance Committee Creditor Reporting System database, Canada committed some 11.5 million constant US dollars in space-related official development assistance over the 2010-23 period, (Figure 5.4). Commitments applying satellite technologies and data targeted the following sectors: “Other multi-sector” (zoonotic disease outbreak prevention, disaster risk reduction), general environmental protection and agriculture.

Canada: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025_[2]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

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- OECD (2025), *Creditor Reporting System (CRS)*, database, <https://data-explorer.oecd.org>. [2]

France

France is one of the world's leading space-faring nations and a central actor in Europe's space ecosystem. In 1965, it became the third country, after the Soviet Union and the United States, to place a satellite in orbit with its own launcher, using the Diamant rocket to launch Astérix. It hosts Europe's spaceport in Kourou, French Guiana, whose near-equatorial location supports efficient access to a wide range of orbits and underpins European launch autonomy.

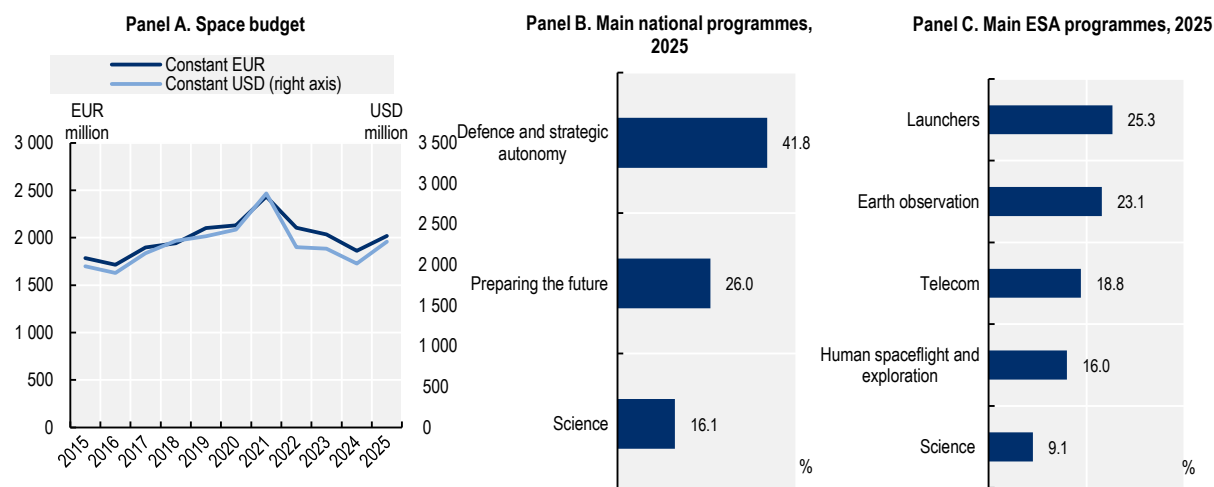
France at a glance

Indicator	France	OECD average
First satellite in orbit	Astérix (1965)	
First successful orbital launch	1965 (Diamant A)	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	104 (75%)	
Number of satellites launched since 2019 (June 2026)	98 (7th)	
Number of spaceports	1 (Kourou Space Centre in French Guiana)	
Institutional civil space budget as a share of gross domestic product	0.082%	0.065%
Institutional civil space budget per capita	40.2	35.4
Space literature fractional counts per 100 000 inhabitants	1.7	2.1

France is also among the European Space Agency's largest contributors and has strong capabilities across launch systems, satellite manufacturing, earth observation, telecommunications, space science and defence, with an active ecosystem of space startups. In June 2026, France ranked sixth globally in satellite ownership, with around 75% of active satellites commercially operated, and seventh for satellites launched since 2019. While recent orbital activity is mainly commercially led, France maintains a broad portfolio of government satellites for civil, security and military applications.

Evolution of France's civil government space budget

In constant EUR and USD million, base year 2015



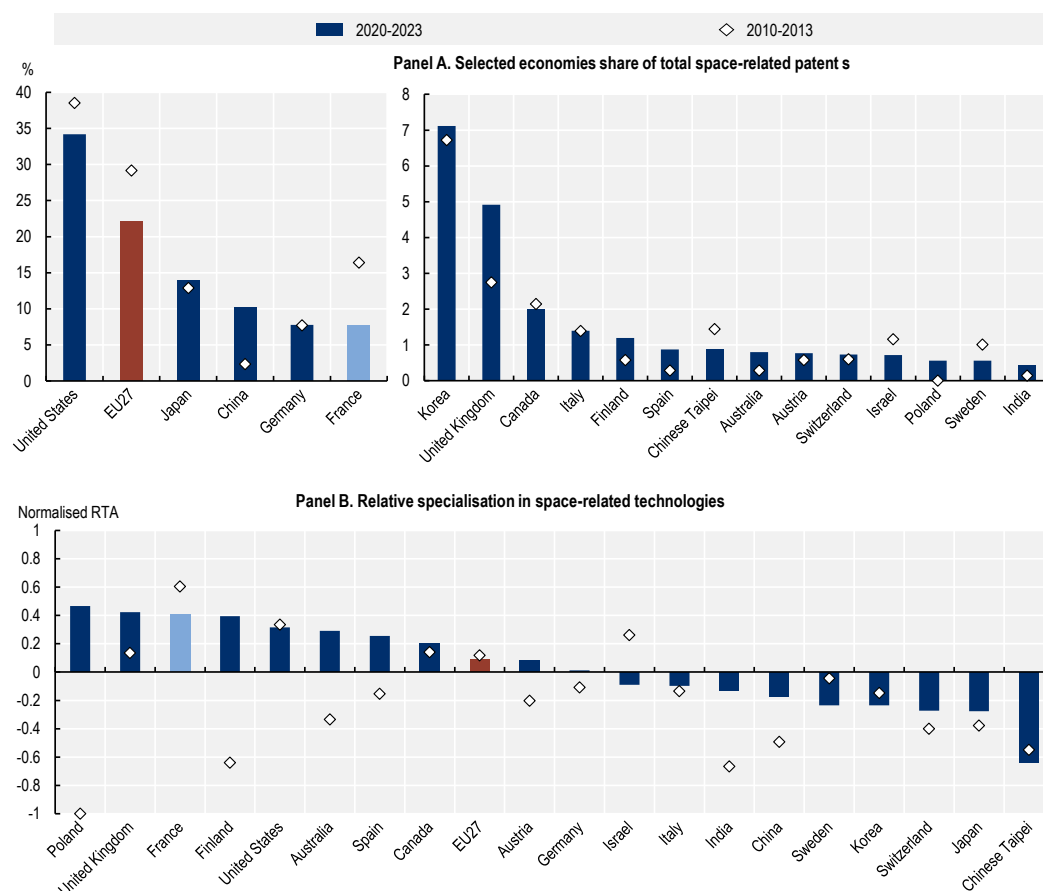
Note: Budget estimates include national activities and contributions to the European Space Agency, Eumetsat and as well as estimated contributions to the European Union Space Programme.

Source: OECD analysis based on institutional sources.

In 2025, the French civilian government space budget amounted to USD 2 760.8 million (EUR 2 443.2 million), including national and multilateral activities, contributions to ESA and Eumetsat, as well as estimated contributions to the European Union (EU) Space Programme, which is mainly based on each country's share of the EU's aggregated gross national income (Figure 5.5). This is a slight increase compared with 2024 and pre-COVID-19 levels in real terms. Earth observation and launchers were the main budget posts in the ESA contributions, whereas the national programme focused on maintaining the strategic autonomy of France and Europe, and R&D promotion. Unveiled on 12 November 2025, the new National Space Strategy reaffirmed France's key priorities: sovereignty, autonomous access to space, industrial competitiveness, defence, the environmental transition, and European cooperation. Earlier in June 2025, the decarbonisation roadmap for the French space sector set out formal objectives for reducing its carbon footprint.

France: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

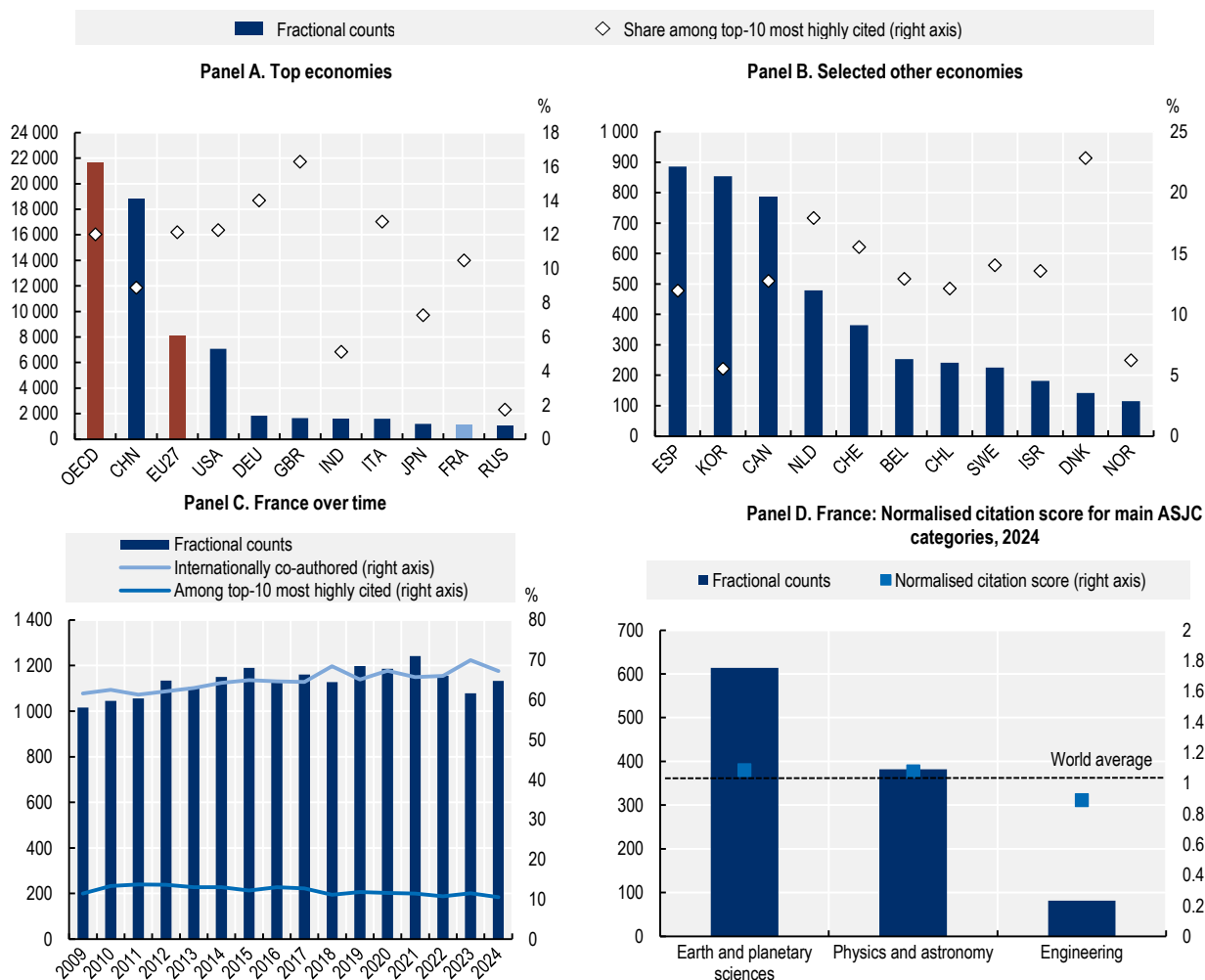
Source: OECD (2026^[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

France is currently reassessing its measurement of the space sector to better capture all relevant activities. The French space sector is both broad and high-tech, covering all industry segments from manufacturing to satellite data/signal exploitation. The country also includes several assemblers at the top of the space manufacturing value chain. The biggest industrial cluster can be found in the southwestern part of the country, with strong links to aeronautics. There is a growing number of space startups in France, assisted by two ESA business incubation centres and the Connect by CNES support programme. From 2026 onwards, the Guiana Space Centre will have a dedicated complex for private micro and mini launchers.

France accounted for 7.7% of global space-related patent applications in 2020-23, compared with 16.4% in 2010-13, as shown in Figure 5.6. Key areas include propulsion and space system control (14% and 10%, respectively, of global patent applications). France has a strong revealed technological advantage (RTA), meaning that its share in space-related patenting is equal to its share in overall patenting activities. France's relative specialisation in space technology patenting has decreased since 2010-13.

France: Space-related scientific output and excellence indicators in 2024

Fractional counts



Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

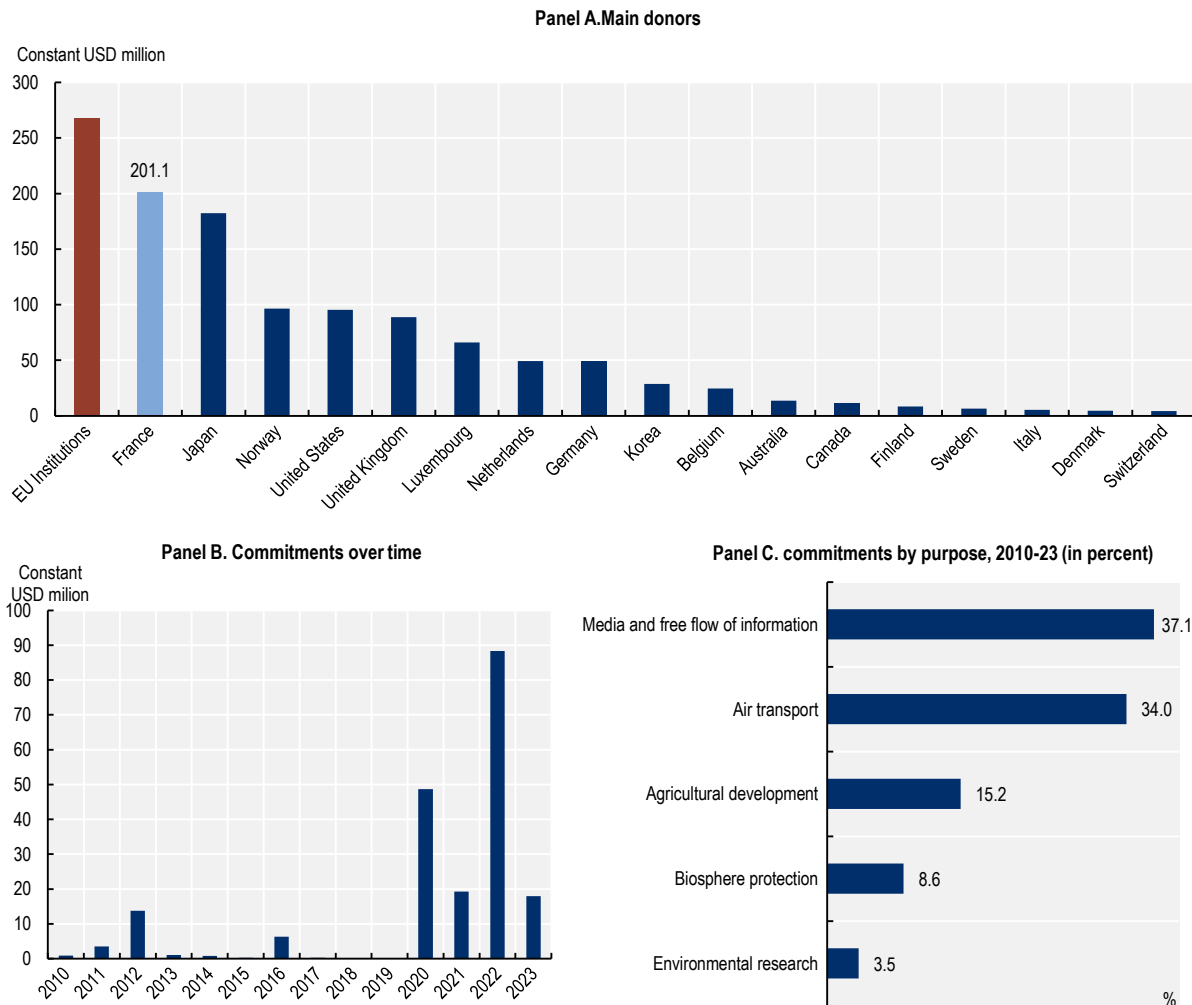
Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

In terms of space-related scientific production [hereafter ‘space literature’], France’s production in fractional counts has remained stable over time but is below the OECD average per 100 000 inhabitants (Figure 5.7). The share of papers that are internationally co-authored has increased since 2009 and is high, above 70%. When looking at citations, a common proxy for scientific excellence, 10.5% of France’s space literature was in the top-10 percentile for citations in 2024. This is below the OECD average of 12%. France’s normalised citation score, comparing citation patterns within specific scientific fields, was at or just below the world average in the All Science Journal Classification (ASJC) categories “earth and planetary sciences” and “physics and astronomy”.

Based on data in the OECD Development Assistance Committee Creditor Reporting System database, France is the world leader in space-related official development assistance, committing some 201 million constant US dollars over the 2010-23 period, (Figure 5.8). Commitments mainly targeted free satellite broadcasting in French-speaking countries (Canal France International) and air transportation (satellite data and signals for improved safety).

France: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[21]), “Creditor Reporting System (CRS)”, OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

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on June 2026).
- OECD (2025), *Creditor Reporting System (CRS)*, database, <https://data-explorer.oecd.rg>. [2]

Germany

Germany is one of Europe's leading space nations, a founding ESA member and its largest contributor in the 2025 ministerial subscriptions. It hosts key European space institutions, including ESA's European Astronaut Centre in Cologne and European Space Operations Centre in Darmstadt, as well as EUMETSAT's headquarters in Darmstadt and the European Southern Observatory's headquarters in Garching.

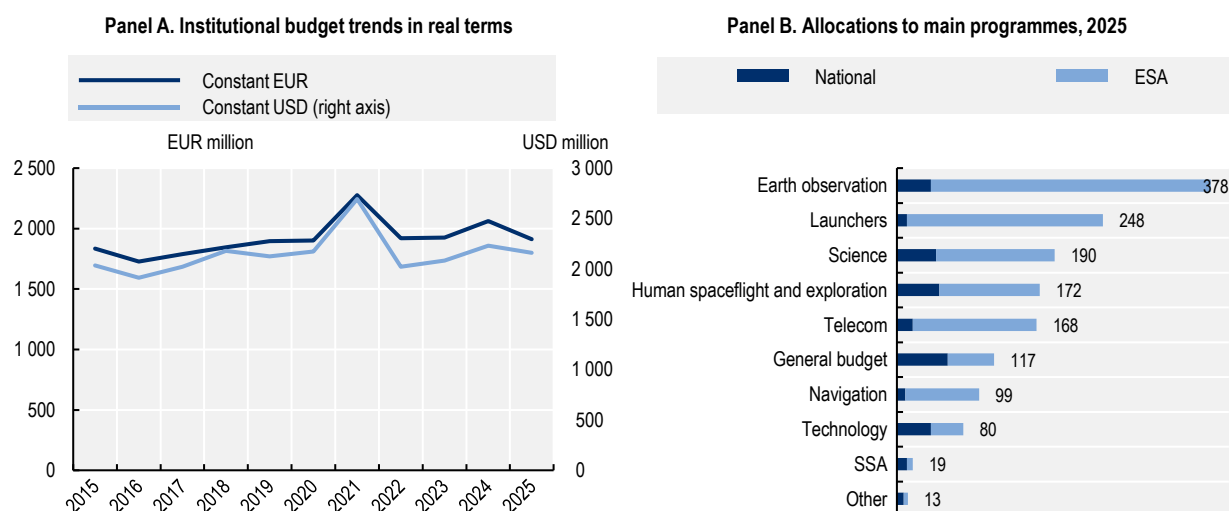
Germany at a glance

Indicator		OECD average
First satellite in orbit	Azur (1969)	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	65 (49%)	
Number of satellites launched since 2019 (June 2026)	83 (8th)	
Space-related workforce (2024)	30 940	
Space-related commercial revenues (2024)	EUR 11.69 billion	
Institutional civil space budget as a share of gross domestic product	0.055%	0.065%
Institutional civil space budget per capita	32.7	33.6
Space literature fractional counts per 100 000 inhabitants	2.2	2.1

Germany also has a dense industrial and innovation base, with strengths in satellite systems, operations, earth observation, exploration, launcher propulsion and downstream applications, supported by ESA Business Incubation Centres in Bavaria, Northern Germany, North Rhine-Westphalia and Hessen/Baden-Württemberg. In terms of orbital assets, Germany ranked eighth globally for active satellites in June 2026, 49% of which were commercially operated, and also eighth for satellites launched since 2019.

Evolution of Germany's civil government space budget

In constant EUR and USD million, base year 2015



Note: Budget estimates include national activities and contributions to the European Space Agency, Eumetsat and as well as estimated contributions to the European Union Space Programme.

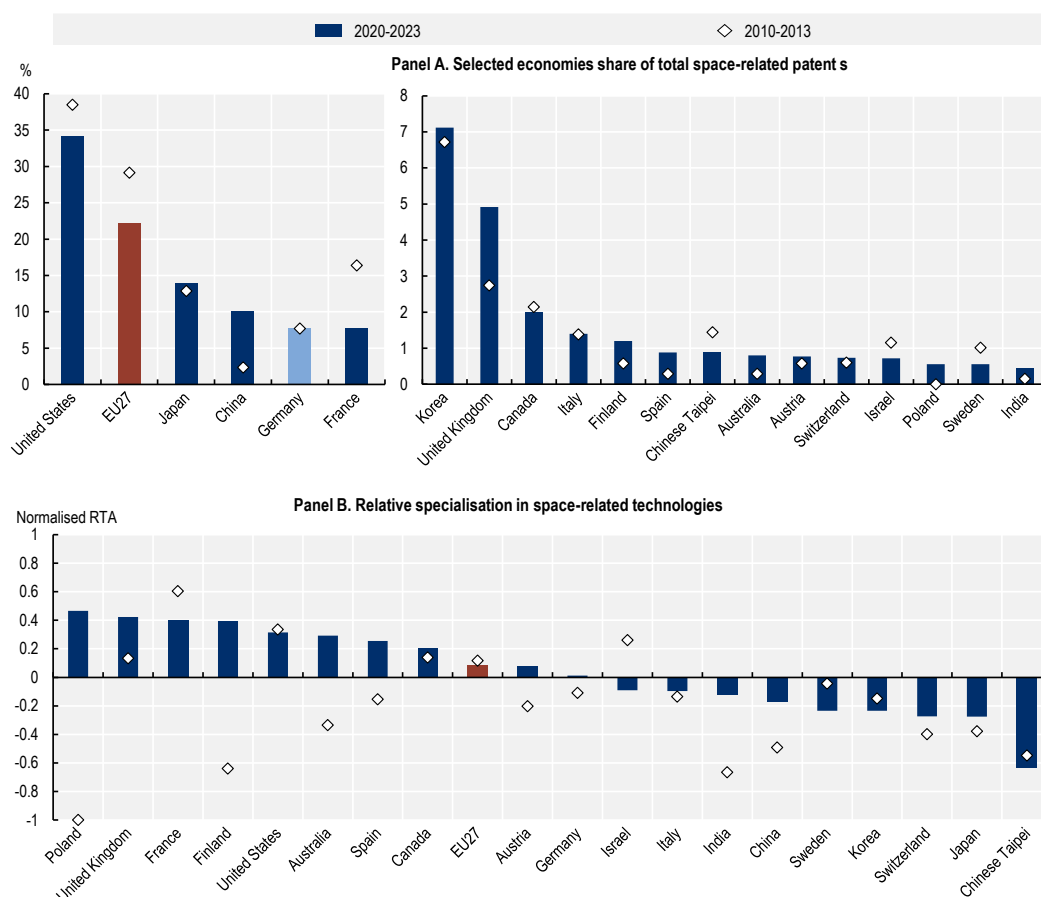
Source: OECD analysis based on institutional sources.

In 2025, the German civilian government space budget reached USD 2 776 million (EUR 2 456.8 million), including national activities, contributions to ESA and Eumetsat, as well as estimated contributions to the European Union (EU) Space Programme, which is mainly based on each country's share of the EU's aggregated gross national income (Figure 5.9). The greatest share of combined national and ESA funding was dedicated to earth observation, followed by launchers. Funding levels have remained stable since 2015. In 2025, Germany announced a major military space funding package of 35 EUR billion (USD 41 billion) up to 2030. This would involve support for new satellite constellations (e.g. early warning, secure communications), space system hardening against cyberattacks, improved redundancies and secure access to space.

Germany's space sector spans across the whole value chain from manufacturing to data/signal exploitation, including both incumbents and start-ups. The sector generated an estimated EUR 11.69 billion in revenues, employing a workforce of 30 940, according to the survey and modelling conducted by the German Space Agency.

Germany: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



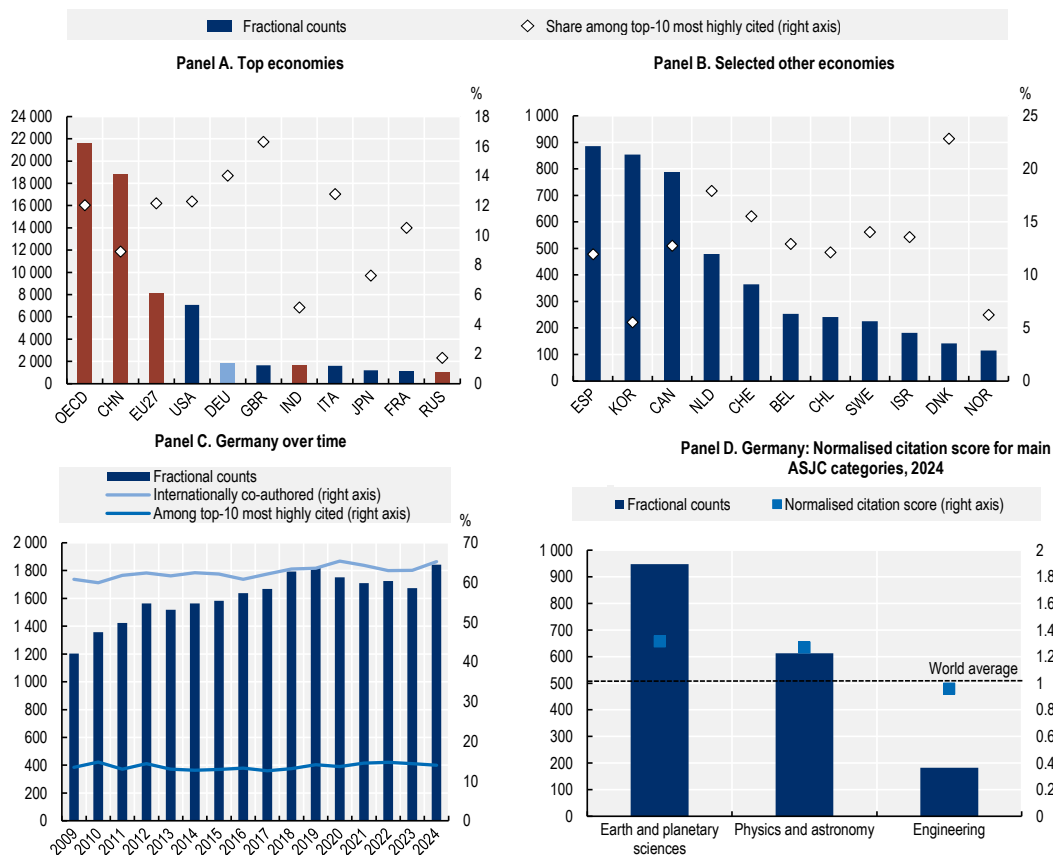
Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD (2026^[1]), STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

Germany accounted for 7.8% of global space-related patent applications in 2020-23, the same as in 2010-13., as shown in Figure 5.10. Germany has a neutral revealed technological advantage (RTA), meaning that its share in space-related patenting is equal to its share in overall patenting activities. Germany's relative specialisation in space technology patenting has increased since 2010-13.

Germany: Space-related scientific output and excellence indicators in 2024

Fractional counts



Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

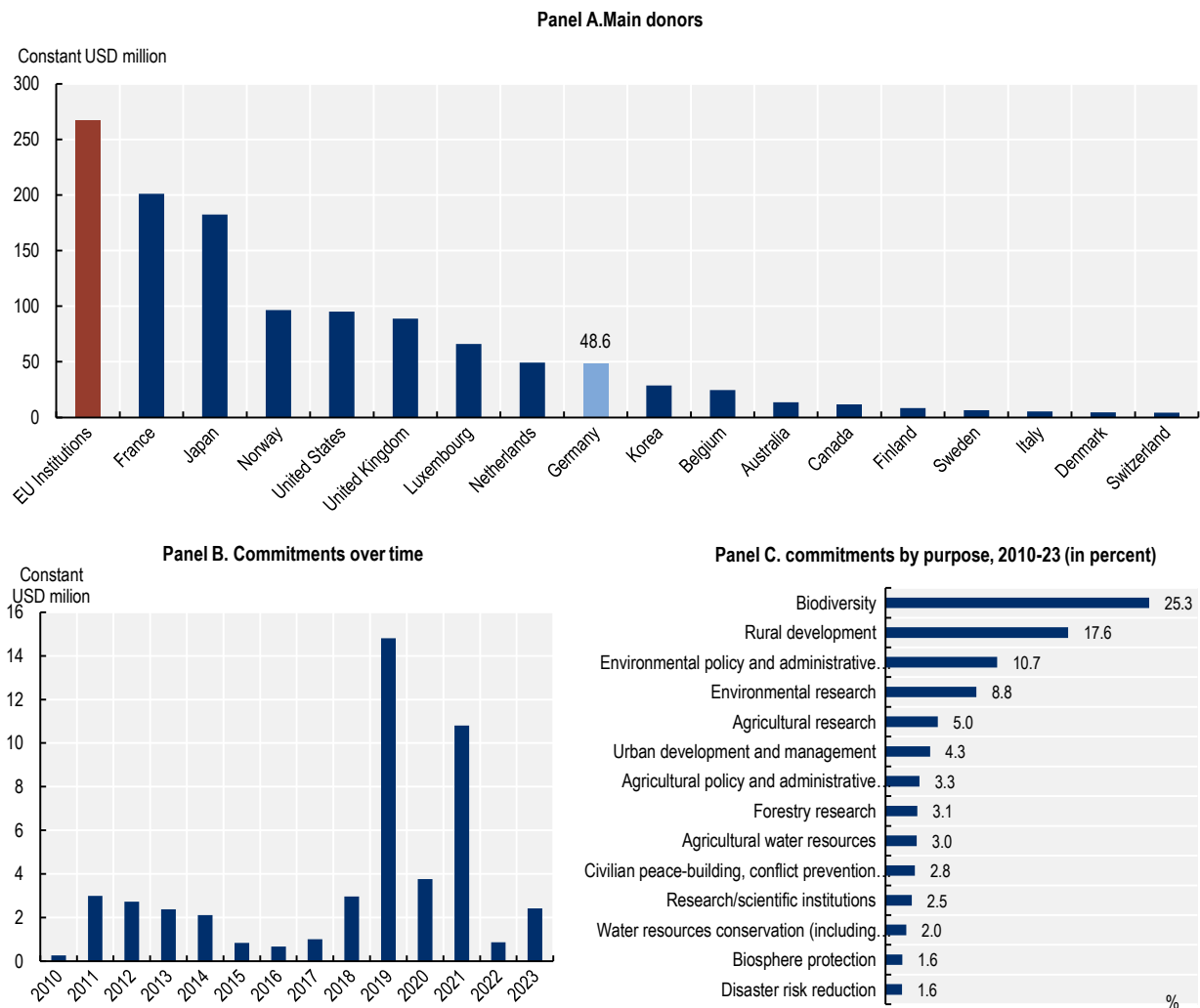
Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, <https://www.elsevier.com/products/scopus>, April 2026.

In terms of space-related scientific production [hereafter 'space literature'], Germany is among the top producers, following China and the United States (Figure 5.11). Production has increased over time, while the share of papers that are internationally co-authored has remained stable and high. When looking at citations, a common proxy for scientific excellence, 14% of Germany's space literature was in the top-10 percentile for citations in 2024. This is above the OECD average of 12%. Germany's normalised citation score, comparing citation patterns within specific scientific fields, is above the world average in the All Science Journal Classification (ASJC) categories "earth and planetary sciences" and "physics and astronomy".

Based on data in the OECD Development Assistance Committee Creditor Reporting System database, Germany committed some 48.6.7 million constant US dollars in space-related official development assistance over the 2010-23 period, (Figure 5.12). Commitments mainly targeted environmental protection (biodiversity, environmental policy, environmental and forestry research) and rural and agricultural development.

Germany: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[2]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

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- OECD (2025), *Creditor Reporting System (CRS)*, database, <https://data-explorer.oecd.rg>. [2]

Italy

Italy is one of Europe's leading space-faring nations. With San Marco 1 in 1964, it became the fifth country to operate a satellite in orbit, and it remains one of ESA's three largest contributors. Italy has broad industrial strengths in space transportation, earth observation, telecommunications, exploration infrastructure and downstream services, including through the Vega/Vega-C launcher family and dual-use earth observation capabilities. ESA's earth observation centre, ESRIN, is located in Frascati and supports earth observation mission operations, data exploitation, Vega-related activities and planetary defence functions.

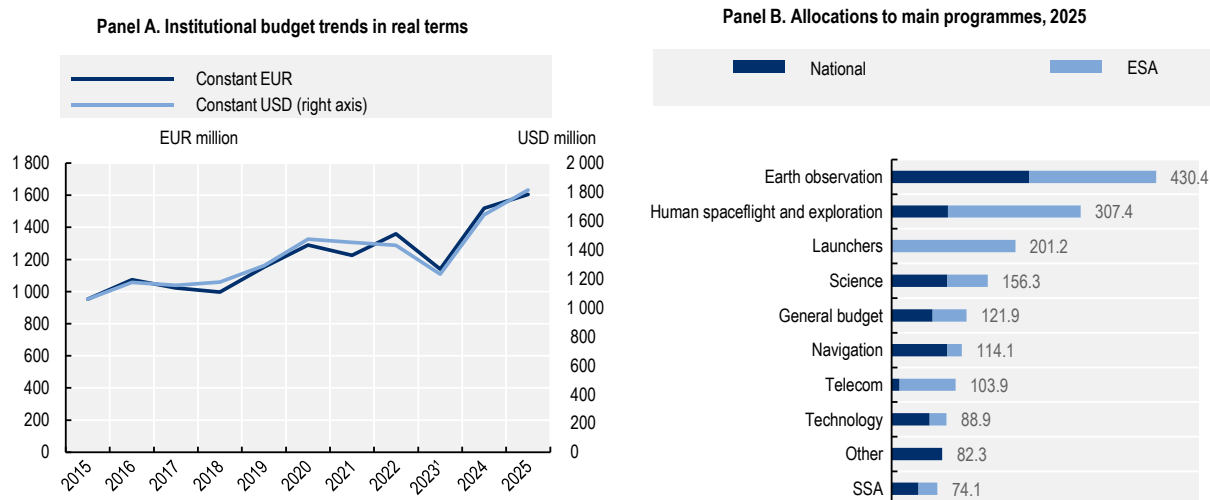
Italy at a glance

Indicator	Italy	OECD average
First satellite in orbit	San Marco 1 (1964)	
Number of operational satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	80 (53%)	
Number of satellites launched since 2019 (June 2026)	104 (6th)	
Institutional civil space budget as a share of gross domestic product	0.089%	0.065%
Institutional civil space budget per capita	38.7	33.6
Space literature fractional counts per 100 000 inhabitants	2.7	2.1

Italy also hosts the European Centre for Medium-Range Weather Forecasts' data centre in Bologna and has expanded its start-up support network to five ESA Business Incubation Centres, in Turin, Rome, Milan, Padua and Brindisi. In June 2026, Italy ranked seventh globally for active satellite ownership, 53% of which was commercially operated, and sixth for satellites launched since 2019.

Evolution of Italy's civil government space budget

In constant EUR and USD million (base year=2015)



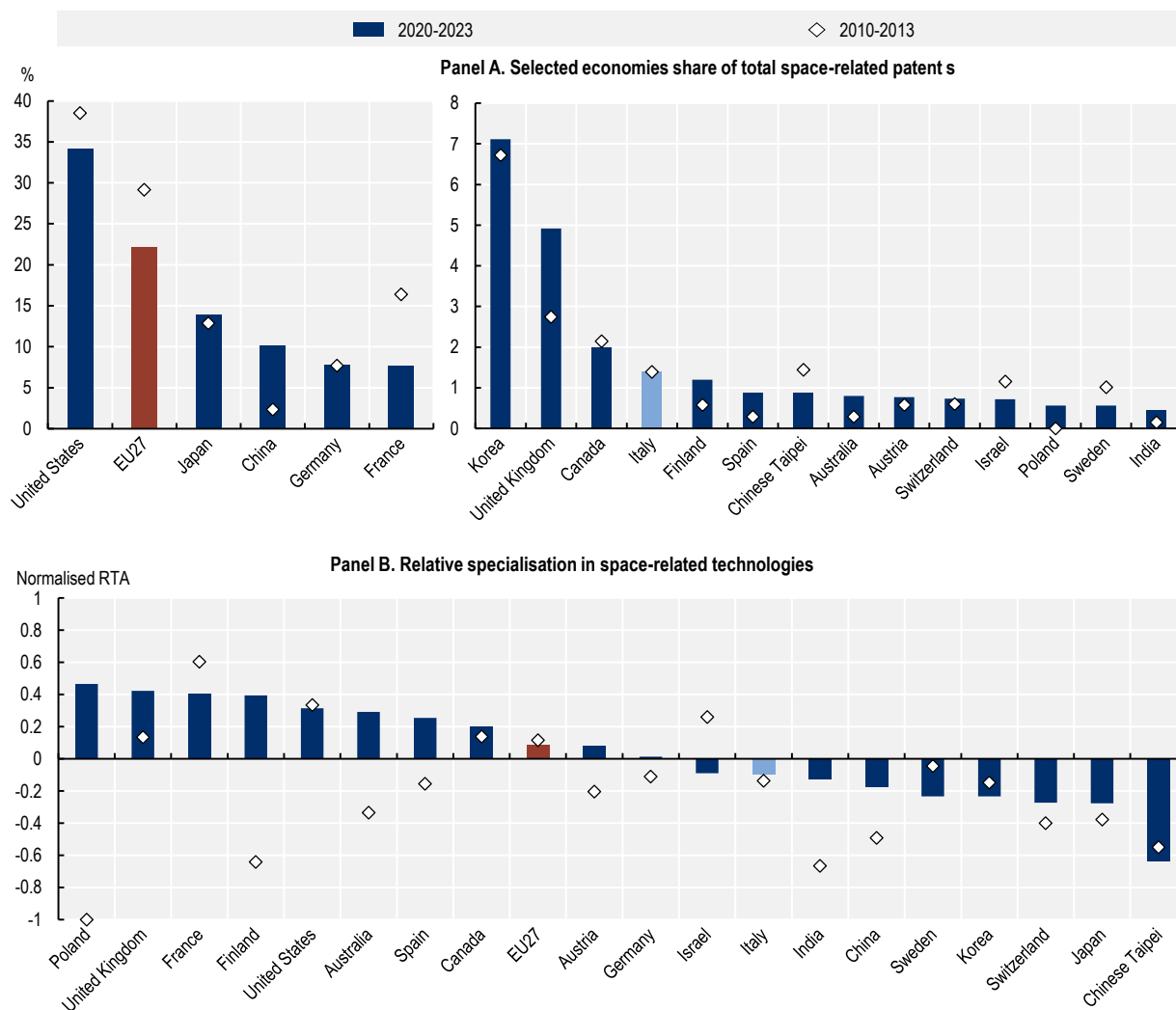
Note: Budget estimates include national activities and contributions to the European Space Agency, Eumetsat and as well as estimated contributions to the European Union Space Programme. 1. Break in series.

Source: OECD analysis based on institutional sources.

In 2025, the Italian civilian space budget reached USD 2 224.1 billion (EUR 1 968.2 billion), an increase from 2023-24 in real terms and above pre-COVID levels (Figure 5.13). This reflects major policy developments, notably the 2022-26 post-COVID-19 economic recovery package, higher commitments to the European Space Agency and new national initiatives. Earth observation was the biggest budget heading in 2025 – Italy is currently deploying its IRIDE earth observation programme, involving observations from six different constellation families. In 2025, the country adopted its first comprehensive space law, providing safeguards for technical, financial and environmental security. The legislation also established a space economy fund, endowed with EUR 35 million for 2025, to support industrial growth and innovation.

Italy: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

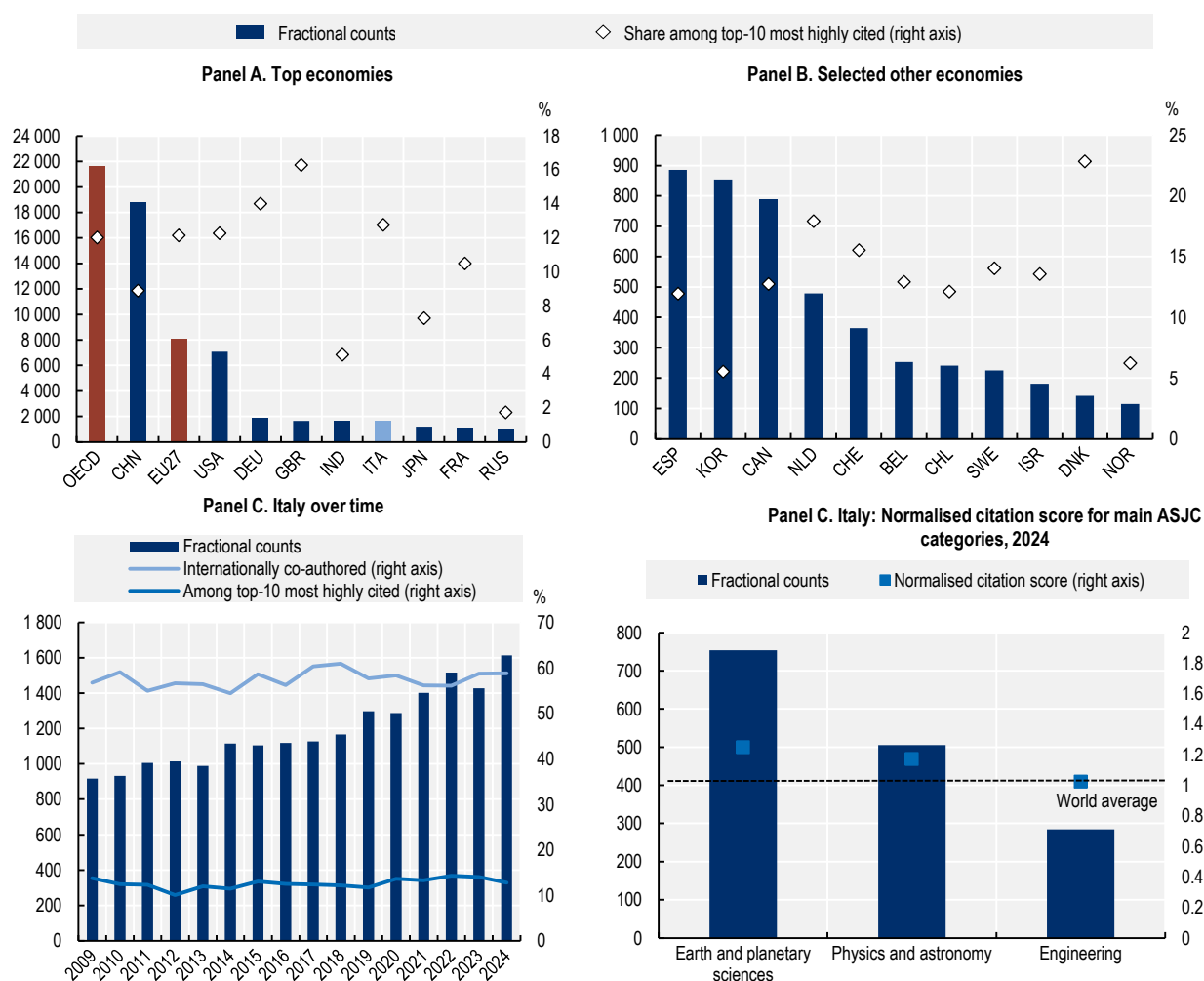
Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

Based on Italy's space economy thematic account launched in 2025, the Italian space economy generated EUR 8 billion in output in 2021, employing some 23 000 workers. This represented EUR 2 billion in gross value added, constituting 0.1% of Italy's GDP. Italy's space industry ecosystem comprises more than 350 companies, ranging from large system Integrators and mid-cap companies to small- and medium-sized enterprises, alongside a rapidly growing start-up community. The capabilities, products, services and technologies developed by the national space sector are showcased in the Italian Space Industry Online Catalogue (available at <https://italianspaceindustry.it>). In addition, a new web portal will soon be launched to further highlight the space-related expertise and competences of Italian research centres and academia.

Italy accounted for 1.7% of global space-related patents applications in the 2020-23 period, the same share as in 2010-13 (Figure 5.14). Italy has a negative revealed technological advantage (RTA), meaning that its share in space-related patenting is smaller than its share in overall patenting activities. Nevertheless, Italy's relative specialisation in space-related patenting has improved compared with 2010–13, reflecting a positive evolution in the country's patenting activity in space technologies.

Italy: Space-related scientific output and excellence indicators in 2024

Fractional counts



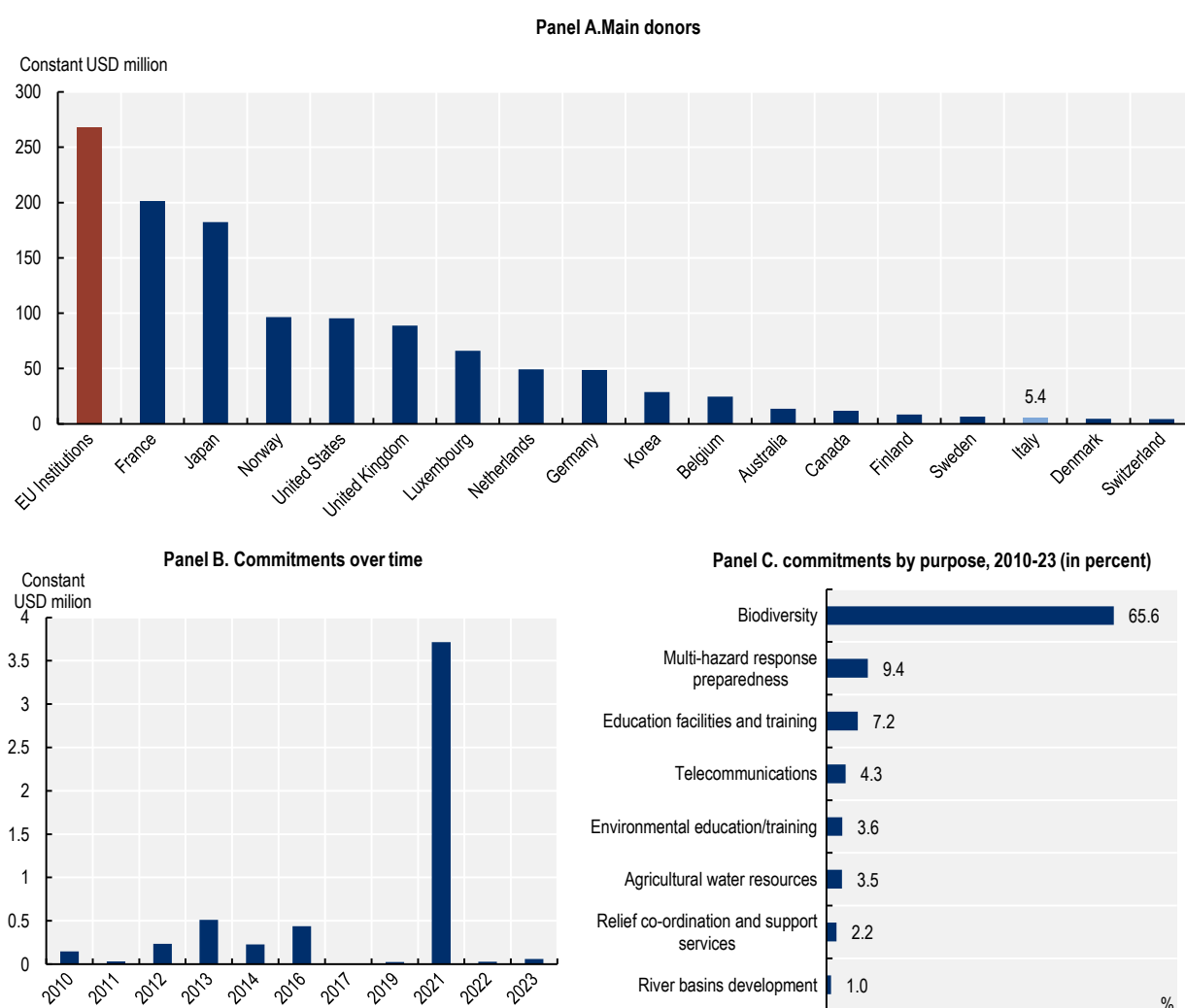
Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.
Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

In terms of space-related scientific production [hereafter ‘space literature’] and excellence, Italy’s production, measured in fractional counts, has increased steadily since 2009 and is above the OECD average (Figure 5.15). The share of internationally co-authored papers has remained stable over the period, at around 60%. When looking at citations, a common proxy for scientific excellence, 12.8% of Italy’s space literature was in the top-10 percentile for citations in 2024. This is above the OECD average of 12%. Italy’s normalised citation score, comparing citation patterns within specific scientific fields, is above the world average in the All Science Journal Classification (ASJC) categories “earth and planetary sciences” and “physics and astronomy”.

Between 2010 and 2023, Italy provided USD 5.4 million in space-related ODA (expressed in constant 2023 dollars), with the majority of funding directed towards environmental protection initiatives (Figure 5.16).

Italy: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[1]), “Creditor Reporting System (CRS)”, OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

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- OECD (2025), *Creditor Reporting System (CRS)*, database, <https://data-explorer.oecd.rg>. [1]

Korea

The Republic of Korea [hereafter 'Korea'] is one of the most ambitious space powers in the OECD area and has rapidly expanded its capabilities since placing its first satellite in orbit in 1992. It is one of 12 economies or regions having demonstrated autonomous launch capability and successfully placed its first lunar orbiter, the Korea Pathfinder Lunar Orbiter (KPLO, named Danuri), into lunar orbit in December 2022.

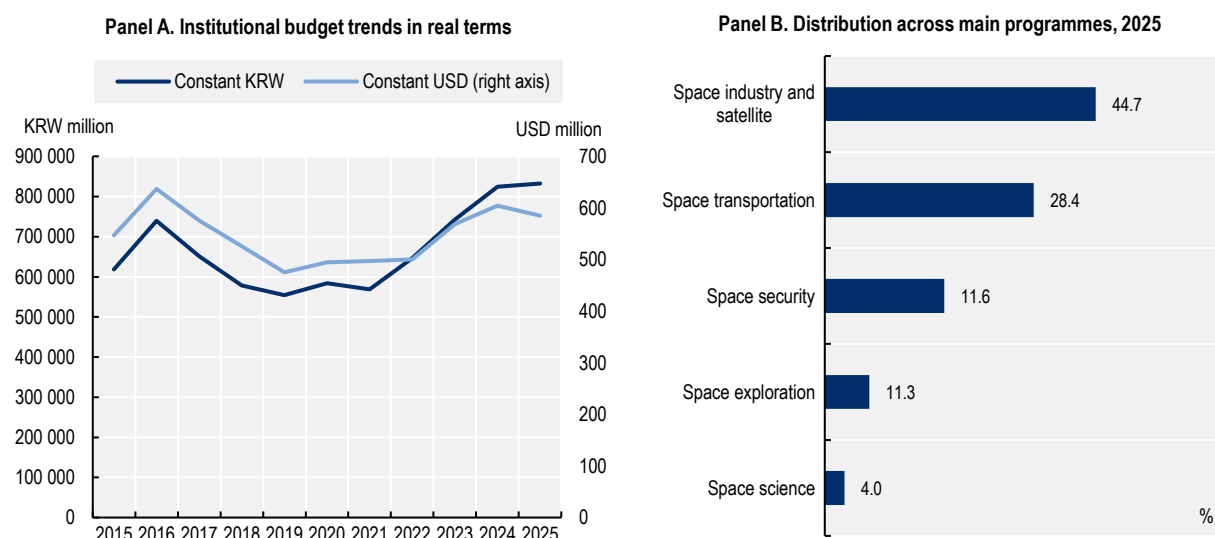
Republic of Korea at a Glance

Indicator	Korea	OECD average
First satellite in orbit	KITSAT-1 (1992)	
First successful orbital launch	2013	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	64 (38%)	
Number of satellites launched since 2019 (June 2026)	63	
Number of spaceports	1 (Naro Space Centre)	
Space-related workforce (2024)	11 147	
Space-related commercial revenues (2024)	KRW 4 356.8 billion	
Institutional civil space budget as a share of gross domestic product	0.089%	0.065%
Institutional civil space budget per capita	38.7	33.6
Space literature fractional counts per 100 000 inhabitants	2.7	2.1

The Korea Augmentation Satellite System, designed to support safe aviation, became fully operational in 2024. In 2024, the Korea Aerospace Administration was established to co-ordinate space policy across ministries and lead efforts in R&D, industrial promotion and international co-operation. As of June 2026, Korea ranked 10th globally in terms of satellite ownership – 38% of which was commercial – and 12th for satellites launched since 2019.

Evolution of Korea's civil government space budget

In constant KRW and USD million, base year 2015

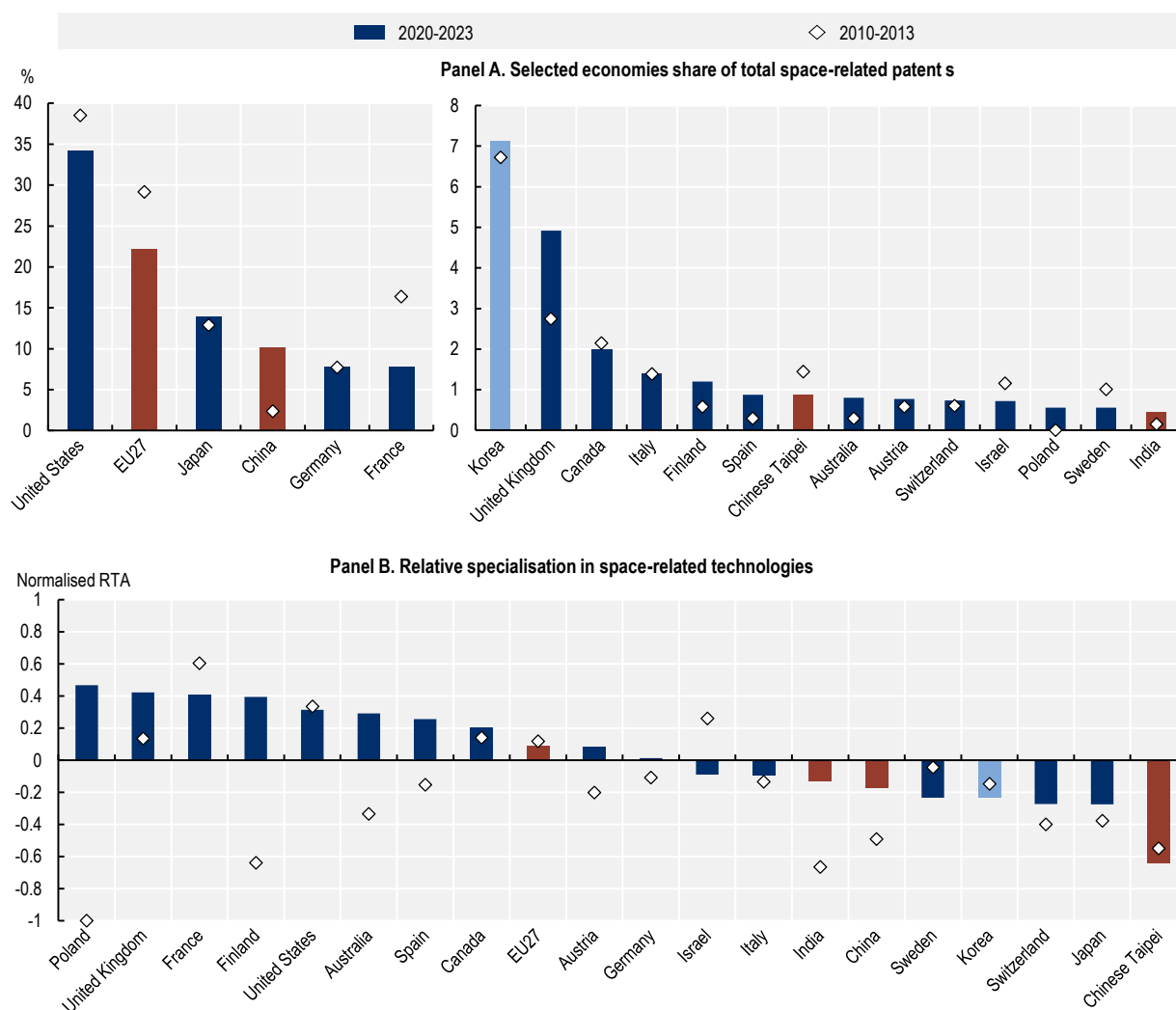


Source: OECD analysis based on institutional sources.

In 2025, Korea's civilian institutional space budget reached USD 719.3 million (KRW 1 023 billion), mainly dedicated to space industry and satellite development, followed by space transportation (Figure 5.17). Key projects include the development of next-generation launchers and a microsatellite constellation for rapid revisit dual-use monitoring of the Korean peninsula and territorial waters. The country is furthermore developing a satellite navigation constellation for autonomous, high-precision positioning, timing and navigation services for deployment in 2035. Finally, building on the data from the Danuri lunar orbiter mission, Korea is preparing a robotic lunar landing in the early 2030s.

Korea: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>). Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

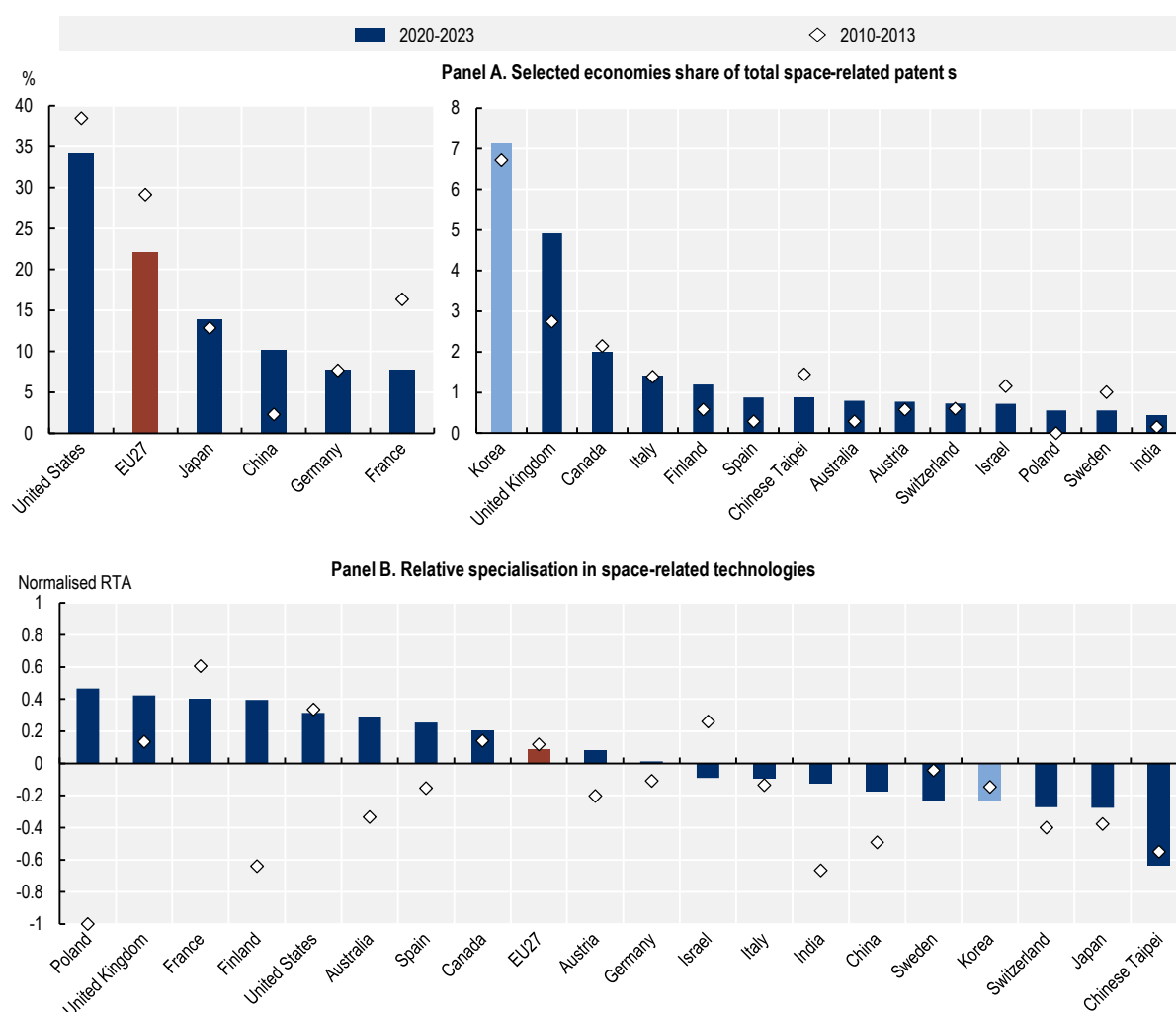
Korea's space sector involves both space manufacturing and satellite services providers. The manufacture of equipment to exploit satellite data and signals plays an important role. Government has traditionally led

space efforts in Korea, but it increasingly seeks to foster commercialisation through domestic product development and increased industry participation in projects. According to the Korea Aerospace Industry Survey, the Korean space sector had a total workforce of 11 147 (including higher education institutions and government agencies) in 2024 and generated commercial revenues of KRW 4 356.8 billion. This is an increase compared to previous years. Satellite services and related equipment manufacture accounted for more than 60% of commercial revenues.

Korea accounted for 7.1% of global space-related patents applications in the 2020-23 period, a small increase compared to 2010-13 (Figures 5.18). Korea has a negative revealed technological advantage (RTA), meaning that its share in space-related patenting is lower than its share in overall patenting activities. Korea's relative specialisation in space technology patenting has decreased since 2010-13.

Korea: Space-related scientific output and excellence indicators in 2024

Fractional counts



Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

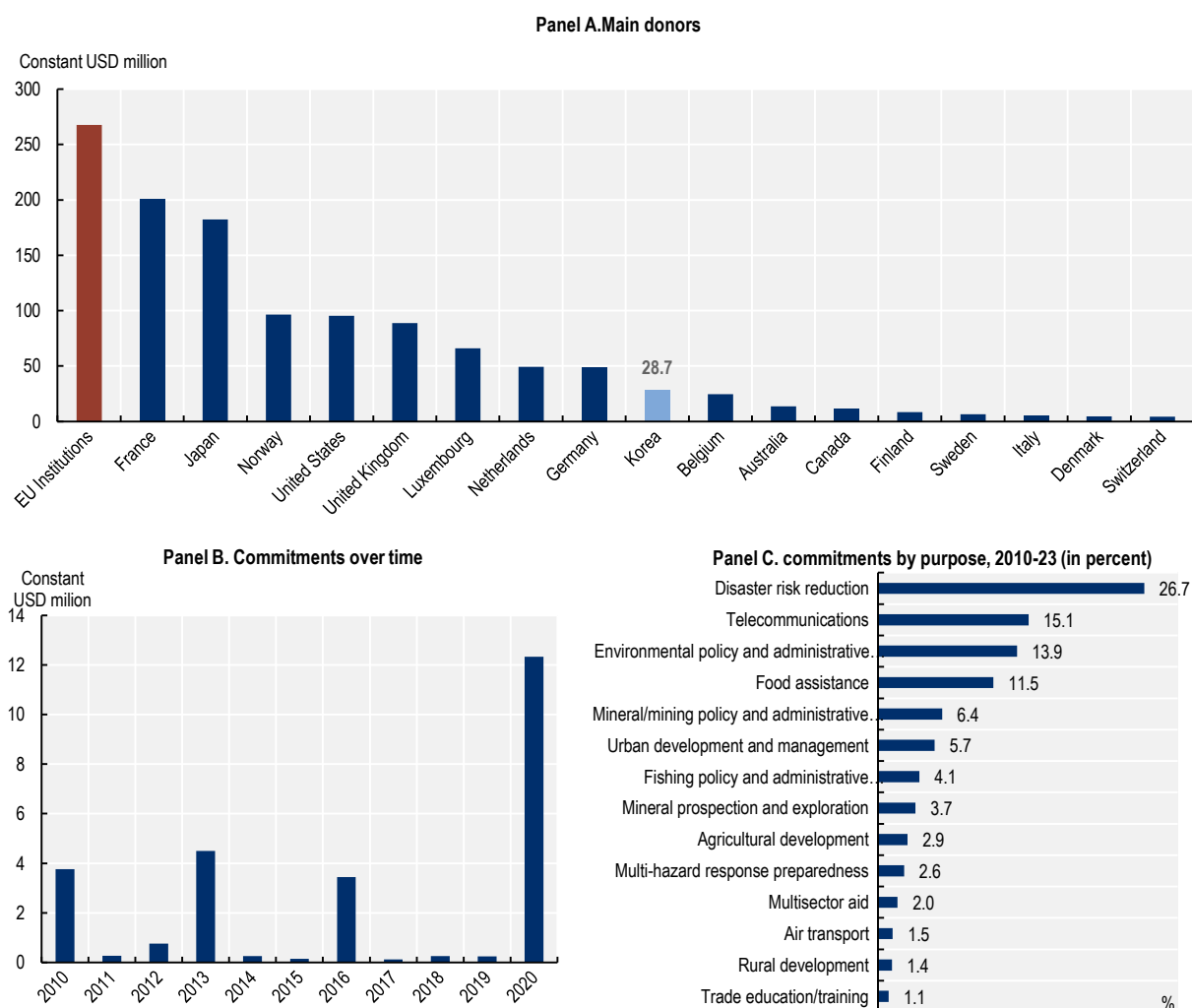
In terms of space-related scientific production and excellence, Korea's space literature (measured in fractional counts) has increased substantially since 2009, and its production per 100 000 inhabitants is

above the OECD average (Figure 5.19). The share of internationally co-authored papers has decreased considerably since 2009 and is now at 25%. When looking at citations, a common proxy for scientific excellence, 5.5% of Korea's space literature was in the top-10 percentile for citations in 2024. This is below the OECD average of 12%. Korea's normalised citation score, comparing citation patterns within specific scientific fields, is below the world average in the All Science Journal Classification (ASJC) categories "earth and planetary sciences", "physics and astronomy", and "engineering".

Based on data in the OECD Development Assistance Committee Creditor Reporting System database, Korea committed some 28.7 million constant US dollars in space-related official development assistance over the 2010-23 period, (Figure 5.20). Commitments mainly targeted disaster risk reduction measures, telecommunication and other economic development projects and environmental protection.

Korea: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[1]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

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The Netherlands

The Netherlands has played an active and important role in European space activities since their inception in the 1960s. It hosts ESTEC, the European Space Agency's technical centre, and was home to the first ESA Business Incubation Centre, established in 2004. Dutch capabilities are particularly strong in precision engineering, scientific instruments and satellite applications. Notable examples include TROPOMI, an atmospheric monitoring instrument launched in 2017 on the European Sentinel-5P satellite, and the planned TANGO mission, designed to monitor point-source greenhouse gas emissions.

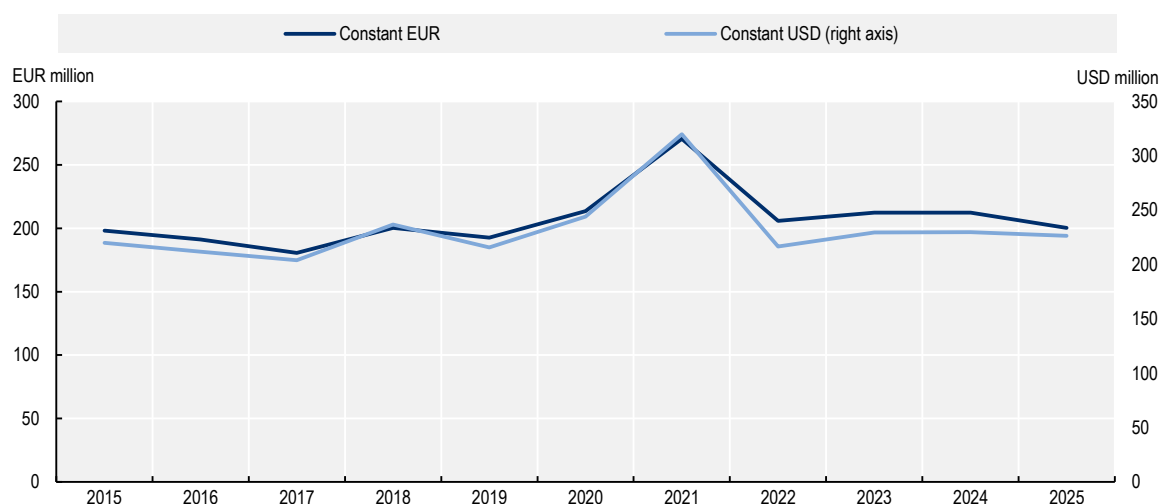
The Netherlands at a glance

Indicator	The Netherlands	OECD average
First satellite in orbit	ANS, 1974	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	7 (43%), rank 39th	
Number of satellites launched since 2019 (June 2026)	8 (39th)	
Space-related workforce		
Space-related commercial revenues		
Institutional civil space budget as a share of gross domestic product	0.023%	0.065%
Institutional civil space budget per capita	17	33.6
Space literature per 100 000 inhabitants	2.7	2.1

As of June 2026, the Netherlands ranked 39th globally in terms of satellite ownership and also 39th for satellites launched since 2019. Around 43% of active Dutch satellites are commercially operated.

Evolution of the Netherlands' civil government space budget

In constant EUR and USD million, base year 2015



Note: Budget estimates include national activities and contributions to the European Space Agency, Eumetsat and as well as estimated contributions to the European Union Space Programme.

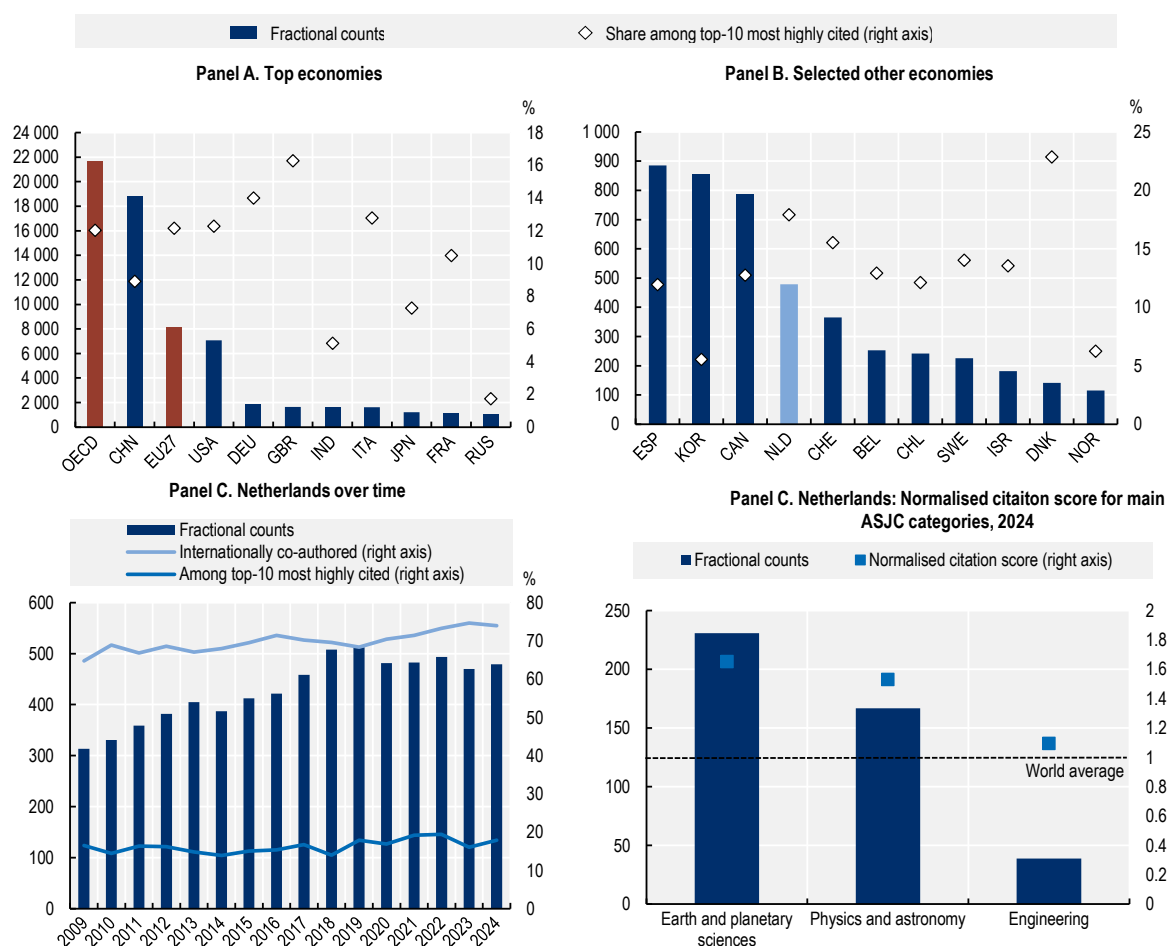
Source: OECD analysis based on institutional sources.

In 2025, the Netherlands' civilian government space budget amounted to USD 304.5 million (EUR 269.4 million), as shown in Figure 5.21. This includes not only contributions to the ESA and

Eumetsat but also estimated contributions to the European Union (EU) Space Programme, which is mainly based on each country's share of the EU's aggregated gross national income. ESA contributions represent the main bulk of the civilian budget. Meanwhile, the Dutch armed forces have launched several satellites since 2021. The Netherlands Space Command listed investment needs of EUR 25-100 million in the 2022-26 timeframe. The Dutch government presented its long-term space agenda in 2024, highlighting the growing strategic and economic importance of space and identifying six key priorities: Security; climate; science; satellite data; economic growth; and international collaboration. The Dutch space industry spans manufacturing, operations and satellite data and signal exploitation. According to the most recent available data from 2018, the sector employed a commercial workforce of 2 150 full-time equivalents, generating about EUR 669 million in production value. The study highlighted the important role of government institutes and higher education institutions, representing another 3 884 full-time equivalents and EUR 525 million in production value. A Dutch particularity compared with most other OECD countries is the importance of global satellite navigation system (GNSS) equipment production in the economy.

Netherlands: Space-related scientific output and excellence indicators in 2024

Fractional counts



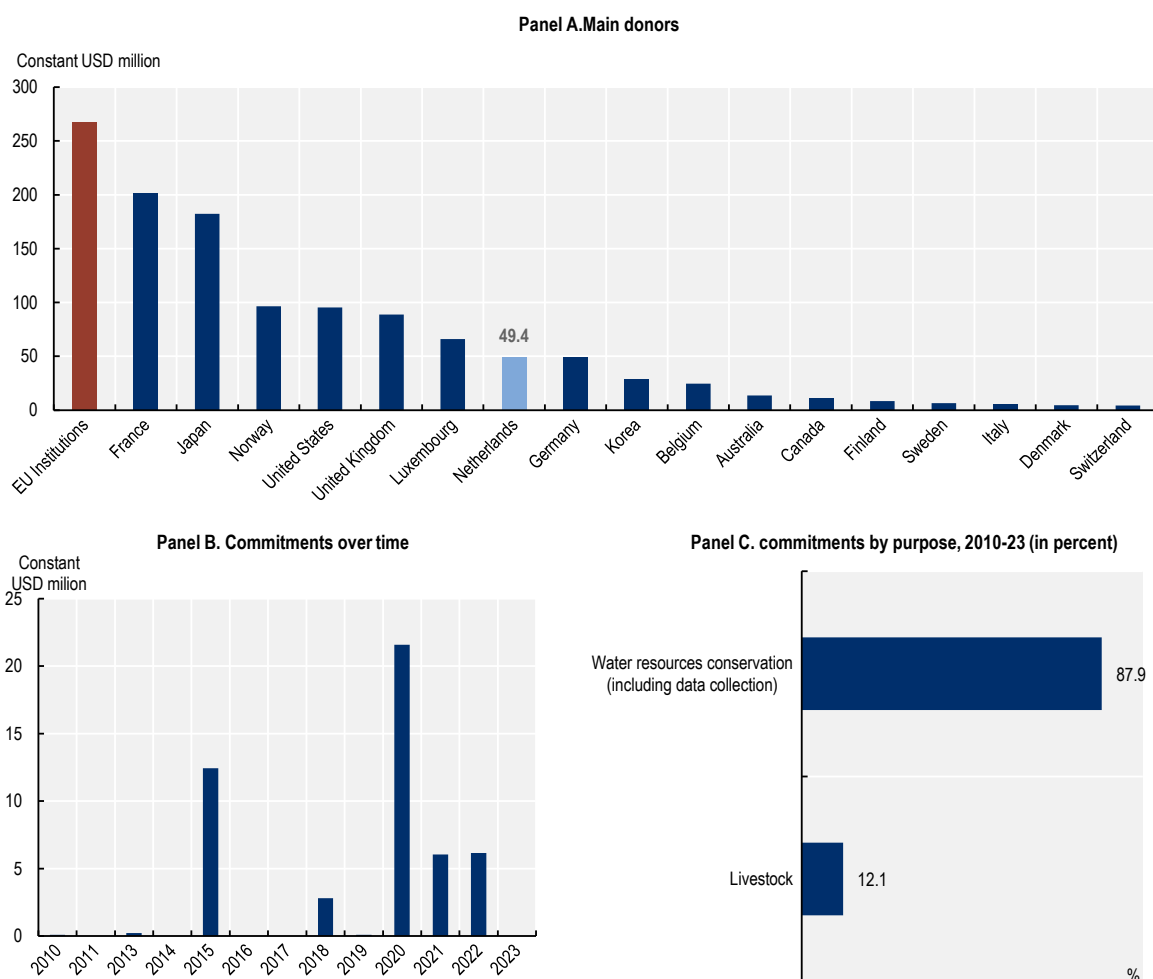
Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database. Analysis only conducted on 100 fractional counts or more.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

The Netherlands is a top performer in space-related scientific production [hereafter ‘space literature’] (Figure 5.22). Production is high in relative terms, above the OECD average, and more than 70% of literature is internationally co-authored. From an excellence perspective, using citation rates as a proxy, 17.9% of Dutch space literature was in the top-10 percentile most highly cited in 2024, the second in the sample after Denmark, and based on a considerably higher annual volume of documents. The Netherlands’ normalised citation score, comparing citation patterns within specific scientific fields, ranked above the world average in the All Science Journal Classification (ASJC) categories “earth and planetary sciences”, “physics and astronomy” and “engineering”.

Netherlands: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[11]), “Creditor Reporting System (CRS)”, OECD.stat (database), <https://stats.oecd.org/index.aspx?DataSetCode=CRS1> (accessed July 2025).

Based on data in the OECD Development Assistance Committee CRS database, the Netherlands is among the top-10 contributors of space-related official development assistance (Figure 5.23). The country committed 49.46 constant US dollars over the 2010-23 period, mainly to projects associated with water conservation and food production. This is linked to the Geodata for Agriculture and Water (G4AW) programme, established in 2013, which used satellite data to provide timely information on weather,

climate, hazards and crop conditions, so food producers could make better decisions, improve productivity, optimise their use of water, seeds and fertiliser, and strengthen their resilience to climate risks and pests.

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 (accessed on 23 June 2026).
- OECD (2026), *STI Micro-data Lab: Intellectual Property Database*, <http://oe.cd/ipstats> (accessed [3]
 on June 2026).
- OECD (2025), *Creditor Reporting System (CRS)*, database, <https://data-explorer.oecd.org>. [1]

Norway

Norway has been active in space through ESA membership since 1987, with capabilities shaped by its geography and national needs, notably in satellite communications, earth observation, navigation, maritime monitoring and Arctic services. Its high-latitude infrastructure is an important asset, including Svalbard ground stations for polar-orbiting satellites and Galileo, and Andøya, which has supported sounding-rocket activities since 1962. Norwegian firms have contributed equipment to several ESA and US missions, while recent orbital activity has been led by Space Norway, a wholly state-owned company responsible for safety-critical space infrastructure. Its Arctic Satellite Broadband Mission provides high-elliptical-orbit connectivity for the Norwegian Armed Forces, the US Space Force and Viasat, extending broadband coverage in the High North. Commercial momentum has increased with the opening of Andøya Spaceport in 2023, positioned as a European launch site for small and medium-sized satellites, with Isar Aerospace as an early launch customer.

Norway at a glance

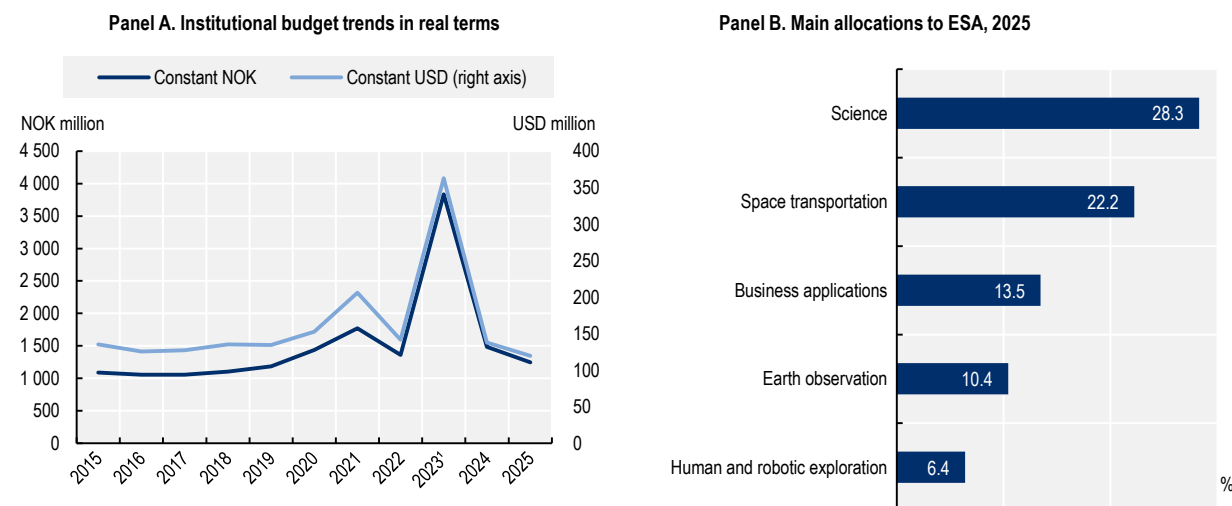
Indicator	Norway	OECD average
First satellite in orbit	Thor 1, 1989	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	18 (16.7%), rank 23rd	
Number of satellites launched since 2019 (June 2026)	12 (29th)	
Number of spaceports	1 (Andøya Spaceport)	
Space-related workforce (2021)	2 200	
Space-related commercial revenues (2021)	NOK 11 billion	
Institutional civil space budget as a share of gross domestic product	0.032%	0.065%
Institutional civil space budget per capita	29.1	33.6
Space literature per 100 000 inhabitants	2.1	2.1

Norway also supports space entrepreneurship through ESA BIC Norway, launched in 2018 and operating through a national network of incubator partners. In June 2026, Norway ranked 23rd globally in satellite ownership and 29th for satellites launched since 2019.

In 2025, Norway's civilian government space budget reached USD 164.8 million (NOK 1 713) million, which is similar to recent years in real terms (Figure 5.24). Contributions to the European Space Agency and European Union programmes represent the bulk of the budget but national activities are on the rise, including both orbital and terrestrial infrastructure. The 2023 budget spike refers to an equity injection in Space Norway to purchase Telenor's geostationary telecommunications satellites. In 2024 the two satellites in the Arctic Broadband Satellite mission were launched, jointly operated with the US Department of Defence and Inmarsat/Viasat. As noted in the introduction, a commercial spaceport was opened in 2023 in the northern part of Norway.

Evolution of Norway's civil government space budget

In constant NOK and USD million, base year 2015



Note: Budget estimates include national activities and contributions to the European Space Agency, Eumetsat and the European Union (Copernicus, Galileo/EGNOS). 1. The 2023 spike reflects an equity injection to Space Norway, to finance the acquisition of Telenor Satellite and support Arctic satellite infrastructure.

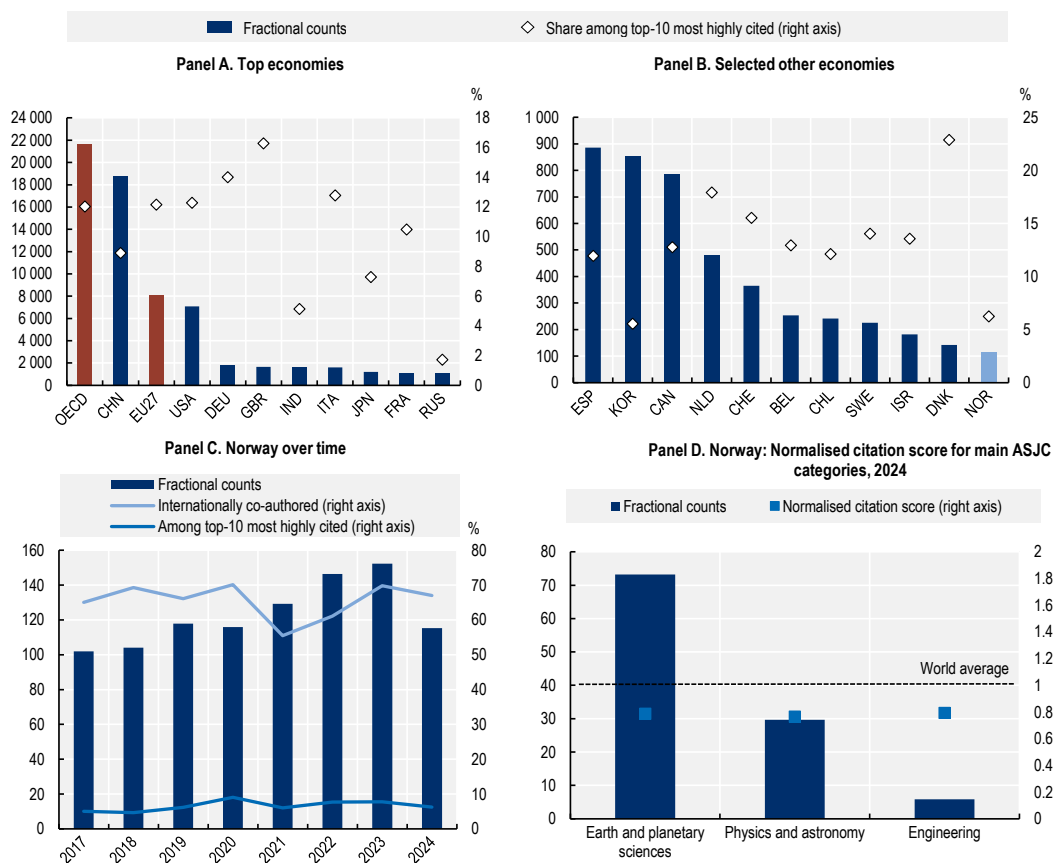
Source: OECD analysis based on institutional sources.

According to Norway's latest key figures report, conducted in 2022-23, Norwegian companies generated NOK 11 billion in revenues with a commercial workforce of 2 200 in 2021, including 138 actors with activities in the space sector. Activity was oriented towards satellite operations and data/signal exploitation. The sector relies heavily on exports (81% average export rate) and is concentrated –top-5 actors accounted for more than 70% of revenues. Employment was concentrated in the southern and northern parts of Norway. It is worth noting the strategic and economic importance of the Svalsat ground station for satellite communications in the Svalbard archipelago. It is the world's northernmost and biggest polar satellite ground station and had more than 150 antennas from public and private space organisations in 2025.

Norway's space-related production of scientific documents [hereafter 'space literature'] is characterised by high level international collaboration, with close to 70% of annual production internationally co-authored (Figure 5.25). Fractional counts for 2024 were higher than 2018 levels but lower than those in 2019-23. Controlled for population, Norway produces at OECD average levels. In terms of excellence, the share of domestic literature that is in the top-10 percentile for citations increased between 2017 and 2024 but is below the OECD average. Norway's normalised citation score, comparing citation patterns within specific scientific fields, also ranked below the world average in the All Science Journal Classification (ASJC) categories "earth and planetary sciences", "physics and astronomy" and "engineering".

Norway: Space-related scientific output and excellence indicators in 2024

Fractional counts



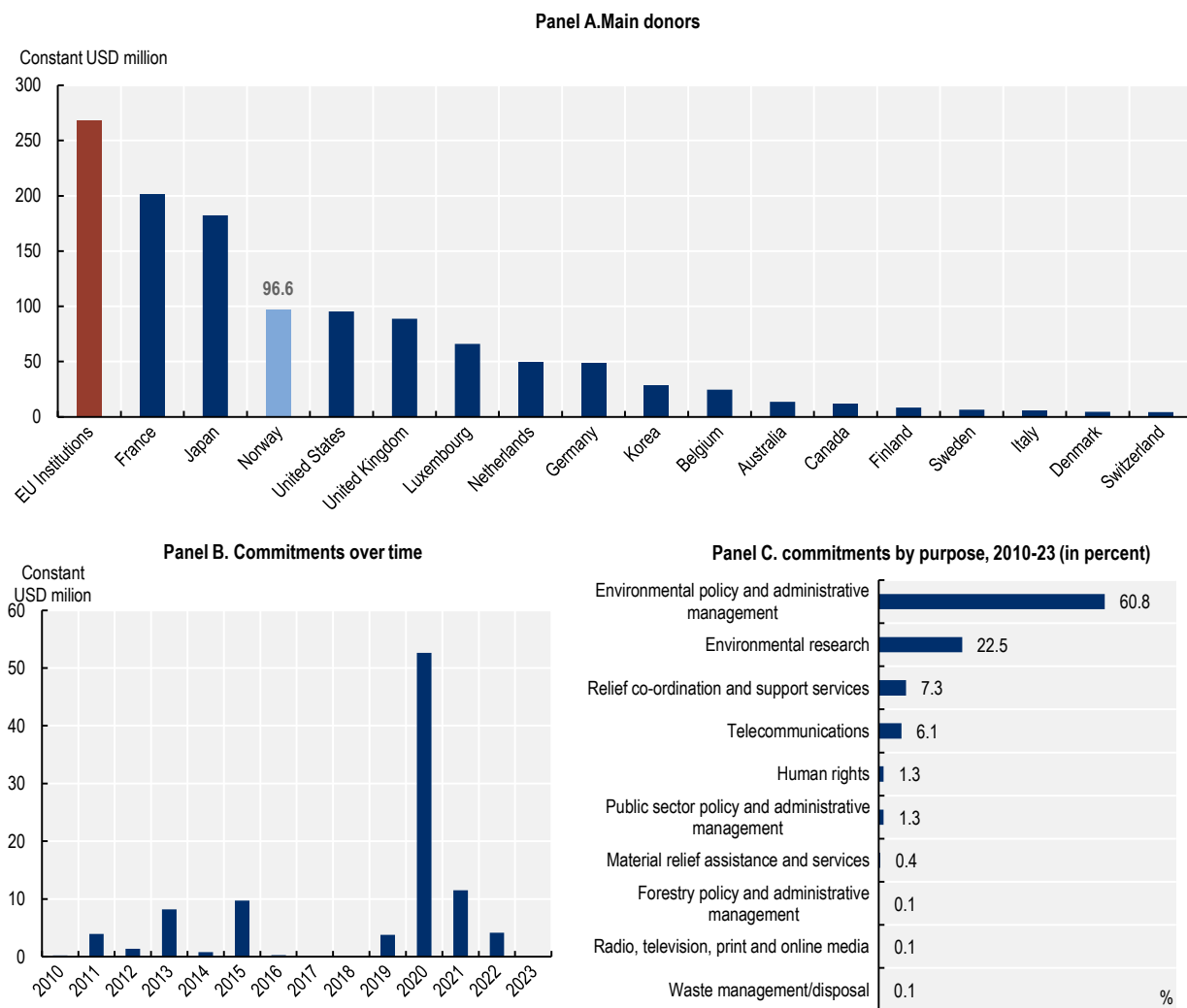
Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database. Analysis only conducted on 100 fractional counts or more.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

Based on data in the OECD Development Assistance Committee Creditor Reporting System database, Norway is one of the top contributors of space-related official development assistance (Figure 5.26). Norway committed 96.6 million constant US dollars over the 2010-23 period, mainly to environmental protection. A significant share is related to the Norwegian International Climate and Forest Initiative's investments in satellite-based monitoring of tropical forests, including the Satellite Data Programme launched in 2020.

Norway: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[1]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

References

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Switzerland

Switzerland is an important supplier of high-precision technologies, instruments and components for international missions and a founding member of the European Space Agency. Its strengths lie in space science, precision engineering, atomic clocks, launcher structures and scientific payloads, with commercial Swiss technologies used in programmes such as Galileo, Ariane and Vega.

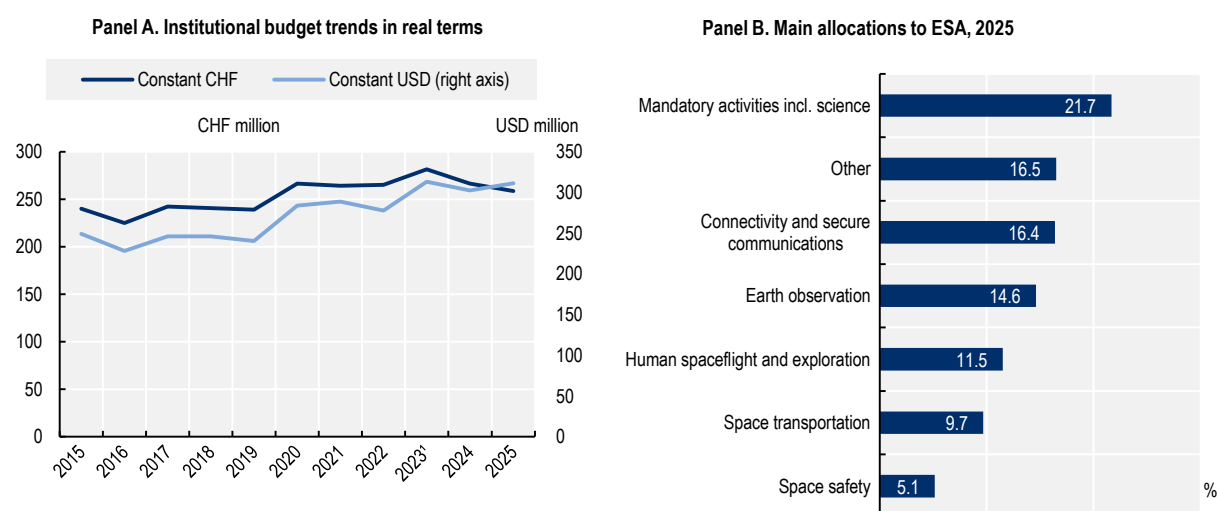
Switzerland at a glance

Indicator	Switzerland	OECD average
First satellite in orbit	SwissCube-1, 2009	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	13 (77%)	
Number of satellites launched since 2019 (June 2026)	26 (22nd)	
Institutional civil space budget as a share of gross domestic product	0.032%	0.065%
Institutional civil space budget per capita	36.9	33.6
Space literature per 100 000 inhabitants	4.1	2.1

The country combines strong academic capabilities with a specialised industrial base and start-up ecosystem, supported since 2016 by ESA BIC Switzerland, operated by the technical university ETH in Zurich. Orbital activity is mainly commercial or academic. Switzerland's first satellite, SwissCube-1, was developed by the Lausanne technical university EPFL with students and partners and launched in 2009 to study atmospheric airglow. In June 2026, Switzerland ranked 25th globally for satellite ownership and 22nd for satellites launched since 2019.

Evolution of Switzerland's civil government space budget

In constant CHF and USD million, base year 2015



Note: The budget includes national complementary activities, contributions to the European Space Agency, to Eumetsat and to the Galileo/EGNOS programme.

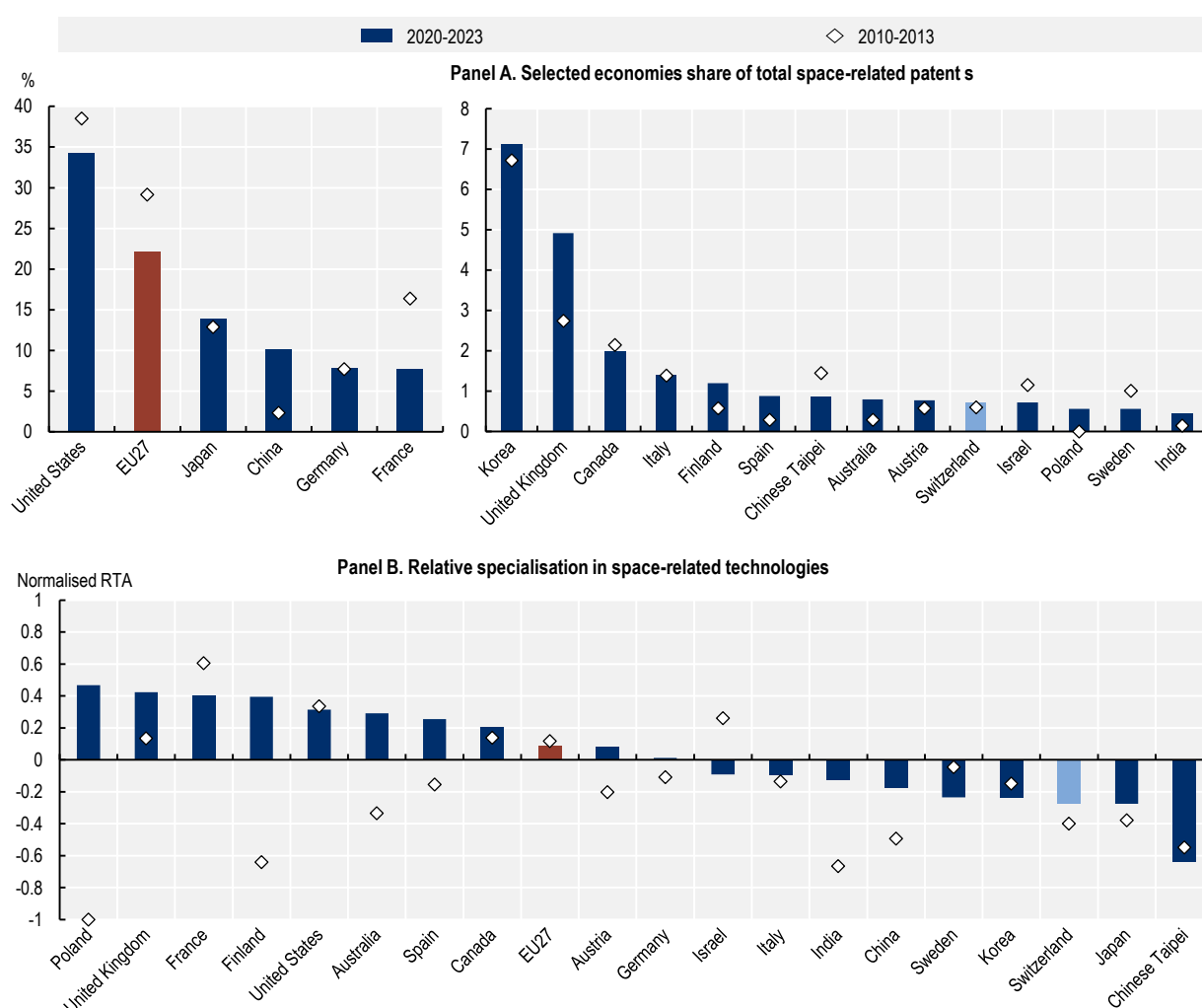
Source: OECD analysis based on institutional sources.

In 2025, Switzerland's government space budget reached USD 334 million (CHF 278.4 million), representing 0.032% of GDP (Figure 5.27). This is a slight decrease compared with 2020-24, but higher than 2019 levels. The budget is dominated by contributions to the European Space Agency, with connectivity and secure communications and earth observation as the two main optional programmes.

Swiss space industry has traditionally focused on space manufacturing (e.g. fairings and atomic clocks), but several satellite operators have emerged since 2010, often relying on in-house manufacturing of miniaturised satellites. Applications include for example internet-of-things and orbital debris removal. The Swiss company ClearSpace will lead the European Clearspace-1 mission scheduled for 2028, removing a satellite from orbit.

Switzerland: Patens for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

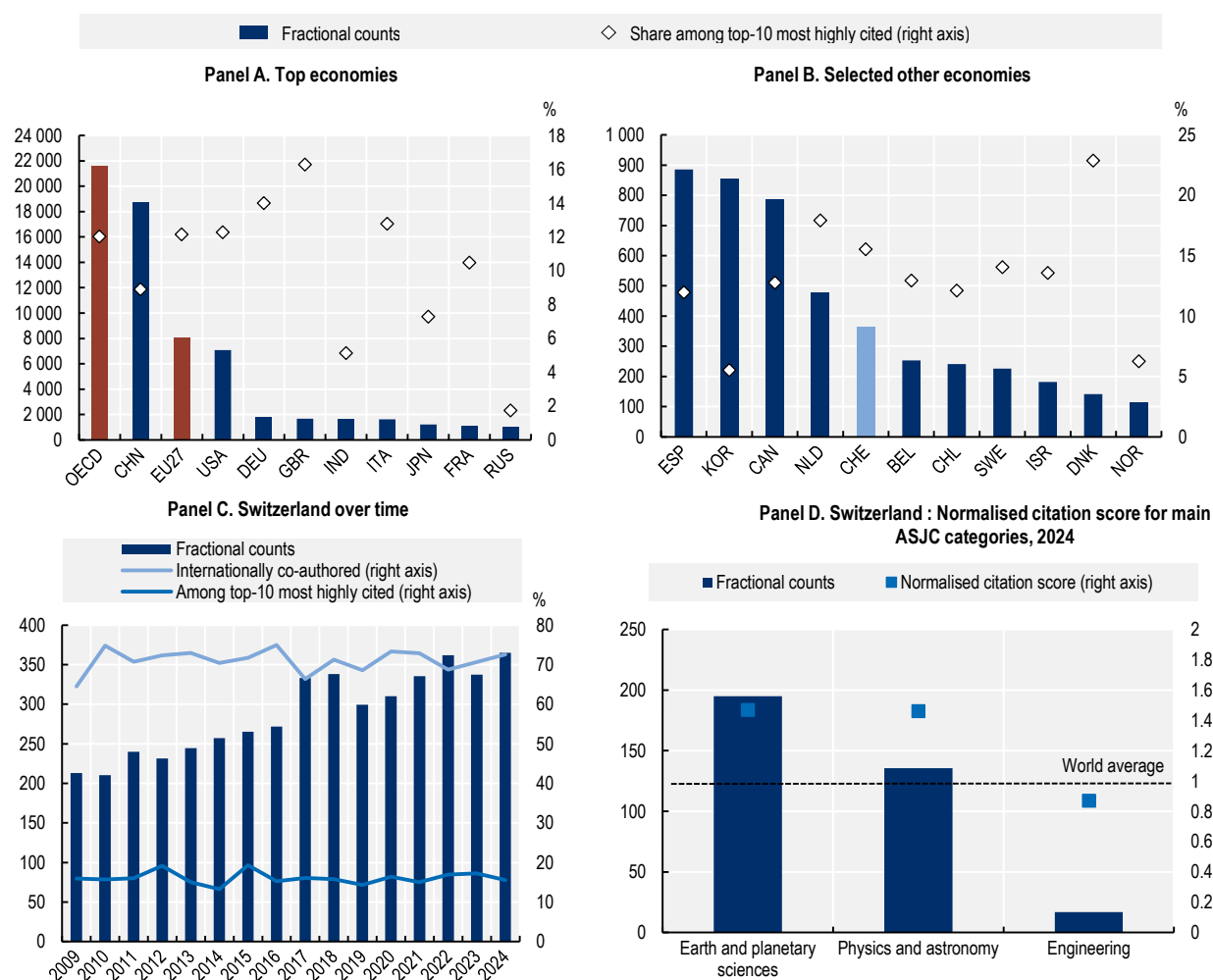
Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

Switzerland accounted for 0.7% of global space-related patent applications in the 2020-23 period (Figure 5.28), which is a small increase compared to 2010-13. Its share in space-related patenting is lower than its global share, giving it a negative revealed technological advantage (RTA). Switzerland's space-related RTA has decreased since 2010-13.

In terms of space-related scientific production and excellence, Switzerland's space literature (measured in fractional counts) has increased since 2009, while its high share of internationally co-authored papers has remained stable and high (Figure 5.29). Its production per 100 000 inhabitants is twice the OECD average. When looking at citations, a common proxy for scientific excellence, 15.5% of Switzerland's space literature was in the top-10 percentile for citations in 2024. This is above the OECD average of 12%. Switzerland's normalised citation score, comparing citation patterns within specific scientific fields, is above the world average in the All Science Journal Classification (ASJC) categories "earth and planetary sciences", and "physics and astronomy".

Switzerland: Space-related scientific output and excellence indicators in 2024

Fractional counts



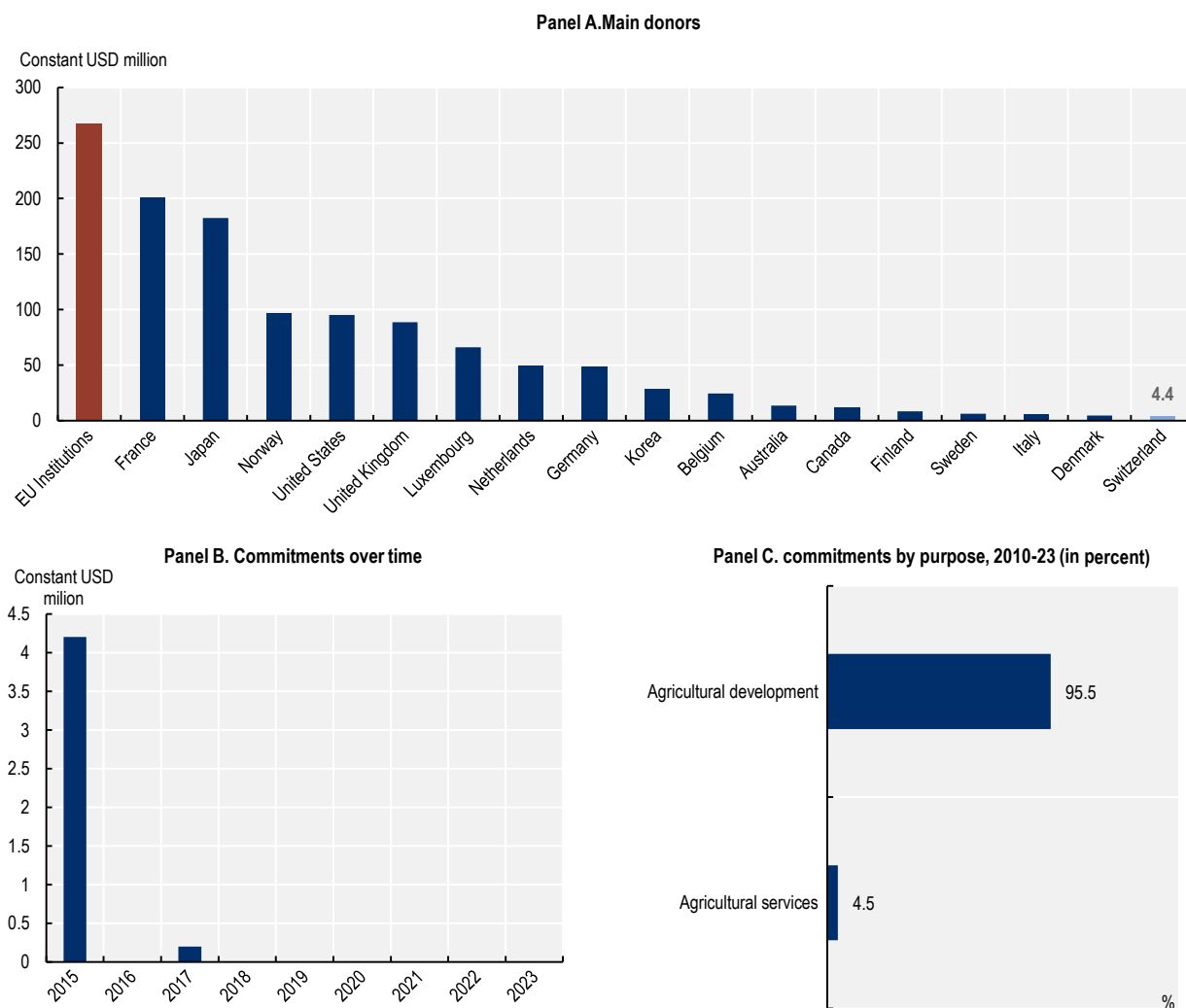
Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

Based on data in the OECD Development Assistance Committee Creditor Reporting System database, Switzerland committed some 4.4 million constant US dollars in space-related official development assistance over the 2010-23 period, mainly targeting the development of agriculture (Figure 5.30).

Switzerland: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[1]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

References

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- OECD (2025), *Creditor Reporting System (CRS)*, database, <https://data-explorer.oecd.rg>. [1]

United Kingdom

The United Kingdom has a long-standing role in spaceflight. It was the third country to operate a satellite, with Ariel-1 launched in 1962, after the Soviet Union and the United States, and the sixth country to place a satellite in orbit using a domestically developed launch vehicle, when Prospero was launched by Black Arrow in 1971.

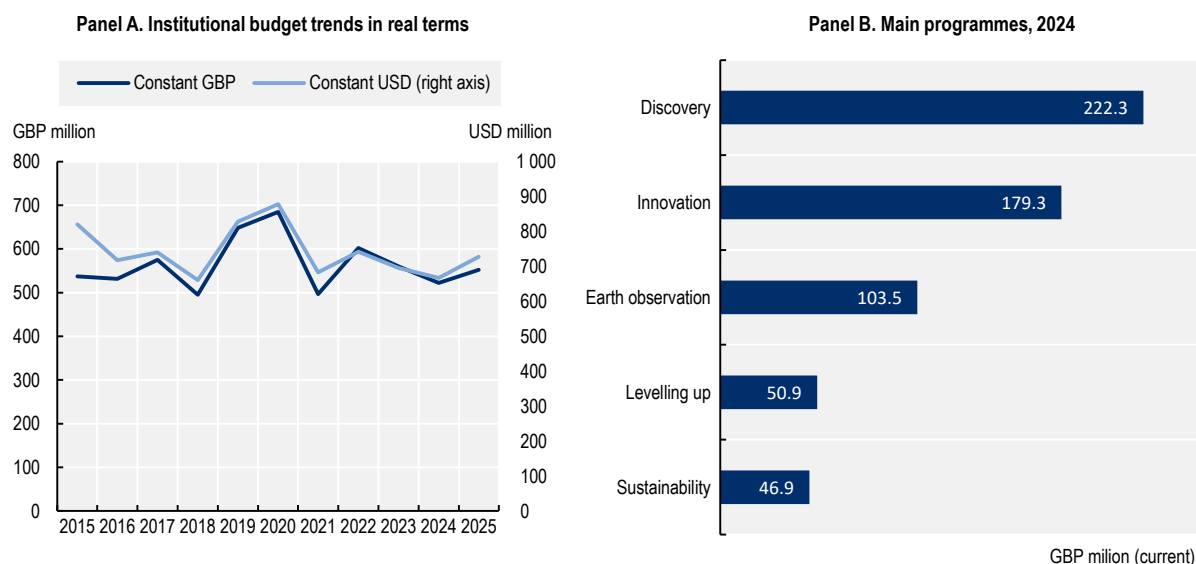
The United Kingdom at a glance

Indicator	United Kingdom	OECD average
First satellite in orbit	Ariel 1 (1962)	
First successful orbital launch	1972 (Black Arrow)	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	757 (98%)	
Number of satellites launched since 2019 (June 2026)	729 (3rd)	
Number of spaceports	2 licensed (Cornwall and SaxaVord),	
Space-related workforce (2022)	55 550 (full-time-equivalents)	
Space-related commercial revenues (2022)	GBP 18.6 billion	
Institutional civil space budget as a share of gross domestic product	0.025%	0.065%
Institutional civil space budget per capita	4.5	33.6
Space literature fractional counts per 100 000 inhabitants	2.4	2.1

The United Kingdom is also a founding member and major contributor to the European Space Agency (ESA), and hosts ESA's European Centre for Space Applications and Telecommunications at Harwell, a hub for satellite communications, applications, climate-related space services and commercialisation.

Evolution of United Kingdom's civil government space budget

In constant GBP and USD million (base year= 2015)



Note: Budget estimates include national activities and contributions to the European Space Agency, Eumetsat and the European Union (Copernicus, Galileo/EGNOS). The United Kingdom contributed to the EU Space Programme as an EU member until 2020. It rejoined the EU Copernicus programme in 2024.

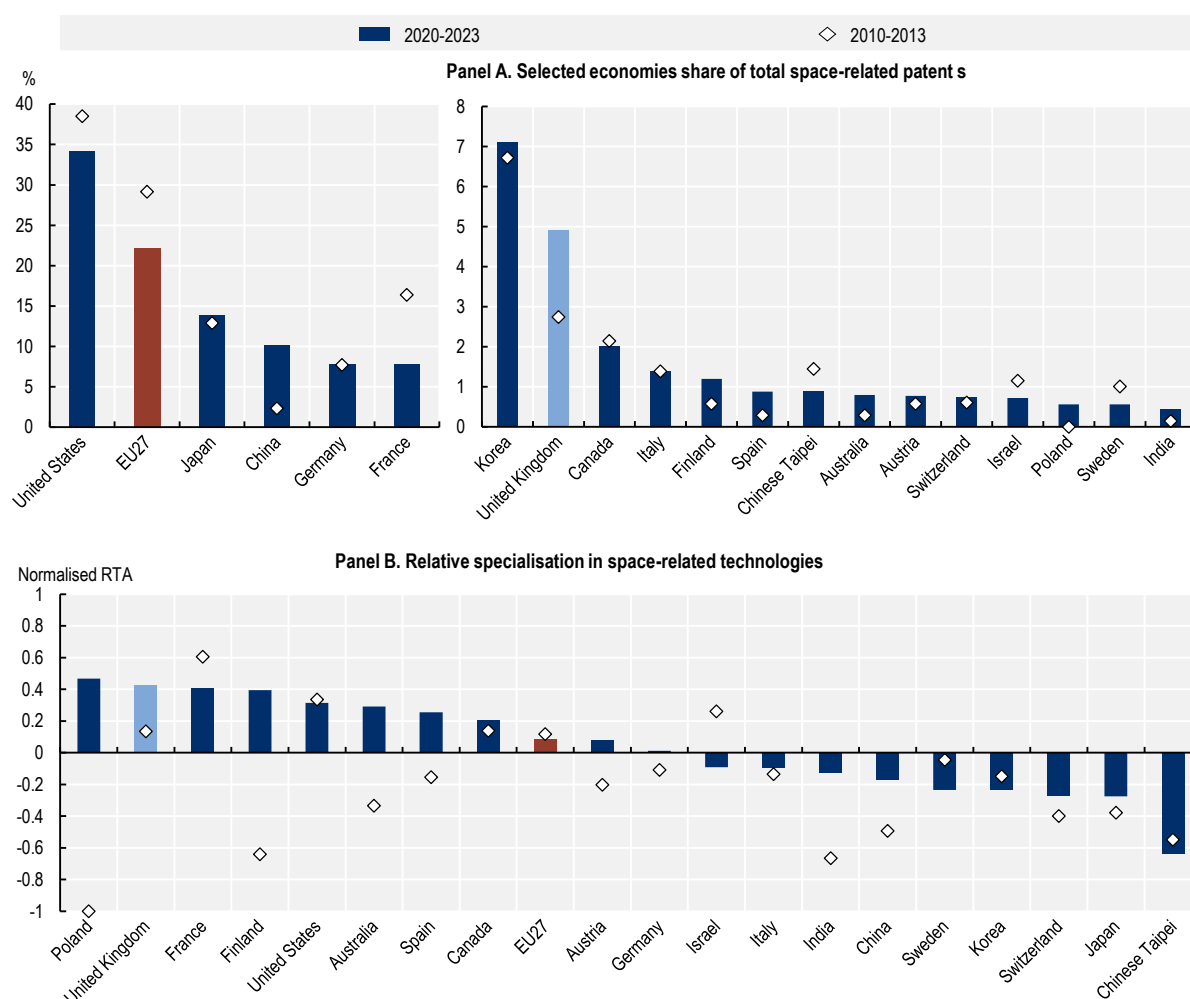
Source: OECD analysis based on institutional sources.

The UK space sector has particular strengths in small satellites, satellite communications, spacecraft manufacturing, operations, space science and downstream services. In orbital terms, the United Kingdom ranked third globally for satellite ownership in June 2026, predominantly reflecting commercial operators, and also third for satellites launched since 2019.

In 2025, the UK civilian space budget amounted to USD 1 003.7 million (GBP 762.3 million), including contributions to national activities, ESA, Eumetsat and selected EU space programmes (Copernicus), as shown in Figure 5.31. Key priorities include “discovery” (mainly collaborative science and exploration missions), “innovation” (technology development, applications and ecosystem support) and “earth observation”. Several developments affected the budget over the 2015-25 period, including COVID-19 and the formal withdrawal from the European Union in 2020.

United Kingdom: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



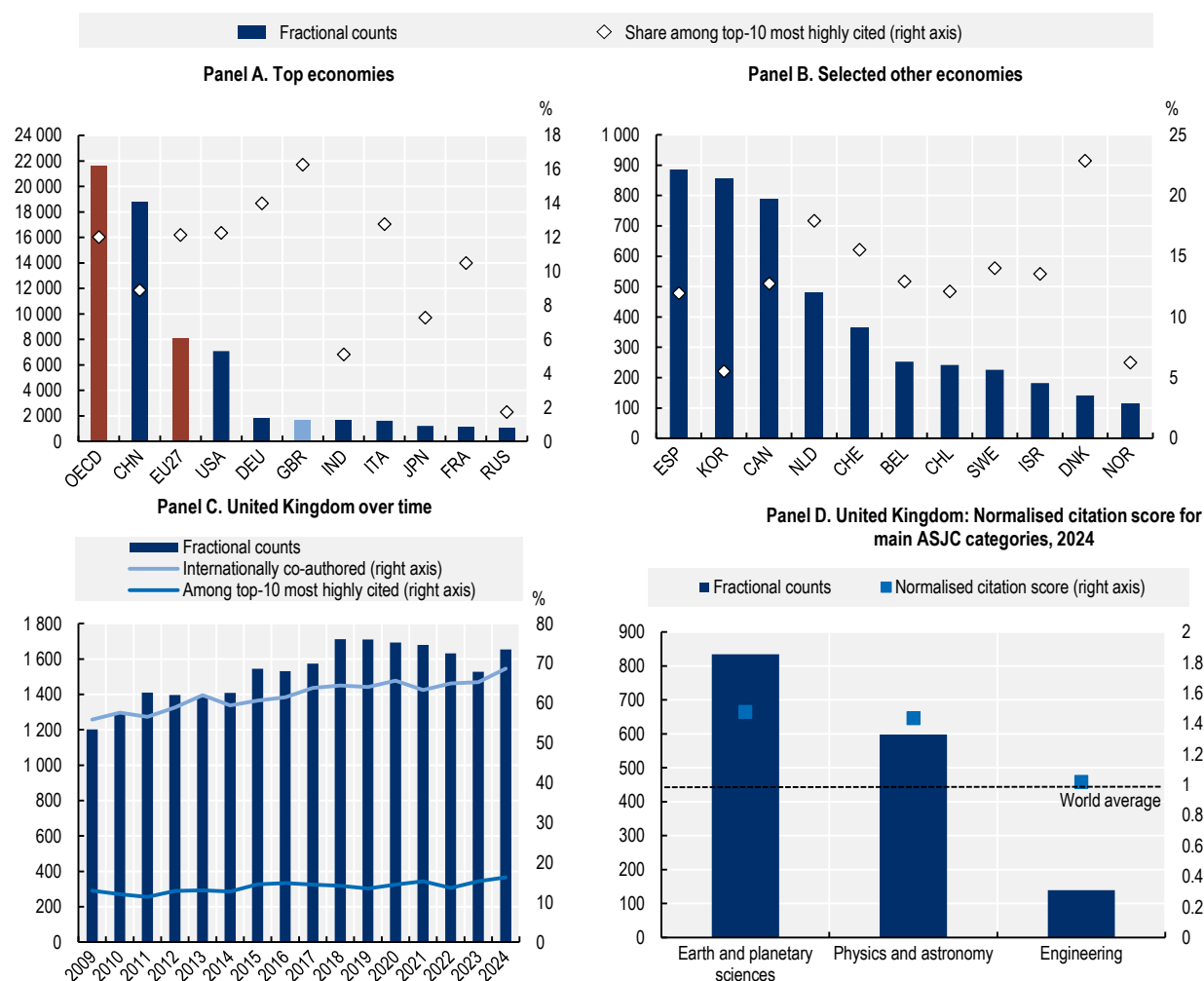
Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

According to the UK survey on the size and health of the space industry, the sector employed 55 550 full-time-equivalents in 2022 and generated an income of GBP 18.6 billion. Direct-to-home broadcasting services accounted for almost half of the income, but its share has been trending downwards since 2010.

The United Kingdom accounted for 4.9% of global space-related patents applications in the 2020-23 period almost a doubling since 2010-13 (Figure 5.32), mainly filed by commercial actors in the fields of space debris, life support and space-related quantum. The United Kingdom has a strong positive revealed technological advantage (RTA), meaning that its share in space-related patenting is higher than its share in overall patenting activities. UK relative specialisation in space technology patenting has considerably increased compared with 2010-13.

United Kingdom: Space-related scientific output and excellence indicators in 2024



Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

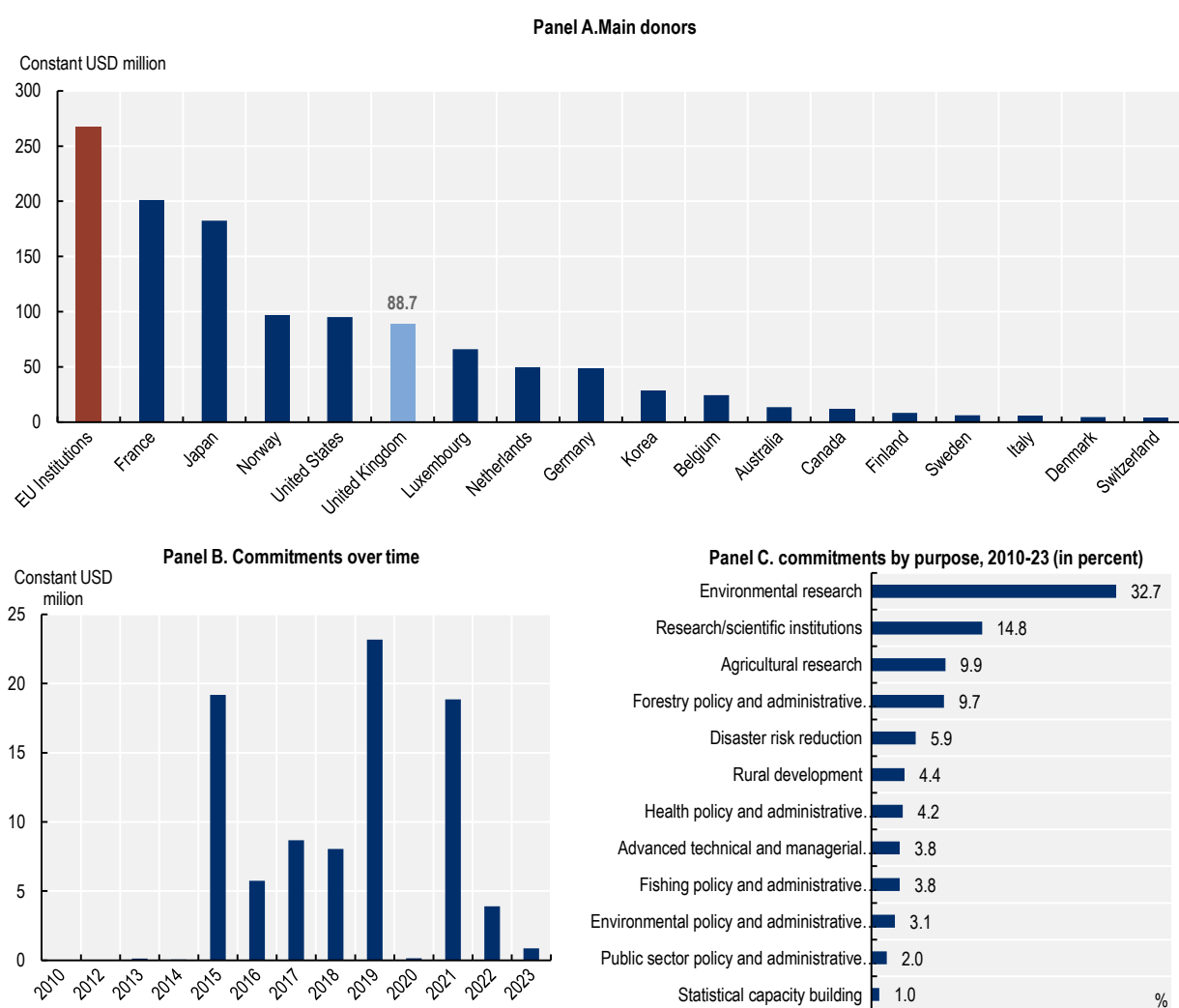
The United Kingdom is a top performer in terms of space-related scientific production [hereafter 'space literature'] and excellence. Production has increased steadily since 2009 (measured in fractional counts) and is above the OECD average (Figure 5.33). The share of internationally co-authored papers has also increased steadily, approaching 70%. When looking at citations, a common proxy for scientific excellence, 16.3% of UK space literature was in the top-10 percentile for citations in 2024. This is well above the OECD

average of 12%. The United Kingdom's normalised citation score, comparing citation patterns within specific scientific fields, is above the world average in the All Science Journal Classification (ASJC) categories "earth and planetary sciences", "physics and astronomy", and "engineering".

The United Kingdom committed 88.7 million of constant 2023 dollars to space-related official development assistance (ODA) over the 2010-23 period (Figure 5.34). The commitments were mainly allocated to environmental research (32.7%), research institutions (14.8%) and agricultural research (9.9%). The five-year International Partnership Programme, funded by the Global Challenges Research Fund and run by the UK Space Agency, was established in 2016 in to employ UK expertise in satellite technology and data services to help tackle development challenges.

United Kingdom: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025^[1]), "Creditor Reporting System (CRS)", OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1>.

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- McDowell, J. (2026), *Jonathan's Space Report*, <https://www.planet4589.org/index.html> (accessed on 23 June 2026). [3]
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United States

The United States is one of the world's key space actors, with a significant share of orbital assets supported by a dynamic commercial launch sector and long-standing technological and scientific capabilities. It became the second economy to operate a satellite in orbit in 1958 and remains the only country to have conducted crewed lunar landings, beginning in 1969.

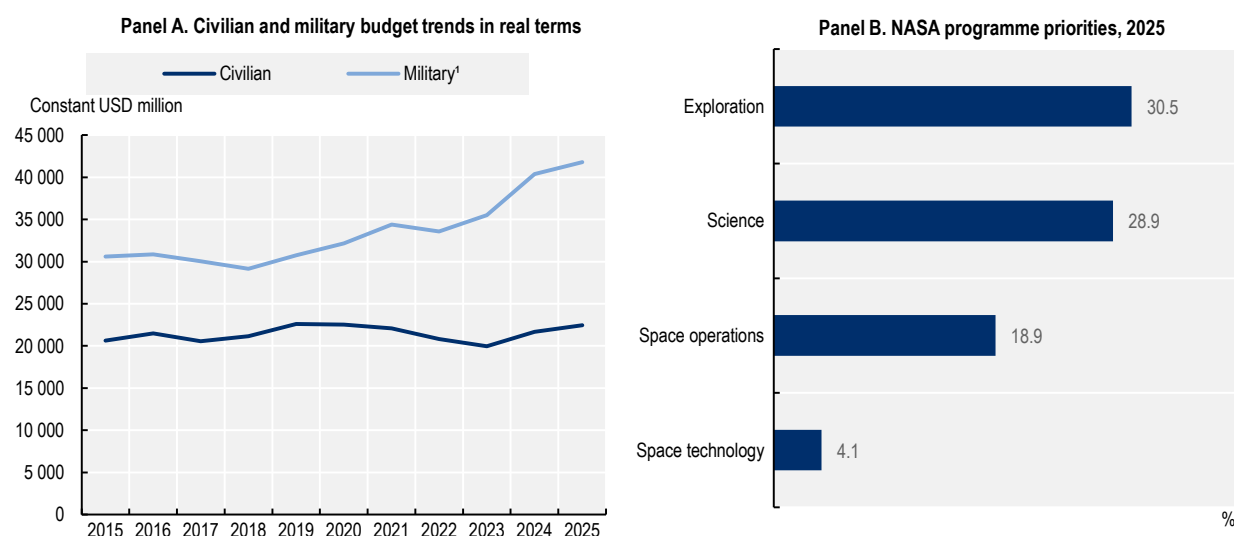
The United States at a glance

Indicator	United States	OECD average
First satellite in orbit	Explorer 1 (1958)	
First successful orbital launch	1958 (Juno 1)	
Number of satellites in orbit, as of 23 June 2026; (Share of satellites that are privately owned)	12 436 (94%)	
Number of satellites launched since 2019 (June 2026)	14 618	
Number of spaceports	3 federal, 13 commercially licensed (two of which are co-located on a federal range), 2 exclusive-use	
Institutional civil space budget as a share of gross domestic product	0.091%	0.065%
Institutional civil space budget per capita	81.9	33.6
Space literature fractional counts per 100 000 inhabitants	2.1	2.1

The United States also has the largest government space programme globally, a well-developed space finance ecosystem, including substantial access to venture capital. In addition, 13 US spaceports are licensed for launch or re-entry activities.

Evolution of the civil and military government space budgets of the United States

In constant USD million



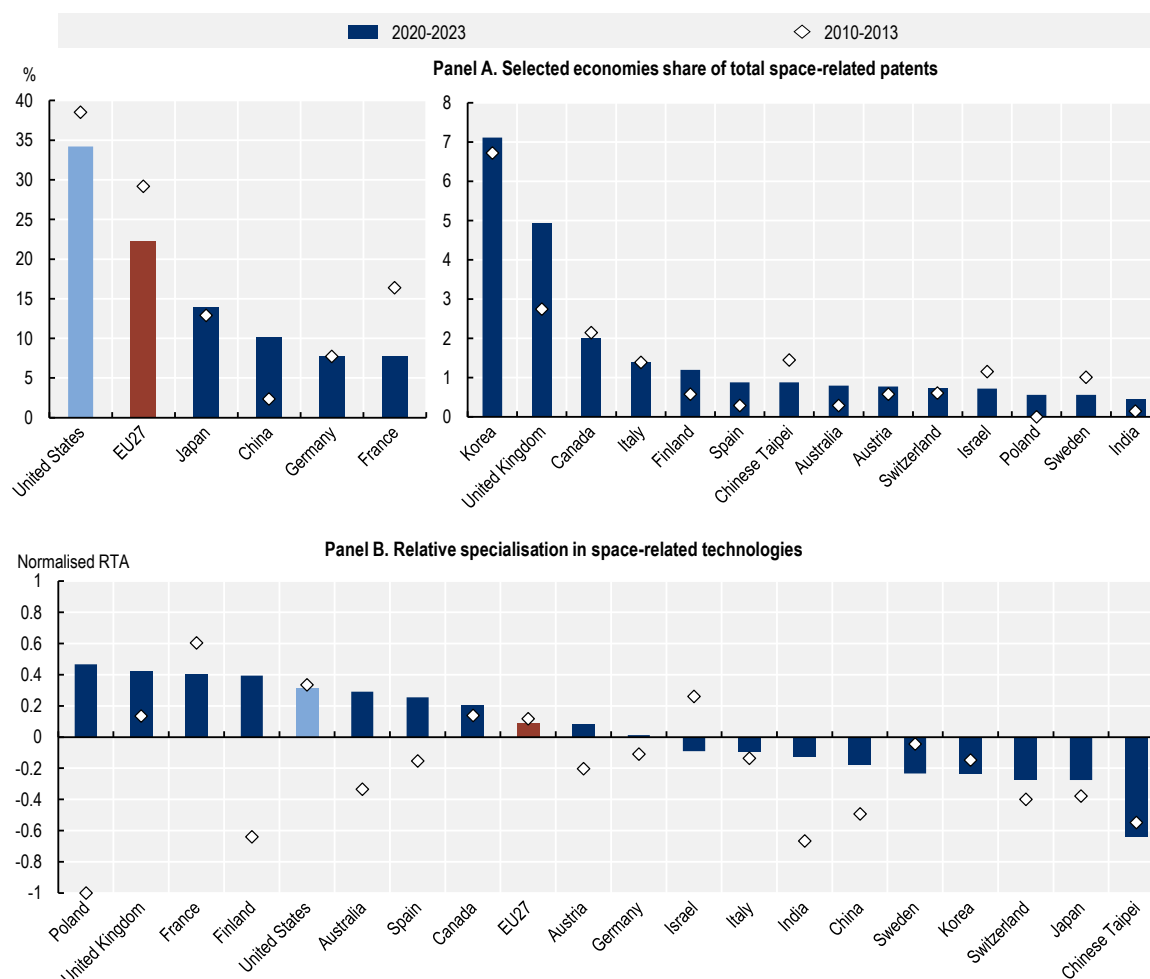
Note: The creation of the US Space Force in 2019 may have led to changes in the reporting of military space budgets. 1. Includes only budget provisions for which information is publicly available.

Source: OECD analysis based on institutional sources.

The United States space budget was estimated at USD 53.3 billion in 2025, excluding classified activities (Figure 5.35). Civilian and dual-use activities accounted for 54.8% of the total. Civilian space activities are implemented across several government departments and agencies, primarily the National Aeronautics and Space Administration (NASA) and the Department of Commerce, notably through the Office of Space Commerce for commercial space policy and the National Oceanic and Atmospheric Administration for weather satellites. Other key actors include the US Geological Survey for Landsat; the Federal Aviation Administration for commercial space transportation; the Department of Energy for nuclear power systems; the Department of Agriculture for applications such as smart agriculture and forest-fire management; and the National Science Foundation for space science and related infrastructure. NASA's main areas of activity include exploration, science, space operations, including the International Space Station, and technology development. Under the Artemis programme, NASA aims to establish a sustained human presence on the Moon by the early 2030s.

United States: Patents for space-related technologies

IP5 patent families, by priority date and applicant's location, using fractional counts



Note: IP5 patent families correspond to patent families filed in at least two offices worldwide, including at least one of the Five IP largest offices (IP5, i.e. the European Patent Office, EPO; the Japan Patent Office, JPO; the Korean Intellectual Property Office, KIPO; The China National Intellectual Property Administration, CNIPA; and the US Patent and Trademark Office, USPTO). Patents related to space-related technologies have been identified by the EPO using the EPO-PATSTAT 2025b data (<https://www.epo.org/patstat>).

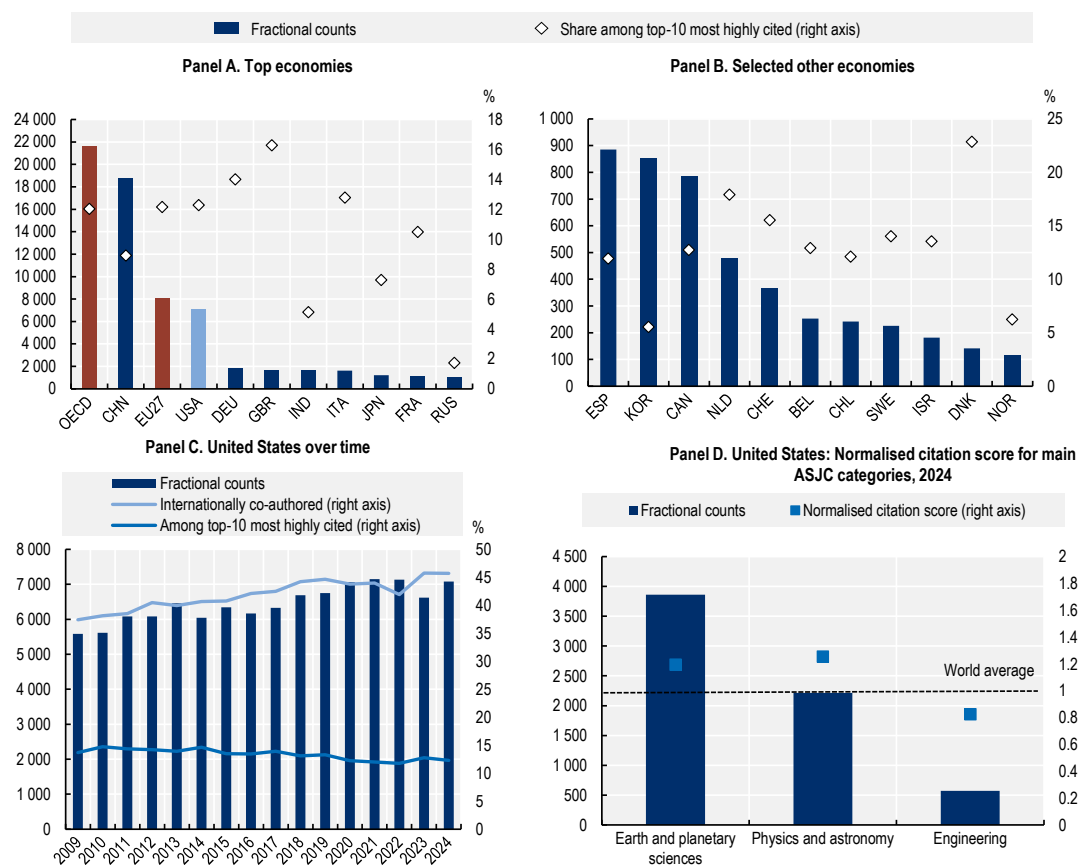
Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, June 2026.

The United States was the first country to create a space economy thematic account within the framework of national accounts, to robustly track the US space economy over time and compare it with other sectors. In 2023, private industries in the United States space economy generated an estimated USD 192.4 billion in gross output and USD 106.9 billion in value added. The space economy as a whole accounted for USD 142.5 billion in value added, equivalent to 0.5% of US GDP. The largest private-sector segments included manufacturing, covering satellites, ground equipment, positioning, navigation and timing equipment, and space vehicles, as well as direct-to-home broadcasting and satellite telecommunications. The US Bureau of Economic Analysis thematic account also includes government activity, which made a contribution to GDP broadly comparable to that of space manufacturing, at around 0.1% of US GDP, reflecting significant research and development activity in both defence and non-defence space programmes.

The United States accounted for 34.3% of global space-related patents applications in the 2020-23 period, a small decrease compared to 2010-13 (Figures 5.36), mainly filed by commercial actors in the fields of space electrical power, space system control and life support. The United States has a positive revealed technological advantage (RTA), meaning that its share in space-related patenting is higher than its share in overall patenting activities. US relative specialisation in space technology patenting has remained stable compared with 2010-13.

United States: Space-related scientific output and excellence indicators in 2024

Fractional counts



Note: Perimeter of documents based on filtered journal titles and All Science Journal Classification codes in the Scopus database.

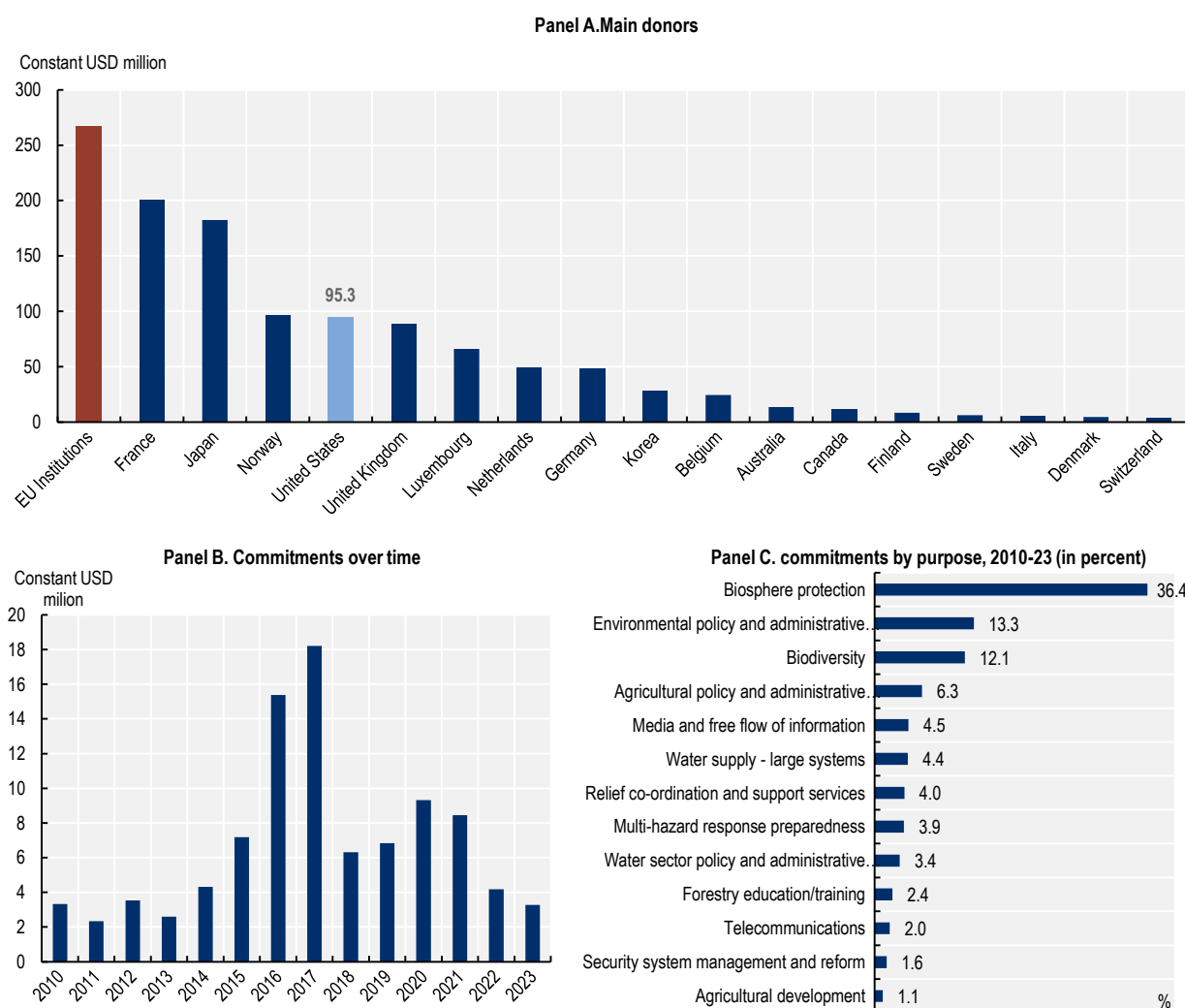
Source: OECD calculations based on Scopus Custom Data, Elsevier, Version 1.2026, April 2026.

In terms of space-related scientific production and excellence, US space literature (measured in fractional counts) has increased steadily since 2009 (Figure 5.37). The share of internationally co-authored papers has also increased over the period but is lower than the OECD average. When looking at citations, 12.7% of US space literature was in the top-10 percentile for citations in 2024. United States' normalised citation score, comparing citation patterns within specific scientific fields, above the world average in the All Science Journal Classification (ASJC) categories “earth and planetary sciences” and “physics and astronomy”.

Over the 2010-23 period, the United States allocated around USD 95.3 million in inflation-adjusted official development assistance for space-related activities (Figure 5.38). Commitments mainly targeted environmental protection, agricultural policy and water management.

United States: Space-related official development assistance

In constant USD million (base year=2023)



Source: Calculations based on OECD (2025⁽¹⁾), “Creditor Reporting System (CRS)”, OECD.stat (database), <https://stats.oecd.org/Index.aspx?DataSetCode=CRS1> (accessed July 2025).

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The Space Economy at a Glance 2026

As space activities expand and become embedded in everyday economic and public-service functions, governments need better evidence to understand their benefits, risks and policy implications. *The Space Economy at a Glance 2026* provides a compact and comparative assessment of recent developments across the space economy. Space systems support communications, positioning and timing, earth observation, weather monitoring, disaster response, transport, agriculture, finance, science, security and a growing range of commercial services. Drawing on OECD indicators and policy analysis, the report examines the public and private forces shaping this transformation, including government space budgets, civil space R&D, private finance, entrepreneurship, scientific and technological capabilities, market dynamics, satellite-enabled applications and emerging approaches to measuring the economic footprint of space activities. It shows how governments remain central to market creation and strategic capability-building, even as commercial actors increasingly operate and shape orbital activity. The report also highlights the policy challenges associated with expanding space use, including orbital congestion, debris, market concentration, supply-chain dependencies, access to essential data and signals, and the resilience of space infrastructure. The report supports governments in designing innovative, competitive, secure and responsible space ecosystems that deliver broad public value for economies and societies.



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